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Msys functions

Synopsis

The functions prefixed with Msys make up the System module. The System module supports the allocation and inquiry of systems. The system control represents a physical board, most commonly an imaging frame grabber. The system control also allows you to access the graphics controller and Host CPU. You can use the system control to specify the physical device to access, and to set some system-wide settings. Once the system control is configured, you can add other components to your application, such as a digitizer control, to control specific aspects of the device.

Functions

MsysAlloc
MsysControl
MsysControlFeature
MsysFree
MsysGetHookInfo
MsysHookFunction
MsysInquire
MsysInquireFeature

MsysAlloc

Synopsis

Allocate an Aurora Imaging Library system.

Syntax

```
AIL_ID MsysAlloc(
    AIL_ID ContextAppId,                //in
    AIL_CONST_TEXT_PTR SystemDescriptor, //in
    AIL_INT SystemNum,                  //in
    AIL_INT64 InitFlag,                 //in
    AIL_ID *SysIdPtr                    //out
)
```

Description

This function allocates an Aurora Imaging Library system so that it can be used by subsequent Aurora Imaging Library functions. This function can allocate an Aurora Imaging Library system, which consists of a Zebra imaging board (or third-party board), the Host CPU and memory, and any available graphics controller. Alternatively, this function can allocate a Host-type system, which consists of the Host CPU and memory, and any available graphics controller.

A system must be allocated before any buffers, displays, or digitizer can be allocated on it. Before allocating a system, an application must be allocated, using MappAlloc() or MappAllocDefault(). To use the default system, you must allocate it using M_SYSTEM_DEFAULT.

Note, upon allocation of an application, a default Host system is automatically allocated. Rather than using **MsysAlloc**() to allocate a Host system, you can use this default Host system, by specifying M_DEFAULT_HOST wherever a Host system identifier is required.

After allocating the system, you should check if the operation was successful, using MappGetError() or by verifying that the system identifier returned is not **M_NULL** (or *nullptr* if M_UNIQUE_ID was specified).

When the system is no longer required, release it using MsysFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the system's lifetime and you must not manually free it.

Parameters

ContextAppId

Specifies the identifier of the application context to use.

For specifying the application context	
Value	Description
M_DEFAULT	Specifies the current application context.
Application Context Identifier	Specifies the application context identifier.

SystemDescriptor INQ

Specifies the type of system to allocate. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of system to allocate																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_SYSTEM_CLARITY_UHD	Allocates an Aurora Imaging Library Clarity UHD system.	c														
M_SYSTEM_CONCORD_POE	Allocates an Aurora Imaging Library Concord POE system. > Note that to use Zebra Concord PoE for acquisition, you must allocate and use an Aurora Imaging Library GigE Vision system (M_SYSTEM_GIGE_VISION) instead; refer to information denoted for a GigE Vision system. You only need to allocate and use an Aurora Imaging Library Concord PoE system (M_SYSTEM_CONCORD_POE) to use the other functionality on the board and to inquire about the board itself. So information in this reference, for use with an Aurora Imaging Library Concord PoE system, is denoted Aurora Imaging Library Concord PoE with ToE since it is typically only applicable to this model of the board. If the information is applicable to the Zebra Concord PoE base model, it will be explicitly specified. For more information, see the Using an Aurora Imaging Library Concord PoE system with the Zebra Concord PoE base model section of the Zebra Concord PoE chapter in the Hardware-specific Notes.		d												
M_SYSTEM_DEFAULT	Specifies the default system. This value is set during Aurora Imaging Library installation and using the Aurora Imaging Configurator utility. Note that the end-user can always change this value.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_SYSTEM_GENTL	Allocates an Aurora Imaging Library GenTL system. This allocation opens general communication with a GenTL Producer (library).			e												
M_SYSTEM_GEVIQ	Allocates an Aurora Imaging Library GevIQ system.				f											

M_SYSTEM_GIGE_VISION	Allocates an Aurora Imaging Library GigE Vision system. This allocation opens general communication with all the GigE Vision-compliant cameras (or devices) found on your subnet (through one or more Gigabit Ethernet network adapters in your computer).					g										
M_SYSTEM_HOST	Specifies a Host system. Note that a Host system has no hardware-supported acquisition capabilities, however some acquisition capabilities are available when using a simulated digitizer, allocated using MdigAlloc() with M_EMULATED.						a									
M_SYSTEM_INDIO	Allocates an Aurora Imaging Library Indio system. If you need to capture images from a GigE Vision camera, you will need to allocate M_SYSTEM_GIGE_VISION as well.							h								
M_SYSTEM_IRIS_GTX	Allocates an Aurora Imaging Library Iris GTX system.								i							
M_SYSTEM_RADIENTEVCL	Allocates an Aurora Imaging Library Radient eV-CL system.										j					
M_SYSTEM_RAPIXOCL	Allocates an Aurora Imaging Library Rapixo CL system.											k				
M_SYSTEM_RAPIXOCOF	Allocates an Aurora Imaging Library Rapixo CoF system.												l			
M_SYSTEM_RAPIXOCXP	Allocates an Aurora Imaging Library Rapixo CXP system.													m		
M_SYSTEM_USB3_VISION	Allocates an Aurora Imaging Library USB3 Vision system. This allocation opens general communication with all the USB3 Vision-compliant cameras (or devices) connected to your computer.														n	
M_SYSTEM_V4L2	Allocates a Video4Linux2 system. This allocation opens general communication with all the V4L2-compliant cameras (or devices) connected to your computer.															b

		c	d	e	f	g	a	h	i		j	k	l	m	n	b
AIL_TEXT("*dailshm://[Passkey:]localhost[:Port]/AILSystemType*") 1	Allocates a DAIL remote system for a separate process on the local computer, using the DAIL SHM protocol. This protocol allows you to communicate between a client and server process on the same computer using shared memory. To allocate a DAIL remote system on a computer, that computer must have a valid DAIL installation. When specifying the string that indicates the local computer: 1. If required, replace *[Passkey]* with the passkey for the DAIL server on the local computer. The passkey is an alphanumeric string of up to 16 characters in length. This passkey is set by the Aurora Imaging Configurator utility in the server settings page. If a passkey was not set for the DAIL server, omit the passkey and the ":" that follows it. 2. Following "localhost:", replace *[Port]* with the port that the local computer should access to initiate new connections with the DAIL server, unless the default connection port is the listening port of the DAIL server; in which case, omit the port number and the ":" that follows it. 3. Replace *AILSystemType* with any valid Aurora Imaging Library system type listed in this table, such as M_SYSTEM_RAPIXOCXP. The "://" and "/" must be included. > The DAIL SHM protocol only supports connections to localhost. Prior to allocating a DAIL remote system, certain conditions must be met. For more information, see Chapter 63: Distributed Aurora Imaging Library.															
	System specific															

		Note that to allocate this system, the Distributed Aurora Imaging Library server cannot be running as a service on the remote computer; you must either set it to run at logon, or start it manually. To set it to run at logon, open the Aurora Imaging Configurator utility and select **Run at every logon with user credentials** in the **Server Settings** pane, found under the **Distributed Aurora Imaging Library** item. To start it manually, you must logon to the remote computer and, from the Aurora Imaging Configurator utility, click on the **Start Server** button, found in the **Server Settings** pane under the **Distributed Aurora Imaging Library** item. For more information, see the Setting up the Distributed Aurora Imaging Library server on remote computers subsection of the Preparing computers for Distributed Aurora Imaging Library section of Chapter 63: Distributed Aurora Imaging Library.	c													
AIL_TEXT("*dailtcp://[Passkey:]RemoteComputerName[:Port]/AILSystemType*") 1		Allocates a DAIL remote system on a remote or local computer, using the TCP/IP protocol. To allocate a DAIL remote system on a computer, that computer must have a valid DAIL installation. When specifying the string that indicates the remote computer (or local computer for a DAIL server also running locally): 1. If required, replace *[Passkey]* with the passkey for the remote computer. The passkey is an alphanumeric string of up to 16 characters in length. This passkey is set by the Aurora Imaging Configurator utility in the server settings page. If a passkey was not set for the remote computer, omit the passkey and the ":" that follows it. If you are attempting to connect to a local computer, specify the passkey for the DAIL server on the local computer. 2. Replace *RemoteComputerName* with the remote computer's name or IP address; Aurora Imaging	c	d	e	f	g	a	h	i		j	k	l	m	n	b

	Library supports both IPv4 and IPv6 addresses. A typical IPv4 string has the format n.n.n.n, where *n* is a number between 0 and 255. A typical IPv6 string has the format x:x:x:x:x:x:x:x, where *x* is a hexadecimal number between 0000 and FFFF. If you are supplying an IPv6 address, you must use square brackets to separate the address from the port. If you are attempting to connect to the local computer, set this to localhost. 3. If required, replace *[Port]* with the port that the local computer should access on the remote computer to initiate new connections, unless the default connection port is the listening port of the DAIL server on the remote computer; in which case, omit the port number and the ":" that follows it. If you are attempting to connect to a local computer, specify the listening port of the DAIL server on the local computer. 4. Replace *AILSystemType* with any valid Aurora Imaging Library system type listed in this table, such as M_SYSTEM_RAPIXOCXP. The "://" and "/" must be included. Prior to allocating a DAIL remote system, certain conditions must be met. For more information, see Chapter 63: Distributed Aurora Imaging Library.														
	System specific														

		Note that to allocate this system, the Distributed Aurora Imaging Library server cannot be running as a service on the remote computer; you must either set it to run at logon, or start it manually. To set it to run at logon, open the Aurora Imaging Configurator utility and select **Run at every logon with user credentials** in the **Server Settings** pane, found under the **Distributed Aurora Imaging Library** item. To start it manually, you must logon to the remote computer and, from the Aurora Imaging Configurator utility, click on the **Start Server** button, found in the **Server Settings** pane under the **Distributed Aurora Imaging Library** item. For more information, see the Setting up the Distributed Aurora Imaging Library server on remote computers subsection of the Preparing computers for Distributed Aurora Imaging Library section of Chapter 63: Distributed Aurora Imaging Library.	c														

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

SystemNum INQ

Specifies the number (rank) or user-defined name of the target board of the specified system type. This parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the number of the target board																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default board. > The default board is set in the Aurora Imaging Configurator utility.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DEV*n* +	Specifies the device number (rank) of the board. You can set *n* to one of the following values: 0 <= *n* <=15.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															
	Note that only M_DEV0 is valid.					g									n	b

AIL_TEXT("*BoardIdentifierString*") [1]	Specifies the user-defined name of the board. When allocating the board with a name, the InitFlag must be set to M_DEVICE_NAME. The name must also have been previously assigned and written to the board. Typically, this is done from a different executable that allocates a system for the board using M_DEVn, and then assigns a name to the board using MsysControl() with M_DEVICE_NAME. Note, if using a C compiler, you must pass the value as M_PTR_TO_AIL_INT(AIL_TEXT("*BoardIdentifierString*")).		d														

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Combination value for M_DEVn.

You can add the following value to the above-mentioned value to identify the GenTL library for which to allocate an Aurora Imaging Library GenTL system.

All third-party proprietary GenTL libraries installed on your computer are sorted and indexed by Aurora Imaging Library. Use the GenTL Producer index number to identify the library that your Aurora Imaging Library system should use to communicate with the hardware devices on the specified transportation layer. These libraries are indexed and sorted for reference by Aurora Imaging Library (the GenTL Consumer). In all cases, the same number and type of libraries should be installed on every computer that will run your application; the order in which they are installed, however, is not important.

To determine the number of GenTL libraries installed on your computer, use MappInquire() with M_GENTL_PRODUCER_COUNT or use the **General Default Values** page of the Aurora Imaging Configurator utility.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For identifying the instance of the GenTL library to use																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GENTL_PRODUCER(AIL_INT IndexValue)	Specifies the GenTL Producer (library) for which to allocate this Aurora Imaging Library GenTL system.			e												
	Parameters															
	IndexValue Specifies the index of the GenTL Producer.			e												
	0 <= *Value* <= 127 — Specifies the index.			e												

InitFlag INQ

Specifies how to perform the allocation.

To specify the type of initialization to perform on the selected system, set this parameter to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of initialization setup																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default value.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															

		Same as M_COMPLETE.	c	d		f	g	a	h	i		j	k	l	m	n	
		Same as M_MIXED.			e												
		Specifies to initialize the Video4Linux2 system to its default state.															b
M_CL	Specifies to initialize Camera Link transport layer technology.				e												
M_COMPLETE +	Specifies to initialize the system completely; the system is initialized to its default state and any required resident software is downloaded. At least one complete initialization is necessary after you power-up your system.		c	d		f	g	a	h	i		j	k	l	m	n	
M_CXP	Specifies to initialize CoaXPress transport layer technology.				e												
M_GEV	Specifies to initialize Ethernet transport layer technology.				e												
M_MIXED	Specifies to initialize the transport layer technology specified by the GenTL Producer (library).				e												
M_U3V	Specifies to initialize USB transport layer technology.				e												

Combination value that can be used alone or as combination value for M_COMPLETE.

You can use the following value on its own, or add it to the above-mentioned value, to set that the target board is identified using its user-defined name.

> This InitFlag is also available for the Zebra Concord PoE base model.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying that the target board is identified using its user-defined name																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEVICE_NAME	Specifies that the SystemNum parameter identifies the target board using its user-defined name.		d													

SysIdPtr

Specifies the address of the variable in which to write the system identifier or specifies the data type that the function should use to return the system identifier.

For retrieving the identifier or specifying how to return it	
Value	Description
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated system; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated system; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_SYS_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the system (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.
Address in which to write the identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated system. If allocation fails, **M_NULL** is written as the identifier.

Return value

The returned value is the system identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_SYS_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Remark

- If you are creating a DLL that includes a call to **MsysAlloc**(), ensure that the call is not made from the `DllMain()` function, because **MsysAlloc**() might load a required DLL and you cannot load a DLL from `DllMain()`. If necessary, call **MsysAlloc**() from an initialization function in your DLL instead.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MsysControl

Synopsis

Control a system setting.

Syntax

```
void MsysControl(
    AIL_ID SysId,           //out
    AIL_INT64 ControlType,  //in
    AIL_DOUBLE ControlValue //in
)
```

Description

This function allows you to control the specified system setting.

> Note that, when a control type has only one supported control value on a given system, it is not documented in this function because it cannot be changed to another value. Instead, it can only be inquired.

To inquire the current value of a particular system setting, use MsysInquire().

You can also interactively control and test most of the system settings in real-time, using Aurora Imaging Intellicam's Feature Browser.

> Note that presented above is an example prototype of this function. You can call MsysControl() with the same prototype as any of the type-specific functions below, without using the function's type-specific name; that is, use MsysControl() and not, for example, MsysControlInt64(). The data is automatically cast into the appropriate type. This is true except when using a C compiler. When using a C compiler, you must always call this function with its type-specific name (you cannot use the standard name of this function).

Parameters

SysId

Specifies the system identifier. This parameter should be set to one of the following values:

For the system identifier	
Value	Description
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().

ControlType

Specifies the type of setting to control.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the new value to assign to the system setting specified by the ControlType parameter.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **ControlType** and **ControlValue** parameters.

- For general system settings
- For discovering connected devices
- For routing I/O signals and setting their mode
- For resetting CXP connection error counts
- For setting the state of specified user-bits in a static-user-output register
- For controlling the settings of a timer
- For controlling the settings of a rotary decoder
- For controlling the settings of a data latch associated with rotary encoders
- For UART settings
- For configuring and sending an action command to a GigE Vision camera
- For specifying a Trigger-over-Ethernet packet (action command or GigE Vision software trigger) for transmission using a ToE module

The following control types allow you to control the general system settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For general system settings																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_ALLOCATION_OVERSCAN		Sets whether image buffers, allocated on the system, are allocated by default with an overscan region. Specify the size of the overscan region using M_ALLOCATION_OVERSCAN_SIZE. Note that you can override this setting when allocating a buffer using the M_ALLOCATION_OVERSCAN attribute. INQ															
	M_DEFAULT	Same as M_ENABLE.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	M_DISABLE	Specifies that image buffers allocated on the system will have no overscan region.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	M_ENABLE	Specifies that image buffers are allocated on the system with an overscan region.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_ALLOCATION_OVERSCAN_SIZE		Sets the size of the overscan region, added around all subsequently allocated image buffers (MbufAlloc...()). The overscan settings previously allocated image buffers are not changed. For more information, see the Buffer overscan region section of Chapter 23: Data buffers. To enable or disable the allocation of an overscan region, change the setting of M_ALLOCATION_OVERSCAN. INQ	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default size of the overscan region.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
		System specific															
		The default size is 8 pixels.	c		e		g	a		i		j	k	l	m	n	b

Value	Specifies the size of the overscan region, in pixels. For example, if you specify a size of 2, a 2-pixel overscan border is added around subsequently allocated image buffers.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DEBUG_LOG_INFO	Logs a user-defined event in Aurora Imaging Gecho Viewer. The event will appear in its thread's timeline. This control can be useful for debugging or monitoring specific sections of code or internal data for your application.				f	g	a					k	l	m	n	
Value	Specifies the value that will be logged for the user-defined event.				f	g	a					k	l	m	n	
M_DEFAULT_PITCH_BYTE_MULTIPLE	Sets the pitch (or stride) multiple (in bytes) for the buffers allocated on the system. The pitch is the number of bytes between the beginnings of any two adjacent rows of the buffer's data. The pitch multiple is the factor by which the pitch of a buffer must be divisible. When a buffer is allocated, if the specified width is not a multiple of this value, padding is added to the buffer so that the buffer's pitch meets this constraint. You would typically change this value when you need to reduce the padding of your buffer. Note that, allocating a buffer with an overscan region can also be a cause of buffer padding. For more information, refer to M_ALLOCATION_OVERSCAN and the Accessing an Aurora Imaging Library buffer directly section of Chapter 23: Data buffers. When dealing with an on-board operation, changing the pitch multiple might force on-board operations to be performed off-board due to hardware limitations, thereby increasing the amount of time the operation might take. INQ	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value for the pitch multiple. The default value is highly Aurora Imaging Library system-dependent. Always inquire the current pitch multiple before setting it, using MsysInquire() with M_DEFAULT_PITCH_BYTE_MULTIPLE.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
Value	Specifies the pitch multiple, in bytes.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DEVICE_NAME	Sets a user-defined name for the board. The user-defined name is written to the board, making it persistent, and allows the board to be allocated using this user-defined name in the future. The name is typically assigned and written to the board from a separate executable that allocates a system for the board with M_DEVn and then calls **MsysControl**() with the control type (M_DEVICE_NAME). This control type is also available for the Zebra Concord PoE base model.		d													

	AIL_TEXT("*DeviceName*") 1	Specifies the name of the board. Note that this must be a unique name with a string that can be up to M_DEVICE_NAME_MAX_SIZE characters. > Note, if using a C compiler, you must pass the value as M_PTR_TO_AIL_INT(AIL_TEXT("*DeviceName*")).		d												
M_GC_FEATURE_BROWSER +		[This is only applicable to Windows] Sets whether to open or close a dialog box that allows you to view and edit the GenTL SFNC-compliant system and interface configuration information interactively. This window is referred to as the Feature Browser.			e											
	M_DEFAULT	Same as M_OPEN.			e											
	M_CLOSE	Closes Feature Browser.			e											
	M_OPEN +	Opens Feature Browser. You must specify a combination value from the following table: • To set whether the Feature Browser should be synchronous or asynchronous			e											
M_GC_FEATURE_EXECUTE_POLLING_MODE		Sets whether the executable feature is executed synchronously or asynchronously. INQ			e	f	g						l	m		
	M_DEFAULT	Same as M_MANUAL.			e	f	g						l	m		
	M_AUTOMATIC	Specifies that the specific executable camera feature is executed synchronously. When in automatic mode, the executable feature is executed synchronously. Aurora Imaging Library polls the feature to establish if the executable feature has completed, returning control only when the operation is complete. Get the polling interval using M...InquireFeature() with M_FEATURE_POLLING_INTERVAL.			e	f	g						l	m		
	M_MANUAL	Specifies that the executable camera feature is executed asynchronously. When in manual mode, the executable feature is executed asynchronously, and control is returned once the executable operation begins. To determine whether the executable feature completed, use M...InquireFeature() with M_FEATURE_EXECUTE_COMPLETED.			e	f	g						l	m		
M_LED_USER		Sets the color of the user LED on your Zebra Iris GTX. INQ							i							
	M_DEFAULT	Same as M_OFF.							i							

	M_GREEN	Specifies to turn the user LED green.							i							
	M_OFF	Specifies to turn the user LED off.							i							
	M_ORANGE	Specifies to turn the user LED orange.							i							
	M_RED	Specifies to turn the user LED red.							i							
M_MODIFIED_BUFFER_HOOK_MODE		Sets whether to run user-defined functions hooked to a buffer modification on separate threads, up to the number of CPU cores present in the computer. This is particularly useful when functions are hooked using MdigProcess(). INQ	c		e	f	g				j	k	l	m	n	b
	M_DEFAULT	Same as M_SINGLE_THREAD.	c		e	f	g				j	k	l	m	n	b
	M_MULTI_THREAD +	Specifies to run user-defined functions hooked to a buffer modification on separate threads. The hooked functions are executed by any available CPU core. Aurora Imaging Library cannot guarantee the processing order of any image on such a computer, because they are executed concurrently. If required, use Microsoft Windows functions to synchronize the threads. Note that by default, this control value will allocate, as necessary, up to 16 threads, or the total number of CPU cores in the computer, whichever is least.	c		e	f	g				j	k	l	m	n	b
	M_SINGLE_THREAD	Specifies that only one thread should be created and that all user-defined functions hooked to buffer modifications are run on the same thread.	c		e	f	g				j	k	l	m	n	b
M_POWER_OVER_CABLE +		Sets whether the board provides power to connected devices. INQ		d				a						m		
		System specific														
		You must specify a combination value from the following table: • To specify which connector or port to control						a						m		
		[This is only applicable to Zebra 4Sight] For a Host system, this sets whether PoE (Power over Ethernet) is enabled on the specified port. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								

		For Zebra Concord PoE, this sets whether PoE (Power over Ethernet) is enabled on all ports. This control type is also available for the Zebra Concord PoE base model.		d													
		For Zebra Rapixo CXP, this sets whether PoCXP (Power over CXP) is automatically enabled when a PoCXP-compliant camera is connected to the specified connector.													m		
	M_DEFAULT	Specifies the default value.		d				a							m		
		System specific															
		[This is only applicable to Zebra 4Sight] For Zebra 4sight EV6/EV7, the default value is set in the UEFI. > On legacy computers, this was referred to as a BIOS.		d				a							m		
		For Zebra Concord PoE, this is the same as M_ON.		d													
		For Zebra Rapixo CXP, this is the same as M_AUTOMATIC.													m		
	M_AUTOMATIC	Specifies to automatically enable or disable PoCXP on this CXP connector, depending on detected device support.													m		
	M_OFF	Specifies not to provide power to connected devices.		d				a							m		
		System specific															
		[This is only applicable to Zebra 4Sight] For Zebra 4sight EV6/EV7, this specifies to disable PoE on the specified port.						a									
		For Zebra Concord PoE, this specifies to disable PoE on all ports.		d													
		For Zebra Rapixo CXP, this specifies to manually disable PoCXP on the specified CXP connector.													m		
	M_ON	Specifies to provide power to connected devices.		d				a							m		
		System specific															
		[This is only applicable to Zebra 4Sight] For Zebra 4sight EV6/EV7, this specifies to enable PoE on the specified port.						a									

		For Zebra Concord PoE, this specifies to enable PoE on all ports.		d													
		For Zebra Rapixo CXP, this specifies to manually enable PoCXP on the specified CXP connector. You should typically use the default setting (M_AUTOMATIC) which specifies to automatically detect whether the connected camera supports PoCXP. If you manually enable PoCXP and connect a device that is not fully CXP-compliant, the device might be damaged.												m		
	M_RESET	Specifies to reset an over- or under-current condition. To learn whether there is an over- or under-current condition, use MsysInquire() with M_POWER_OVER_CABLE_STATUS. > Typically, to prevent the over- or under-current condition from immediately recurring, you should first restart PoCXP for this connector by setting this control type to M_OFF and then returning it to its previous setting (M_AUTOMATIC is recommended). If the condition persists, discontinue using PoCXP with your camera by leaving this control type set to M_OFF.												m		
M_THREAD_MODE		Sets whether threads allocated on the system can execute in asynchronous mode. If you specify that threads on the system can execute in asynchronous mode, the MthrControl() M_THREAD_MODE control type setting of each of its threads is taken into account; otherwise, their M_THREAD_MODE control type setting is ignored. INQ	c	d	e	f	g	a	h	i		j	k	l	m	n	b
		System specific															
		This inquire type is also available for the Zebra Concord PoE base model.		d													
	M_DEFAULT	Specifies the default value.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
		System specific															
		Same as M_ASYNCHRONOUS.	c		e	f	g					j	k	l	m	n	b
		Same as M_SYNCHRONOUS.						a		i							
	M_ASYNCHRONOUS	Specifies that threads allocated on the system can execute in asynchronous mode. In asynchronous mode, control is returned to the Host immediately after an Aurora Imaging Library function is sent to the processor of the system (when the system and function allow an immediate return).	c		e	f	g			i		j	k	l	m	n	b

	M_SYNCHRONOUS	Specifies that threads allocated on the system can only execute in synchronous mode. In synchronous mode, the execution of an Aurora Imaging Library function sent to the processor of a system must be completed (execution terminated) before returning control to the Host.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_TIMEOUT		Sets the maximum amount of time for the Host to wait for a synchronous function to return before generating a time-out error, in sec. Note that this value only works with functions that are performed on-board, such as MbufCopy() when copying from (or to) on-board memory or functions using an on-board processing FPGA. INQ	c	d	e	f	g			i		j	k	l	m	n	b
		System specific															
		This inquire type is also available for the Zebra Concord PoE base model.		d													
	M_DEFAULT	Specifies the default value.	c	d	e	f	g			i		j	k	l	m	n	b
		System specific															
		The default value is 20 sec.	c		e		g			i		j	k	l	m	n	b
	M_INFINITE	Waits indefinitely.								i							
	Value	Specifies the time to wait, in sec.	c	d	e	f	g			i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Combination values for M_POWER_OVER_CABLE.

You must add one of the following values to the above-mentioned value to specify which connector or port to control.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which connector or port to control																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ALL	Specifies to control all relevant connectors or ports on your board or industrial computer.						a							m		
M_CONNECTION*n*	Specifies to control connector *n*, where *n* corresponds to a physical connector or port on your board or industrial computer.						a							m		
	System specific															
	[This is only applicable to Zebra 4Sight] For a Zebra 4sight EV6/EV7, *n* can be a value from 1 to 4, depending on the number of PoE-enabled Ethernet ports on the version of your industrial computer.						a									

		For Zebra Rapixo CXP, *n* can be a value from 0 to 3, depending on the number of CXP connectors on the version of your board.													m			

Combination values for M_GC_FEATURE_BROWSER.

You can add one of the following values to the above-mentioned value to specify which instance of the GenTL interface configuration file (XML file) is associated with the feature.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which configuration information file to access																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GENTL_INTERFACE_NUMBER(AIL_INT IndexValue)	Specifies which instance of the GenTL interface configuration file is associated with the feature.			e												
	Parameters															
	IndexValue Specifies the index of the GenTL interface configuration file.			e												
	0 <= *Value* <= *MsysInquire*(M_GENTL_INTERFACE_COUNT) Specifies the index.			e												
M_GENTL_SYSTEM	Specifies to display the GenTL system configuration information.			e												

Combination values for [when ControlType=M_GC_FEATURE_BROWSER: M_OPEN].

You must add one of the following values to the above-mentioned value to set whether the Feature Browser should be synchronous or asynchronous.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying whether Feature Browser should be synchronous or asynchronous																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ASYNCHRONOUS	Specifies that this function returns immediately once a Feature Browser window opens.			e												
M_SYNCHRONOUS	Specifies that this function is blocked until a Feature Browser window closes.			e												

Combination value for [when ControlType=M_MODIFIED_BUFFER_HOOK_MODE: M_MULTI_THREAD].

You can add the following value to the above-mentioned value to set the number of hook threads to allocate.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the number of hook threads																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Value >= 1	Specifies the number of hook threads to allocate. Note that this value cannot exceed the number of CPU cores available.			e		g					j	k	l	m	n	b

The following control type allows you to refresh the list of discovered devices.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For discovering connected devices			(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlType		Description															
	ControlValue																
M_DISCOVER_DEVICE		Refresh the list of discovered devices that can be accessed by the specified system (for example, all cameras connected to a frame grabber, or all GigE Vision devices accessible on the local subnet). After you have refreshed the list, you can inquire information about connected devices using MsysInquire() with settings from For discovering connected devices. Note that this does not affect which devices are available for you to allocate using MdigAlloc(); you do not need to discover a device before allocating it.															
		System specific															
		This can trigger an M_CAMERA_PRESENT event.				f	g									n	
	M_DEFAULT	Specifies the default mode of operation for M_DISCOVER_DEVICE.			e	f	g							l	m	n	b

The following control types allow you to set the mode and the routing of your Zebra imaging board's I/O signals (such as, auxiliary signal). Once the routing and mode are determined for an I/O signal, the Aurora Imaging Library function that you should use to act upon an input signal or setup the source of an output signal depends on the functionality. For example, you can use **MsysControl**() with M_USER_BIT... control types, MdigControl() with M_GRAB_TRIGGER... control types, or MdigControl() with M_TIMER... control types.

Note that for other Zebra imaging boards that have auxiliary I/O signals, but are not supported with the constants below, see MdigControl().

Each Zebra imaging board and Aurora Imaging Library driver has its own list of limitations regarding the signals you can control with this function. While general limitations are listed in the table below, for a complete list of the available signals and their limitations, see the Connectors and signal names section of the Aurora Imaging Library Hardware-specific Notes chapter for your Zebra imaging board or Aurora Imaging Library driver.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For routing I/O signals and setting their mode			(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlType		Description															
	ControlValue																
M_IO_DEBOUNCE_TIME +		Sets the amount of time that the specified auxiliary input signal is debounced. INQ You must specify a combination value from the following table: • To set the type and number of the I/O signal to affect															
		System specific															

		[This is only applicable to Zebra 4Sight] When a signal is debounced on Zebra 4Sight EV6/EV7, after detecting a valid input signal edge any other transitions are considered noise and are suppressed for the specified time, to ignore any additional transitions caused by the contact bounce of mechanical switches.					a								
		When a signal is debounced on Zebra Iris GTX, after detecting a valid input signal edge any other transitions are considered noise and are suppressed for the specified time, to ignore any additional transitions caused by the contact bounce of mechanical switches. To remove problems occurring at the edge of a signal, use M_IO_GLITCH_FILTER_STATE.							i						
Value >= 0	Specifies the minimum amount of time to ignore any additional signal transitions after accepting a signal transition, in nsec.		d				a	h	i						
		System specific													
		[This is only applicable to Zebra 4Sight] In the case of Zebra 4Sight EV6/EV7, the maximum value is 167772150. Note that the number must be a multiple of 30.					a								
		In the case of Zebra Iris GTX, the maximum value is 268435440. Note that the number must be a multiple of 16.							i						
M_IO_GLITCH_FILTER_STATE +	Sets whether to enable a glitch filter on input signals. A glitch is an unexpected signal transition of a short duration. Enabling the glitch filter will eliminate glitches of less than 500 nsec on all input signals. INQ You must specify a combination value from the following table: • To set the type and number of the I/O signal to affect		d				a	h	i						
		System specific													
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.					a								
M_DEFAULT	Same as M_DISABLE.		d				a	h	i						
M_DISABLE	Specifies not to use a glitch filter.		d				a	h	i						
M_ENABLE	Specifies to use a glitch filter.		d				a	h	i						

M_IO_INTERRUPT_ACTIVATION +	Sets the signal transition upon which to generate an interrupt, if interrupt generation has been enabled for the specified I/O signal. Use M_IO_INTERRUPT_STATE to enable interrupt generation. Note that this only applies to input signals. Note that this control type only has an effect when M_IO_INTERRUPT_STATE is enabled. INQ You must specify a combination value from the following table: • To set the type and number of the I/O signal to affect		d				a	h	i							
	System specific															
	This control type is only available with auxiliary I/O signals 3-6.								i							
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_EDGE_RISING.		d				a	h	i							
M_ANY_EDGE	Specifies to generate an interrupt upon both a low-to-high and a high-to-low signal transition.						a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_EDGE_FALLING	Specifies that an interrupt will be generated upon a high-to-low signal transition.		d				a	h	i							
M_EDGE_RISING	Specifies that an interrupt will be generated upon a low-to-high signal transition.		d				a	h	i							
M_IO_INTERRUPT_STATE +	Sets whether to generate an interrupt upon the specified transition of the I/O signal. Use M_IO_INTERRUPT_ACTIVATION to specify the transition. INQ You must specify a combination value from the following table: • To set the type and number of the I/O signal to affect		d				a	h	i							
	System specific															
	This control type is only available with auxiliary I/O signals 3-6.								i							

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_DISABLE.		d				a	h	i							
M_DISABLE	Specifies not to generate an interrupt.		d				a	h	i							
M_ENABLE	Specifies to generate an interrupt.		d				a	h	i							
M_IO_INVERTER +	Sets whether the specified I/O signal should be inverted. This causes the low portion of the signal (the delay period) to be high and the high portion of the signal (the active portion) to be low. INQ You must specify a combination value from the following table: • To set the type and number of the I/O signal to affect		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_DISABLE.		d				a	h	i							
M_DISABLE	Specifies not to invert the specified I/O signal.		d				a	h	i							
M_ENABLE	Specifies to invert the specified I/O signal.		d				a	h	i							
M_IO_SOURCE +	Sets the type of signal to route to an output signal, or a bidirectional signal set to output mode. INQ You must specify a combination value from the following table: • To set the type and number of the I/O signal to affect		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_USER_BITn.		d				a	h	i							
M_EXPOSURE	Specifies to route the exposure signal of the camera.								i							

	M_GRAB_TRIGGER_READY	Specifies to route the internal grab trigger ready signal.								i						
	M_IO_COMMAND_LIST*n* +	Specifies to route a bit of the I/O command register of I/O command list *n*, where *n* is the number of the I/O command list. You must specify a combination value from the following table: • To specify which I/O command register bit to use		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be either 1 or 2. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
		For Zebra Iris GTX, *n* must be 1.								i						
	M_ROTARY_ENCODER*n*	Specifies to route the output of rotary encoder *n*, where *n* is the number of rotary encoders available.		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be either 1 or 2. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
		For Zebra Iris GTX, *n* must be 1.								i						
	M_TIMER_STROBE	Specifies to route the internal timer strobe signal.								i						
	M_TIMER*n*	Specifies to route the output of timer *n*, where *n* is the number of timers available.		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. For Zebra 4Sight EV6/EV7, *n* can be a value from 1 to 16.						a								
		For Zebra Iris GTX, *n* can be a value from 1 to 8.								i						
	M_USER_BIT*n*	Specifies to route the state of bit n of the main static-user-output register, where *n* is the bit number.		d				a	h	i						
		System specific														

			[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
			For Zebra Iris GTX, *n* can be a value from 0 to 2.								i							

Combination value for the values listed in For routing I/O signals and setting their mode.

You must add the following value to the above-mentioned values to set the type and number of the I/O signal to affect.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type and number of the I/O signal to affect																		
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_AUX_IO*n*	Specifies to affect auxiliary signal n, where *n* is the signal number. For a list of the available auxiliary I/O signals, see the Connectors and signal names section of the Aurora Imaging Library Hardware-specific Notes chapter for your Zebra imaging board.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									

The following is used to reset CXP errors.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For resetting CXP connection error counts																		
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																	
M_TL_ERROR_CORRECTED_COUNT +		Resets the count of corrected duplicate character errors in the CXP control words. INQ You must specify a combination value from the following table: • To specify the CXP input connector																
	Value = 0	Specifies to reset the count of corrected duplicate character errors in the CXP control words. > Note that if a non-zero value is specified, an error is generated.												l	m			
M_TL_ERROR_COUNT +		Resets the count of all errors encountered. INQ You must specify a combination value from the following table: • To specify the CXP input connector												l	m			

	Value = 0	Specifies to reset the count of all errors encountered. > Note that if a non-zero value is specified, an error is generated.												I	m		
M_TL_ERROR_CTRL_CRC_COUNT +		Resets the count of CRC errors detected in a control packet. INQ You must specify a combination value from the following table: • To specify the CXP input connector												I	m		
	Value = 0	Specifies to reset the count of CRC errors detected in a control packet. > Note that if a non-zero value is specified, an error is generated.												I	m		
M_TL_ERROR_DATA_CRC_COUNT +		Resets the count of CRC errors detected in a data packet. INQ You must specify a combination value from the following table: • To specify the CXP input connector												I	m		
	Value = 0	Specifies to reset the count of CRC errors detected in a data packet. > Note that if a non-zero value is specified, an error is generated.												I	m		
M_TL_ERROR_ENCODING_COUNT +		Resets the count of protocol encoding errors detected. INQ You must specify a combination value from the following table: • To specify the CXP input connector												I	m		
	Value = 0	Specifies to reset the count of protocol encoding errors detected. > Note that if a non-zero value is specified, an error is generated.												I	m		
M_TL_ERROR_EVENT_CRC_COUNT +		Resets the count of CRC errors detected in an event packet. INQ You must specify a combination value from the following table: • To specify the CXP input connector												I	m		

	Value = 0	Specifies to reset the count of CRC errors detected in an event packet. Note that if a non-zero value is specified, an error is generated.												l	m		
M_TL_ERROR_LOCK_LOSS_COUNT +		Resets the count of lock losses encountered. INQ You must specify a combination value from the following table: • To specify the CXP input connector												l	m		
	Value = 0	Specifies to reset the count of lock losses encountered. Note that if a non-zero value is specified, an error is generated.												l	m		
M_TL_ERROR_UNCORRECTED_COUNT +		Resets the count of uncorrected duplicate character errors in the CXP control words. INQ You must specify a combination value from the following table: • To specify the CXP input connector												l	m		
	Value = 0	Specifies to reset the count of uncorrected duplicate character errors in the CXP control words. Note that if a non-zero value is specified, an error is generated.												l	m		

Combination values for the values listed in For resetting CXP connection error counts.

You must add one of the following values to the above-mentioned values to specify the CXP input connector.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the CXP input connector																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ALL	Specifies all CXP input connectors.												l	m		
M_CONNECTION*n*	Specifies the connection made at CXP input connector *n*, where *n* is the CXP input connector number from 0 to 3.												l	m		

The following control types and control values allow you to set the bits in a static-user-output register. You can route the bits to output signals or I/O signals set to output; to do so use M_IO_SOURCE with M_USER_BIT...). To establish which user-bits can be routed to a specific signal, see the connectors and signal names section of the Aurora Imaging Library Hardware-specific Notes chapter for your Zebra imaging board.

[Zebra 4Sight]

This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.

> Note that for other Zebra imaging boards that have user-bits, but are not supported with the constants below, see MdigControl().

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For setting the state of specified user-bits in a static-user-output register			(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlType		Description															
	ControlValue																
M_USER_BIT_STATE +		Sets the state of the specified bit of a static-user-output register. INQ You must specify a combination value from the following table: • To specify the bit in a static-user-output register to affect															
		System specific															
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
	M_OFF	Specifies that the specified bit is set to off.		d				a	h	i							
	M_ON	Specifies that the specified bit is set to on.		d				a	h	i							
M_USER_BIT_STATE_ALL		Sets the state of all the bits in the main static-user-output register. INQ		d				a	h	i							
		System specific															
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
	Value	Specifies a bit-encoded value that establishes the value of all the bits of the specified static-user-output register. It is recommended to specify the value in hexadecimal notation (0x), so that it is more legible to what you are setting each bit of the register.		d				a	h	i							

Combination value for M_USER_BIT_STATE.

You must add the following value to the above-mentioned value to specify the bit in a static-user-output register to affect.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the bit in a static-user-output register to affect		(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Combination value	Description															
M_USER_BIT*n*	Specifies to affect bit n of the main static-user-output register.		d				a	h	i							

The following control types and control values specify the settings for controlling timers and the signals generated from a timer (timer output signals). For more information, see the Timers and coordinating events section of Chapter 56: I/O signals and communicating with external devices.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling the settings of a timer

ControlType / ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_TIMER_ARM +	Sets whether to enable the timer arming mechanism. If timer arming is enabled, then the timer will ignore its trigger signal (M_TIMER_TRIGGER_SOURCE) until a signal transition specified using M_TIMER_ARM_ACTIVATION occurs on the signal specified using M_TIMER_ARM_SOURCE. INQ You must specify a combination value from the following table: • To specify which on-board timer to control															
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_DISABLE.		d				a	h	i							
M_DISABLE	Specifies that timer arming is disabled.		d				a	h	i							
M_ENABLE	Specifies that timer arming is enabled.		d				a	h	i							
M_TIMER_ARM_ACTIVATION +	Sets the signal transition upon which to arm the timer, if timer arming is enabled. Use M_TIMER_ARM to enable timer arming. Use M_TIMER_ARM_SOURCE to set which input signal will arm the timer. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_EDGE_RISING.		d				a	h	i							
M_ANY_EDGE	Specifies that the timer will be armed by both a high-to-low and a low-to-high signal transition.		d				a	h	i							

	M_EDGE_FALLING	Specifies that the timer will be armed upon a high-to-low signal transition.		d				a	h	i						
	M_EDGE_RISING	Specifies that the timer will be armed upon a low-to-high signal transition.		d				a	h	i						
	M_LEVEL_HIGH	Specifies that a timer is continuously armed during a high signal polarity.		d				a	h	i						
	M_LEVEL_LOW	Specifies that a timer is continuously armed during a low signal polarity.		d				a	h	i						
M_TIMER_ARM_SOFTWARE +		Issues a software trigger to arm the timer. To use this setting, the timer's arm source must be set to software (M_TIMER_ARM_SOURCE set to M_SOFTWARE). You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	M_ACTIVATE	Specifies the default behavior.		d				a	h	i						
M_TIMER_ARM_SOURCE +		Sets which input signal will arm the timer, if timer arming is enabled. Use M_TIMER_ARM to enable timer arming. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the source signal used to arm the specified timer, where *n* is the number of the auxiliary signal.		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be a value from 8 to 15.						a								

		With Zebra Iris GTX, *n* can be a value from 3 to 6.							i							
	M_EXPOSURE	Specifies to use the exposure signal as the trigger source.							i							
	M_GRAB_TRIGGER_READY	Specifies to route the internal grab trigger ready signal to the specified signal.							i							
	M_SOFTWARE	Specifies to use software to arm the specified timer.		d			a	h	i							
	M_TIMER_STROBE	Specifies to route the strobe's timer signal to the specified signal.							i							
	M_TIMER*n*	Specifies to use the output signal of the specified timer as the source signal to arm the timer, where *n* is the timer number.		d			a	h	i							
		System specific														
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be a value from 1 to 16.					a									
		With Zebra Iris GTX, *n* can be a value from 1 to 8.							i							
M_TIMER_CLOCK_ACTIVATION +		Sets the edge of the signal that will increment the clock used to control the active portion of the timer's output signal. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d			a	h	i							
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.					a									
	M_DEFAULT	Same as M_EDGE_RISING.		d			a	h	i							
	M_ANY_EDGE	Specifies that the clock will be incremented by both a high-to-low and a low-to-high signal transition.		d			a	h	i							
	M_EDGE_FALLING	Specifies that the clock will be incremented by a high-to-low signal transition.		d			a	h	i							
	M_EDGE_RISING	Specifies that the clock will be incremented by a low-to-high signal transition.		d			a	h	i							

M_TIMER_CLOCK_FREQUENCY +	Sets the frequency of the clock source signal for the active portion of the timer's output signal (M_TIMER_CLOCK_SOURCE). Note that if M_TIMER_CLOCK_SOURCE is set to M_AUX_IOn or M_ROTARY_ENCODERn, and no clock source frequency is specified, the values that are normally specified or returned in units of time will be in number of ticks (clock signal transitions on the clock source). If M_TIMER_CLOCK_SOURCE is set to M_SYSCLK, changing the frequency to any value different than the system clock's frequency will cause an error. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i						
	System specific														
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.					a								
M_UNKNOWN	Specifies that the signal is not periodic or the frequency is unknown.		d				a	h	i						
0 < *Value* <= *Frequency of* M_SYSCLK	Specifies the frequency, in Hz.		d				a	h	i						
M_TIMER_CLOCK_SOURCE +	Sets the source of the clock that drives the active portion of the specified timer's output signal. The clock source used for the delay can be different from the clock source used for the active portion of the timer's output signal. This is useful, for example, if you want to specify the delay in signal transitions and the active portion in nsec. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i						
	System specific														
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.					a								
M_DEFAULT	Same as M_SYSCLK.		d				a	h	i						
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the clock source, where *n* is the number of the auxiliary signal. If the signal is periodic, specify its frequency using M_TIMER_CLOCK_FREQUENCY.		d				a	h	i						

		System specific														
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be a value from 8 to 15.		d				a	h							
		With Zebra Iris GTX, *n* can be a value from 3 to 6.								i						
	M_ROTARY_ENCODER*n*	Specifies to use rotary decoder *n* as the clock source, where *n* is the number of the rotary decoder. If the signal is periodic, specify its frequency using M_TIMER_CLOCK_FREQUENCY.		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be either 1 or 2.						a								
		With Zebra Iris GTX, *n* must be 1.								i						
	M_SYSCLK	Specifies to use the allocated system's clock source.		d				a	h	i						
M_TIMER_DELAY +		Sets the delay between the timer trigger and the active portion of the timer's output signal. Note, an error is generated if the specified delay cannot be respected. The clock source used for the delay can be different from the clock source used for the active portion of the timer's output signal. This is useful, for example, if you want to specify the delay in signal transitions and the active portion in nsec. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	0 <= *Value* <= *Max. value*	Specifies the delay. If M_TIMER_DELAY_CLOCK_SOURCE is set to M_SYSCLK or a frequency is specified using M_TIMER_DELAY_CLOCK_FREQUENCY, this value is specified in nsec. Otherwise, it is specified in number of clock ticks (signal transitions on the clock source). Use MsysInquire() to determine the maximum possible value.		d				a	h	i						

M_TIMER_DELAY_CLOCK_ACTIVATION +	Sets the signal transition that will increment the clock used to control the delay between the timer's trigger and the active portion of the timer's output signal. If M_TIMER_DELAY_CLOCK_SOURCE is set to M_FOLLOW_TIMER_CLOCK, changing this setting to a value different than the value of M_TIMER_CLOCK_ACTIVATION will cause an error. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_EDGE_RISING.		d				a	h	i							
M_ANY_EDGE	Specifies that the clock will be incremented by both a high-to-low and a low-to-high signal transition.		d				a	h	i							
M_EDGE_FALLING	Specifies that the clock will be incremented by a high-to-low signal transition.		d				a	h	i							
M_EDGE_RISING	Specifies that the clock will be incremented by a low-to-high signal transition.		d				a	h	i							
M_TIMER_DELAY_CLOCK_FREQUENCY +	Sets the frequency of the clock source signal for the delay between the timer's trigger and the active portion of the timer's output signal (M_TIMER_DELAY_CLOCK_SOURCE). Note that if M_TIMER_DELAY_CLOCK_SOURCE is set to M_AUX_IOn or M_ROTARY_ENCODERn, and no clock source frequency is specified, the values that are normally specified or returned in units of time will be in number of ticks. If M_TIMER_DELAY_CLOCK_SOURCE is set to M_SYSCLK, changing the frequency to any value different than system clock's frequency will cause an error. If M_TIMER_DELAY_CLOCK_SOURCE is set to M_FOLLOW_TIMER_CLOCK, changing the frequency to a value different than the value of M_TIMER_CLOCK_FREQUENCY will cause an error. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_UNKNOWN	Specifies that the signal is not periodic or the frequency is unknown.		d				a	h	i							
0 < *Value* <= *Frequency of* M_SYSCLK	Specifies the frequency, in Hz.		d				a	h	i							
M_TIMER_DELAY_CLOCK_SOURCE +	Sets the source of the clock that drives the delay between the timer's trigger and the active portion of the specified timer's output signal. The clock source used for the delay can be different from the clock source used for the active portion of the timer's output signal. This is useful, for example, if you want to specify the delay in signal transitions and the active portion in nsec. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_FOLLOW_TIMER_CLOCK.		d				a	h	i							
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the clock source, where *n* is the number of the auxiliary signal.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be a value from 8 to 15.		d				a	h								
	With Zebra Iris GTX, *n* can be a value from 3 to 6.								i							
M_FOLLOW_TIMER_CLOCK	Specifies to use the clock source specified by M_TIMER_CLOCK_SOURCE. If the clock source is set to this value and M_TIMER_CLOCK_SOURCE is changed, this clock source changes as well. Additionally, any changes to M_TIMER_CLOCK_ACTIVATION and M_TIMER_CLOCK_FREQUENCY will also be reflected in M_TIMER_DELAY_CLOCK_ACTIVATION and M_TIMER_DELAY_CLOCK_FREQUENCY.		d				a	h	i							

	M_ROTARY_ENCODER*n*	Specifies to use rotary decoder *n* as the clock source, where *n* is the number of the rotary decoder.		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be either 1 or 2.						a								
		With Zebra Iris GTX, *n* must be 1.								i						
	M_SYSCLK	Specifies to use the allocated system's clock source.		d				a	h	i						
M_TIMER_DURATION +		Sets the duration for the active portion of the timer's output signal. Note, an error is generated if the specified duration cannot be respected. The clock source used for the delay can be different from the clock source used for the active portion of the timer's output signal. This is useful, for example, if you want to specify the delay in signal transitions and the active portion in nsec. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i						
		System specific														
		This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	0 <= *Value* <= *Max. value*	Specifies the duration of the active portion of the timer output signal. If M_TIMER_CLOCK_SOURCE is set to M_SYSCLK or a frequency is specified using M_TIMER_CLOCK_FREQUENCY, this value is specified in nsec. Otherwise, it is specified in number of clock ticks (signal transitions on the clock source). Use MsysInquire() to determine the maximum possible value.		d				a	h	i						
M_TIMER_OUTPUT_INVERTER +		Sets whether the output of the timer should be inverted. This causes the low portion of the signal (the delay period) to be high and the high portion of the signal (the active portion) to be low. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i						
		System specific														

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_DISABLE.		d				a	h	i							
M_DISABLE	Specifies not to invert the output of the timer.		d				a	h	i							
M_ENABLE	Specifies to invert the output of the timer.		d				a	h	i							
M_TIMER_STATE +	Sets the state of the specified timer. When a timer is enabled, the timer waits for a trigger to be received. To set the source of the trigger, use M_TIMER_TRIGGER_SOURCE. Once the trigger is received, the timer starts by outputting a low signal. This lasts for the duration of the delay period (set using M_TIMER_DELAY). The timer then changes to output a high signal for the duration of the active period (set using M_TIMER_DURATION). To invert this signal, use M_TIMER_OUTPUT_INVERTER. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_DISABLE.		d				a	h	i							
M_DISABLE	Specifies that the timer is disabled.		d				a	h	i							
M_ENABLE	Specifies that the timer is enabled.		d				a	h	i							
M_TIMER_TRIGGER_ACTIVATION +	Sets the signal variation upon which to generate a timer trigger. The timer will be triggered when the specified signal transition occurs on the source signal specified by M_TIMER_TRIGGER_SOURCE. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_EDGE_RISING.		d				a	h	i							
M_ANY_EDGE	Specifies that a timer trigger will be generated both upon a high-to-low and a low-to-high signal transition.		d				a	h	i							
M_EDGE_FALLING	Specifies that a timer trigger will be generated upon a high-to-low signal transition.		d				a	h	i							
M_EDGE_RISING	Specifies that a timer trigger will be generated upon a low-to-high signal transition.		d				a	h	i							
M_LEVEL_HIGH	Specifies that a timer trigger is continuously issued during a high signal polarity.		d				a	h	i							
M_LEVEL_LOW	Specifies that a timer trigger is continuously issued during a low signal polarity.		d				a	h	i							
M_TIMER_TRIGGER_OVERLAP +	Sets how to deal with a new trigger that occurs while the associated timer has not yet expired (both its delay and duration). INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_OFF.		d				a	h	i							
M_LATCH	Specifies that a trigger received, while the associated timer has not expired, will be latched (stored). As soon as the current timer expires, a new trigger is issued by software.		d				a	h	i							
M_OFF	Specifies that a new trigger is ignored.		d				a	h	i							
M_RESET	Specifies that a new trigger automatically resets the timer (regardless of whether it is in its delay or active period) and then restarts the timer. This process will repeat for each new trigger received.		d				a	h	i							

M_TIMER_TRIGGER_SOFTWARE +	Issues a software trigger for the specified timer. To use this setting, the timer's trigger source must be set to software (M_TIMER_TRIGGER_SOURCE set to M_SOFTWARE). You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_ACTIVATE	Specifies the default behavior.		d				a	h	i							
M_TIMER_TRIGGER_SOURCE +	Sets the trigger source for the specified timer. The timer will be triggered when the signal transition specified using M_TIMER_TRIGGER_ACTIVATION occurs on the selected source. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the trigger source for the specified timer, where *n* is the number of the auxiliary signal.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be a value from 8 to 15.						a									
	With Zebra Iris GTX, *n* can be a value from 3 to 6.								i							
M_CONTINUOUS	Specifies to run the specified timer in periodic mode; no actual trigger signal is used. The timer is automatically reset after the timer's duration expires. The timer loops between a delay and an active period.		d				a	h	i							
M_EXPOSURE	Specifies to use the exposure signal as the trigger source.								i							

	M_GRAB_TRIGGER_READY	Specifies to route the internal grab trigger ready signal to the specified signal.								i							
	M_IO_COMMAND_LIST*n* +	Specifies to use a bit of the I/O command register of I/O command list *n*, where *n* is the number of the I/O command list. You must specify a combination value from the following table: • To specify which I/O command register bit to use		d				a	h	i							
		System specific															
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be either 1 or 2.						a									
		With Zebra Iris GTX, *n* must be 1.								i							
	M_ROTARY_ENCODER*n*	Specifies to use rotary decoder *n* as the trigger source, where *n* is the number of the rotary decoder.		d				a	h	i							
		System specific															
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be either 1 or 2.						a									
		For Zebra Iris GTX, *n* must be 1.								i							
	M_SOFTWARE	Specifies to use a software trigger as the trigger source. Use M_TIMER_TRIGGER_SOFTWARE to issue the trigger.		d				a	h	i							
	M_TIMER_STROBE	Specifies to route the strobe's timer signal to the specified signal.								i							
	M_TIMER*n*	Specifies to use the output signal of the specified timer as the trigger source, where *n* is the timer number.		d				a	h	i							
		System specific															
		[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be a value from 1 to 16.						a									
		With Zebra Iris GTX, *n* can be a value from 1 to 8.								i							

M_TIMER_USAGE +	Sets the purpose of the timer. INQ You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_PULSE_GENERATION.		d				a	h	i							
M_PULSE_GENERATION	Specifies the normal use of the timer, as described in the Timers and coordinating events section of Chapter 56: I/O signals and communicating with external devices.		d				a	h	i							
M_PULSE_MEASUREMENT	Specifies to use the timer to measure the duration of the pulse that occurs on the timer's trigger source. Note that when in this mode, the values set using M_TIMER_DELAY_CLOCK_SOURCE, M_TIMER_DELAY, and M_TIMER_DURATION are ignored. When a timer is set in this mode, the duration of pulses that occur on the timer's trigger source (M_TIMER_TRIGGER_SOURCE) are measured with respect to the timer's clock source (M_TIMER_CLOCK_SOURCE). To measure active-high pulses, set M_TIMER_TRIGGER_ACTIVATION to M_LEVEL_HIGH; to measure active-low pulses, set M_TIMER_TRIGGER_ACTIVATION to M_LEVEL_LOW. The duration of the pulse can be inquired using MsysInquire() with M_TIMER_VALUE. Between pulses, you can reset the timer value using **MsysControl**() with M_TIMER_VALUE.		d				a	h	i							
M_TIMER_VALUE +	Resets the timer value. You must specify a combination value from the following table: • To specify which on-board timer to control		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
0	Specifies to reset the timer value to 0.		d				a	h	i							

Combination value for the values listed in For controlling the settings of a timer.

You must add the following value to the above-mentioned values to specify which on-board timer to control.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which on-board timer to control																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_TIMER*n*	Specifies on-board timer *n*, where *n* is the number of the timer. To set the exposure and strobe timers, use MdigControl() with M_EXPOSURE_TIME or M_TIMER_STROBE, respectively.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. For Zebra 4Sight EV6/EV7, *n* can be a value from 1 to 16.						a									
	For Zebra Iris GTX, *n* can be a value from 1 to 8.								i							

The following control types allow you to control the settings of a quadrature decoder with inputs from a rotary or linear encoder.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling the settings of a rotary decoder																
ControlType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlValue																
M_ROTARY_ENCODER_BIT0_SOURCE +	Sets the auxiliary input signal on which to receive bit 0 of the 2-bit Gray code. INQ You must specify a combination value from the following table: • To specify which rotary decoder to set															
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the signal on which to receive bit 0 of the 2-bit Gray code, where *n* is the number of the auxiliary signal.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be a value from 8 to 15.						a									

	With Zebra Iris GTX, *n* can be a value from 3 to 6.							i							
M_ROTARY_ENCODER_BIT1_SOURCE +	Sets the auxiliary input signal on which to receive bit 1 of the 2-bit Gray code. INQ You must specify a combination value from the following table: • To specify which rotary decoder to set		d				a	h	i						
	System specific														
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.		d				a	h							
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the signal on which to receive bit 1 of the 2-bit Gray code, where *n* is the number of the auxiliary signal.		d				a	h	i						
	System specific														
	[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be a value from 8 to 15.						a								
	With Zebra Iris GTX, *n* can be a value from 3 to 6.								i						
M_ROTARY_ENCODER_OUTPUT_MODE +	Sets the rotary decoder's counter value and/or the direction of movement upon which the rotary decoder should output a pulse. The pulse can be used to trigger a timer. To trigger a timer, set M_TIMER_TRIGGER_SOURCE to M_ROTARY_ENCODERn. To decimate (subsample) the rotary decoder output signal before sending it to a timer or a grab controller, set this control type to M_POSITION_TRIGGER and set M_ROTARY_ENCODER_POSITION_TRIGGER to the required decimation value. For more information, refer to the Pixel aspect ratio subsection of the Using quadrature input from a rotary encoder section of Chapter 56: I/O signals and communicating with external devices. INQ You must specify a combination value from the following table: • To specify which rotary decoder to set		d				a	h	i						
	System specific														

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_POSITION_TRIGGER	Specifies to output a pulse upon the trigger generated by M_ROTARY_ENCODER_POSITION_TRIGGER.		d				a	h	i							
M_STEP_ANY	Specifies to output a pulse upon any change in the rotary decoder's counter value (position change in any direction).		d				a	h	i							
M_STEP_BACKWARD	Specifies to output a pulse upon a rotary decoder counter decrement only.		d				a	h	i							
M_STEP_FORWARD	Specifies to output a pulse upon a rotary decoder counter increment only.		d				a	h	i							
M_STEP_FORWARD_NEW_POSITIVE	Specifies to output a pulse upon a rotary decoder counter increment of a new value that has not been reached before. For example, if the counter value is at 10, and is decremented down to 7, the rotary decoder will only output a pulse when the counter is incremented up to 11.		d				a	h	i							
M_ROTARY_ENCODER_POSITION +	Resets the rotary decoder's counter to 0 immediately. To reset the counter to 0 upon a signal, use M_ROTARY_ENCODER_RESET_SOURCE. Note that a call to MsysInquire() with M_ROTARY_ENCODER_POSITION inquires the current value of the rotary decoder counter. You must specify a combination value from the following table: • To specify which rotary decoder to set		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
Value = 0	Implements the default behavior. Note that, if a non-zero value is specified, an error is generated.		d				a	h	i							

M_ROTARY_ENCODER_POSITION_TRIGGER +	Sets the value of the rotary decoder's counter upon which a trigger is generated. You can output this trigger to a timer or a grab controller using M_ROTARY_ENCODER_OUTPUT_MODE set to M_POSITION_TRIGGER. INQ You must specify a combination value from the following table: • To specify which rotary decoder to set		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
0 <= *Value* <= 0*xFFFFFFFF*	Specifies the value of the counter upon which a trigger is generated. If a value beyond the supported range is specified, an error is generated. If you are treating the counter values as a signed range of values (for example, forcing the counter to reset to 0 at 0x80000000) and you want to generate a trigger upon a negative value, specify the equivalent value in the range of 0x80000000 to 0xFFFFFFFF.		d				a	h	i							
M_ROTARY_ENCODER_RESET_ACTIVATION +	Sets the signal transition upon which to reset the rotary decoder's counter to 0. The rotary decoder will reset the counter to 0 when the signal transition occurs on the source signal specified by M_ROTARY_ENCODER_RESET_SOURCE. INQ You must specify a combination value from the following table: • To specify which rotary decoder to set		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_EDGE_RISING.		d				a	h	i							
M_EDGE_FALLING	Specifies to reset the rotary decoder upon a high-to-low signal transition.		d				a	h	i							
M_EDGE_RISING	Specifies to reset the rotary decoder upon a low-to-high signal transition.		d				a	h	i							

M_ROTARY_ENCODER_RESET_SOURCE +	Sets the source signal to use to reset the rotary decoder's counter to 0. The rotary decoder will reset the counter to 0 when the signal transition specified using M_ROTARY_ENCODER_RESET_ACTIVATION occurs on the selected source. INQ You must specify a combination value from the following table: • To specify which rotary decoder to set		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_NULL	Specifies not to reset using a hardware signal source.		d				a	h	i							
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the trigger source for the specified timer, where *n* is the number of the auxiliary signal.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be a value from 8 to 15.						a									
	With Zebra Iris GTX, *n* can be a value from 3 to 6.								i							
M_POSITION_TRIGGER	Specifies to use the trigger signal generated by the rotary decoder when the counter reaches the value specified with M_ROTARY_ENCODER_POSITION_TRIGGER.		d				a	h	i							
M_ROTARY_ENCODER_STATE +	Sets whether to enable the rotary decoder. INQ You must specify a combination value from the following table: • To specify which rotary decoder to set		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_DEFAULT	Same as M_DISABLE.		d				a	h	i							

	M_DISABLE	Specifies that the rotary decoder is disabled.		d				a	h	i							
	M_ENABLE	Specifies that the rotary decoder is enabled.		d				a	h	i							

Combination value for the values listed in For controlling the settings of a rotary decoder.

You must add the following value to the above-mentioned values to specify which rotary decoder to set.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which rotary decoder to set																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ROTARY_ENCODER*n*	Specifies rotary decoder *n*, where *n* is the number of the rotary decoder.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. For Zebra 4Sight EV6/EV7, *n* can be either 1 or 2.						a									

The following control types allow you to control the settings of a data latch associated with the rotary decoders of your system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling the settings of a data latch associated with rotary encoders																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_SYS_DATA_LATCH_STATE +		Sets the state of the specified data latch. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. INQ You must specify a combination value from the following table: • To specify which data latch to affect															
	M_DEFAULT	Same as M_DISABLE.						a									
	M_DISABLE	Specifies that the data latch is disabled.						a									
	M_ENABLE	Specifies that the data latch is enabled. The data latch will store the position couner value of the rotary encoder when its associated event (specified using M_SYS_DATA_LATCH_TRIGGER_ACTIVATION) occurs.						a									

M_SYS_DATA_LATCH_TRIGGER_ACTIVATION +		Sets the trigger signal transition upon which to store the specified information to the specified data latch. To set the signal with which to trigger the data latch, use M_SYS_DATA_LATCH_TRIGGER_SOURCE. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. INQ You must specify a combination value from the following table: • To specify which data latch to affect						a								
	M_DEFAULT	Same as M_EDGE_RISING.						a								
	M_ANY_EDGE	Specifies that the specified information is stored in the data latch both upon a high-to-low and a low-to-high signal transition.						a								
	M_EDGE_FALLING	Specifies that the specified information is stored in the data latch upon a high-to-low signal transition.						a								
	M_EDGE_RISING	Specifies that the specified information is stored in the data latch upon a low-to-high signal transition.						a								
M_SYS_DATA_LATCH_TRIGGER_SOURCE +		Sets what triggers storing the specified information to the specified data latch. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. INQ You must specify a combination value from the following table: • To specify which data latch to affect						a								
	M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the trigger source, where *n* is the number of the auxiliary signal. > In this case, *n* can be a value from 8 to 15. To specify the signal transition, use M_SYS_DATA_LATCH_TRIGGER_ACTIVATION.						a								
	M_TIMER*n*	Specifies to use the output signal of timer *n* as the trigger source, where *n* is the number of the timer. > In this case, *n* can be a value from 1 to 16.						a								

M_SYS_DATA_LATCH_TYPE +		Sets which rotary decoder the specified data latch will store the position counter value of when the data latch is triggered. You can access the stored rotary decoder position within a hook function that was triggered by an auxiliary input or a timer, using MsysGetHookInfo() with M_SYS_DATA_LATCH_VALUE. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. INQ You must specify a combination value from the following table: • To specify which data latch to affect						a									
	M_ROTARY_ENCODER*n*	Specifies to store the value of the counter of rotary decoder *n*, where *n* is a number between 1 and 2. To configure the rotary decoder, use the M_ROTARY_ENCODER... control types.						a									

Combination value for the values listed in For controlling the settings of a data latch associated with rotary encoders.

You must add the following value to the above-mentioned values to specify which data latch to affect.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which data latch to set																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_LATCH*n*	Specifies which data latch to affect, where *n* is 1 or 2.						a									

The following control types and control values specify the settings for UARTs. To specify a particular UART, see the For specifying which UART to control combination value below.

> Note that for other Zebra imaging boards that have a UART, but are not supported with the constants below, see MdigControl().

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For UART settings																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_UART_DATA_SIZE +		Sets the number of data bits per character that are sent or received by the UART. INQ															
	M_DEFAULT	Same as 8.										j	k				
	7	Specifies that the data length is 7 bits.										j	k				
	8	Specifies that the data length is 8 bits.										j	k				

M_UART_FREE +	Stops all Aurora Imaging Library UART operations and frees all UART resources used by Aurora Imaging Library.										j	k				
M_DEFAULT	Specifies the default behavior.										j	k				
M_UART_PARITY +	Sets whether character data is sent or received with a parity bit and how the parity bit is set. The parity bit is an extra data bit (0 or 1) that is added to each character for error checking purposes. INQ										j	k				
M_DEFAULT	Same as M_DISABLE.										j	k				
M_DISABLE	Specifies that no extra bit is added (no parity).										j	k				
M_EVEN	Specifies that the number of 1's will be even.										j	k				
M_ODD	Specifies that the number of 1's will be odd.										j	k				
M_UART_READ_CHAR +	Reads one character from the UART input buffer. If the input buffer is empty, the function will wait for the amount of time specified by M_UART_TIMEOUT. If a time-out occurs, the '?' character will be returned.										j	k				
Value	Specifies the address of the variable in which to save the character read from the UART.										j	k				
M_UART_READ_STRING +	Reads a string of incoming data from the UART. The number of characters to read can be specified with M_UART_READ_STRING_SIZE or M_UART_STRING_DELIMITER. M_UART_TIMEOUT specifies the maximum time to wait between each byte when reading incoming data. Use MsysInquire() with M_UART_BYTES_READ to wait for the read operation to complete and retrieve the actual number of bytes read.										j	k				
Value	Specifies the address of the character array in which to save the string read from the UART. The size of this array must be set to the same value as the M_UART_READ_STRING_MAXIMUM_SIZE control type.										j	k				

M_UART_READ_STRING_MAXIMUM_SIZE +	Sets the maximum length of the string to read. This prevents global protection faults from happening when using the M_UART_READ_STRING control type. INQ										j	k				
Value	Specifies the maximum length of the string, in bytes.										j	k				
M_UART_READ_STRING_SIZE +	Sets the length of the string to read using M_UART_READ_STRING. INQ										j	k				
M_DEFAULT	Specifies the use of M_UART_STRING_DELIMITER to delineate the end of the string to read.										j	k				
Value	Specifies the string length, in bytes.										j	k				
M_UART_SPEED +	Sets the baud rate of the UART. INQ										j	k				
M_DEFAULT	Specifies the default value; the default value is 14400.										j	k				
300;600;1200;1800;2400;4800;7200;9600;14400;19200;38400;57600;115200	Specifies the baud rate of the UART. Note that the maximum baud rate is highly dependent on the amount of computer resources available.										j	k				
M_UART_STOP_BITS +	Sets the number of extra data bit(s) (1 or 2) that are added to each character to indicate the end of the character. INQ										j	k				
M_DEFAULT	Same as 1.										j	k				
1	Specifies that there is 1 stop bit.										j	k				
2	Specifies that there are 2 stop bits.										j	k				
M_UART_STRING_DELIMITER +	Sets the character used to terminate strings of incoming or outgoing data. The delimiter is used but not sent when writing data with M_UART_WRITE_STRING; it is read for incoming data with M_UART_READ_STRING. INQ										j	k				
M_DEFAULT	Specifies the default value; the default value is '\0'.										j	k				
Value	Specifies the character used to terminate strings.										j	k				
M_UART_TIMEOUT +	Sets the maximum time to wait between each byte when reading incoming data. INQ										j	k				

M_DEFAULT	Same as M_INFINITE.										j	k				
M_INFINITE	Waits indefinitely.										j	k				
Value	Specifies the time to wait, in msec.										j	k				
M_UART_WRITE_CHAR +	Sends one character to the UART for transmission.										j	k				
Value	Specifies the character to send.										j	k				
M_UART_WRITE_STRING +	Sends a string of data through the UART for transmission. The number of characters to send can be specified with M_UART_WRITE_STRING_SIZE or M_UART_STRING_DELIMITER. Use MsysInquire() with M_UART_BYTES_WRITTEN to wait for the write operation to complete and retrieve the actual number of bytes written.										j	k				
Value	Specifies the character array.										j	k				
M_UART_WRITE_STRING_SIZE +	Sets the length of the string to be sent to the UART for transmission. INQ										j	k				
M_DEFAULT	Specifies the use of M_UART_STRING_DELIMITER to end the string. The delimiter will not be sent through the UART.										j	k				
Value	Specifies the string length, in bytes.										j	k				

Combination value for the values listed in For UART settings.

You can add the following value to the above-mentioned values to set which UART to control.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which UART to control																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_UART_NB(AIL_INT DeviceToControl)	Specifies which UART to control. Use MsysInquire() with M_UART_PRESENT to determine the number of UARTS on the system.										j	k				
	Parameters															
	DeviceToControl Specifies a device (UART) to control.										j	k				

			M_DEV*n* Specifies the device to control, where *n* can be from 0 and 15.										j	k				
		System specific																
		There are 2 UARTS available on Zebra Radient eV-CL DB and eV-CL DF, 4 on Zebra Radient eV-CL QB, and 1 on Zebra Radient eV-CL SB and SF. Note that, using the CLProtocol will occupy one COM port per allocated digitizer.										j						
		There are 2 UARTS available on Zebra Rapixo CL Pro DB and DF, 4 on Zebra Rapixo CL Pro QB, and 1 on Zebra Rapixo CL Pro SF. Note that, using the CLProtocol will occupy one COM port per allocated digitizer.											k					

The following control types and control values allow you to configure an action command and send it to one or more GigE Vision cameras. Action commands require both an Aurora Imaging Library-side and a camera-side configuration. The following control types and control values configure the Aurora Imaging Library-side of the action command. To configure the camera-side, use MdigControlFeature() with the appropriate feature values. For more information, refer to the Triggering simultaneous actions in multiple GigE Vision cameras section of the Zebra GigE Vision driver chapter in the Hardware-specific Notes.

> To configure features of the camera, use MdigControlFeature() with a digitizer allocated for the camera on an Aurora Imaging Library GigE system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For configuring and sending an action command to a GigE Vision camera																		
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																	

M_GC_ACTION_ACKNOWLEDGE_NUMBER +		Sets the number of acknowledgments that Aurora Imaging Library should expect to receive when the action command is issued. There should be one per target camera performing the action. By default, this number is equal to the number of Aurora Imaging Library digitizers added to the action command. If the number of Aurora Imaging Library digitizers added to the action command does not equal the number of Aurora Imaging Library digitizers that will perform the action, Aurora Imaging Library will return an error (for example, if you set M_GC_ACTION_GROUP_MASK to mask out one or more cameras). To avoid this error, use this control type to set the number of acknowledgments that Aurora Imaging Library should expect. INQ You must specify a combination value from the following table: • To set which action command to control															
	M_DEFAULT	Specifies the number of cameras added to the action command, using M_GC_ACTION_ADD_DEVICE.				f	g										
	Value	Specifies the number of acknowledgments expected.				f	g										
M_GC_ACTION_ADD_DEVICE +		Adds a GigE Vision camera to the list of cameras associated with the action command. You must specify a combination value from the following table: • To set which action command to control				f	g										
	Digitizer identifier	Specifies the digitizer identifier allocated for the GigE Vision camera to add. Note that the digitizer should be allocated on an Aurora Imaging Library GigE Vision system using MdigAlloc().				f	g										
M_GC_ACTION_CLEAR_DEVICES +		Removes all cameras from the list of cameras associated with the action command. You must specify a combination value from the following table: • To set which action command to control				f	g										
	M_DEFAULT	Specifies the default behavior.				f	g										

M_GC_ACTION_DEVICE_KEY +	Sets the action device key for the action command. The action device key identifies the cameras on which the action should be performed. For the camera to accept and perform the action, the device key on your camera and the device key of the action command must match. If they do not match, the camera ignores the action command. To configure the device key on your camera, use MdigControlFeature() with the appropriate GenICam SFNC feature (for example, `ActionDeviceKey`). INQ You must specify a combination value from the following table: • To set which action command to control		d		f	g									
	System specific														
	To configure features of the camera, use MdigControlFeature() with a digitizer allocated for the camera on an Aurora Imaging Library GigE system. If using an Aurora Imaging Library Concord PoE system, this control type is only available for Zebra Concord PoE with ToE. To configure an action command for the Zebra Concord PoE base model, use an Aurora Imaging Library GigE system with this control type.		d												
Value	Specifies the action device key.		d		f	g									
M_GC_ACTION_EXECUTE +	Issues the action command. You must specify a combination value from the following table: • To set which action command to control				f	g									
M_DEFAULT	Specifies the default behavior.				f	g									
M_GC_ACTION_GROUP_KEY +	Sets the action group key for the action command. The action group key identifies the action you want to perform on the camera. For the camera to know which action signal to generate, the group key of the action command must match the group key on your camera. If they do not match, the camera ignores the action command. To configure the group key of the action command on your camera, use MdigControlFeature() with the appropriate GenICam SFNC feature (for example, `ActionGroupKey`). INQ You must specify a combination value from the following table: • To set which action command to control		d		f	g									
	System specific														

		To configure features of the camera, use MdigControlFeature() with a digitizer allocated for the camera on an Aurora Imaging Library GigE system. If using an Aurora Imaging Library Concord PoE system, this control type is only available for Zebra Concord PoE with ToE. To configure an action command for the Zebra Concord PoE base model, use an Aurora Imaging Library GigE system with this control type.		d												
	Value	Specifies the action group key.		d		f	g									
M_GC_ACTION_GROUP_MASK +		Sets the action group mask for the action command. In the case where you need one (or more) cameras to temporarily ignore an action command, you can mask out the action command for the camera by changing the action group mask. For the camera to know if it generates an action signal upon receiving the command, the action group mask of the action command, and the action group mask of the camera, when combined in a bitwise AND operation, must result in a non-zero value. If the result is a zero, the camera ignores the action command. Note that, if you are masking out the action command for the camera, you must change the number of action acknowledgment packets Aurora Imaging Library should expect; otherwise, Aurora Imaging Library will generate an error. To set the number of action acknowledgment packets, use M_GC_ACTION_ACKNOWLEDGE_NUMBER. INQ You must specify a combination value from the following table: • To set which action command to control		d		f	g									
		System specific														
		To configure features of the camera, use MdigControlFeature() with a digitizer allocated for the camera on an Aurora Imaging Library GigE system. If using an Aurora Imaging Library Concord PoE system, this control type is only available for Zebra Concord PoE with ToE. To configure an action command for the Zebra Concord PoE base model, use an Aurora Imaging Library GigE system with this control type.		d												
	Value	Specifies the action group mask.		d		f	g									

M_GC_ACTION_REMOVE_DEVICE +		Removes an Aurora Imaging Library GigE Vision camera from the list of cameras associated with the action command. You must specify a combination value from the following table: • To set which action command to control				f	g										
	Digitizer identifier	Specifies the digitizer identifier for the GigE Vision camera to remove. > Note that the digitizer should be allocated on an Aurora Imaging Library GigE Vision system using MdigAlloc().				f	g										
M_GC_ACTION_TIME +		Sets the time at which the action command should execute. Note that, for scheduled action commands to function, your camera must be capable of using IEEE 1588, and the appropriate feature must be enabled (for example, `GevIEEE1588`). To verify the availability of these features on your camera, use MdigInquire() with M_GC_CONTROL_PROTOCOL_CAPABILITY. It should return M_GC_SCHEDULED_ACTION_SUPPORT or M_GC_IEEE_1588_SUPPORT. Note that, you must enable 1588 on each GigE Vision camera that will receive this action command, using MdigControlFeature(), or Feature Browser (accessible through **MsysControl**() with M_GC_FEATURE_BROWSER) and the appropriate SFNC feature (for example, `GevIEEE1588`). INQ You must specify a combination value from the following table: • To set which action command to control				f	g										
	Value	Specifies the time at which to execute the action command on your camera, relative to the time at which the action command was sent, in sec.				f	g										

The following control types and control values allow you to configure a Trigger-over-Ethernet packet for transmission using a ToE module. The Trigger-over-Ethernet packet can be sent as an action command or a GigE Vision software trigger. Action commands and GigE Vision software triggers require both an Aurora Imaging Library-side and a camera-side configuration. The following control types and control values configure the Aurora Imaging Library-side of the action command or GigE Vision software trigger. To configure the camera-side, you need to use a digitizer allocated for the camera on an Aurora Imaging Library GigE system (MsysAlloc() with M_SYSTEM_GIGE_VISION) and use MdigControlFeature() with the appropriate feature values. For more information, refer to the Triggering simultaneous actions in multiple GigE Vision cameras section of the Zebra GigE Vision driver chapter in the Hardware-specific Notes.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a Trigger-over-Ethernet packet (action command or GigE Vision software trigger) for transmission using a ToE module																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																

M_ADD_DESTINATION +	Adds a GigE Vision camera to the list of cameras associated with the specified action command or GigE Vision software trigger. You must specify a combination value from one of the following tables: • To set which GigE Vision software trigger to control • To set which action command to control																
Digitizer identifier	Specifies the digitizer identifier allocated for the GigE Vision camera to add. Note that the digitizer should be allocated on a GigE Vision system using MdigAlloc().		d														
M_CLEAR_DESTINATIONS +	Removes all cameras from the list of cameras associated with the specified action command or GigE Vision software trigger. You must specify a combination value from one of the following tables: • To set which GigE Vision software trigger to control • To set which action command to control		d														
M_DEFAULT	Specifies the default behavior.		d														
M_GC_TRIGGER_SELECTOR +	Sets the type of GigE Vision software trigger that should take place on the camera upon receiving the ToE packet (for example, FrameStart). For the camera to know what should be triggered upon receiving the packet, this control type must match the trigger selector on your camera. To configure the trigger selector on your camera, use MdigControlFeature() with the appropriate GenICam SFNC feature (for example, TriggerSelector). You must specify a combination value from the following table: • To set which GigE Vision software trigger to control		d														
AIL_TEXT("*FeatureName*") 1	Specifies the type of trigger; see your GigE Vision camera's documentation for a list of available types. Note, if using a C compiler, you must pass the value as M_PTR_TO_AIL_INT(AIL_TEXT("*FeatureName*")).		d														
M_REMOVE_DESTINATION +	Removes a GigE Vision camera from the list of cameras associated with the specifies action command or GigE Vision software trigger. You must specify a combination value from one of the following tables: • To set which GigE Vision software trigger to control • To set which action command to control		d														
Digitizer identifier	Specifies the digitizer identifier allocated for the GigE Vision camera to remove. Note that the digitizer should be allocated on a GigE Vision system using MdigAlloc().		d														

M_TRIGGER_COMMAND +	Issues a software trigger event to prompt the transmission of the specified action command or GigE Vision software trigger from the ToE module. To use this control type, M_TRIGGER_SOURCE must be set to M_SOFTWAREn. Note that this will trigger any action command or GigE Vision software trigger that uses the same software signal (M_SOFTWAREn) as the specified action command/GigE Vision software trigger. For example, if you specify M_TRIGGER_COMMAND + M_GC_ACTION1 and the trigger source of M_GC_ACTION1 is M_SOFTWARE1, all ToE packets that use M_SOFTWARE1 are triggered. You must specify a combination value from one of the following tables: • To set which GigE Vision software trigger to control • To set which action command to control		d													
M_ACTIVATE	Specifies the default behavior.		d														
M_TRIGGER_SOURCE +	Sets the event that will cause the ToE module to send the specified action command or GigE Vision software trigger as a ToE packet. INQ You must specify a combination value from one of the following tables: • To set which GigE Vision software trigger to control • To set which action command to control		d														
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the trigger source, where *n* is the number of the auxiliary signal. > In this case, *n* can be a value from 0 to 5.		d														
M_IO_COMMAND_LIST*n* +	Specifies to use the I/O command list *n*, where *n* is the number of the I/O command list. > In this case, *n* can be a value from 0 to 1. You must specify a combination value from the following table: • To specify which I/O command register bit to use		d														
M_ROTARY_ENCODER*n*	Specifies to use rotary decoder n as the trigger source, where *n* is the number of the rotary decoder. > In this case, *n* can be a value from 1 to 2.		d														
M_SOFTWARE*n*	Specifies to use software as a trigger source to trigger the ToE module, where *n* is the number of the software trigger; *n* can be a value between 1 and 4. Use M_TRIGGER_COMMAND to issue the software trigger event. > In this case, *n* can be a value from 1 to 4.		d														

	M_TIMER*n*	Specifies to use the output signal of the specified timer as the trigger source, where *n* is the number of the timer. In this case, *n* can be a value from 1 to 16.		d													
M_TRIGGER_STATE +		Sets the state of the specified action command or GigE Vision software trigger in the ToE module. INQ You must specify a combination value from one of the following tables: • To set which GigE Vision software trigger to control • To set which action command to control		d													
	M_DISABLE	Specifies the ToE packet is disabled.		d													
	M_ENABLE	Specifies the ToE packet is enabled, and will be transmitted when its associated event (specified using M_TRIGGER_SOURCE) occurs.		d													

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Combination value for the values listed in For configuring and sending an action command to a GigE Vision camera, For specifying a Trigger-over-Ethernet packet (action command or GigE Vision software trigger) for transmission using a ToE module. (exceptions might exist)

You must add the following value to the above-mentioned values to set which action command to control. Alternatively, you can add a value from For specifying which GigE Vision software trigger to control.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which action command to control																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GC_ACTION*n*	Specifies to control action command *n*, where *n* is a value from 0 to 31.		d		f	g										
	System specific															
	If using this control type with Zebra Concord PoE with ToE, it cannot be used with M_GC_TRIGGER_SELECTOR.		d													

Combination value for the values listed in For specifying a Trigger-over-Ethernet packet (action command or GigE Vision software trigger) for transmission using a ToE module.

You must add the following value to the above-mentioned values to set which GigE Vision software trigger to control. Alternatively, you can add a value from For specifying which action command to control.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which GigE Vision software trigger to control																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GC_TRIGGER_SOFTWARE*n*	Specifies to control a GigE Vision software trigger n, where n is a value from 0 to 31. This combination value cannot be used with M_GC_ACTION_DEVICE_KEY, M_GC_ACTION_GROUP_KEY, or M_GC_ACTION_GROUP_MASK.		d													

Combination value for [when ControlType=M_IO_SOURCE: M_IO_COMMAND_LISTn;], [when ControlType=M_TIMER_TRIGGER_SOURCE: M_IO_COMMAND_LISTn;], [when ControlType=M_TRIGGER_SOURCE: M_IO_COMMAND_LISTn].

You must add the following value to the above-mentioned values to specify which I/O command register bit to use.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which I/O command register bit to use																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_IO_COMMAND_BIT*n*	Specifies I/O command register bit *n*, where *n* represents the bit number.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. For Zebra 4Sight EV6/EV7, *n* can be a value from 0 to 7.						a									
	With Zebra Iris GTX, *n* can be a value from 0 to 4.								i							
	With Zebra Concord PoE with ToE, *n* can be a value from 0 to 7.		d					h								

Type-specific versions of the function when using a C compiler

```
void MsysControlDouble (AIL_ID SysId, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

SysId

See SysId of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void MsysControlInt64 (AIL_ID SysId, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

SysId

See SysId of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MsysControlFeature

Synopsis

Controls a feature of the camera.

Syntax

```
void MsysControlFeature(
    AIL_ID SysId,                  //in
    AIL_INT64 ControlType,         //in
    AIL_CONST_TEXT_PTR FeatureName, //in
    AIL_INT64 UserVarType,         //in
    const void *UserVarPtr         //in
)
```

Description

This function allows you to directly control various GenICam standard feature naming convention (SFNC) features and manufacturer-specific features of the camera. For more information, refer to the Using Aurora Imaging Library with GenICam section of Chapter 27: Grabbing with your digitizer.

The features described in this function are primarily available to set existing settings using Aurora Imaging Library code or to set information from which you can build your own interface for the camera device's description file (XML). Aurora Imaging Library provides two versions of an interface that you can use to interactively inquire the camera's features. At design-time, you can use Aurora Imaging Intellicam's Feature Browser. At runtime, you can launch the Feature Browser, using MsysControl() with M_GC_FEATURE_BROWSER. Both versions of the Feature Browser provide a list of available feature names and associated possible values, as well as code snippets with the Aurora Imaging Library functions and Aurora Imaging Library constants associated with the settings selected; you can copy the code snippets to your Aurora Imaging Library application code.

To inquire a camera's manufacturer-specific feature, use MsysInquireFeature().

Parameters

SysId

Specifies the identifier of the system accessing the camera to control. This parameter must be given a valid system identifier, previously allocated using MsysAlloc().

ControlType

Specifies the type of information to control for the specified feature.

See the Parameter associations section for possible values that can be specified.

FeatureName

Specifies the name of the camera feature to control.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the name of the feature																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
AIL_TEXT("*FeatureName*") [1]	Specifies the name of the feature. Note that the feature name is case-sensitive. Refer to your camera's documentation for a list of the features available.			e												b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

UserVarType

Specifies the data type of the address pointed to by the UserVarPtr parameter. If not setting the feature's value (using M_FEATURE_VALUE), set this parameter to **M_DEFAULT**.

Note that, in the case of a string (or a multi-byte register), the size of the array passed to the UserVarPtr parameter must be equal to or larger than the returned information; otherwise an application crash will occur.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the UserVarPtr's data type																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_TYPE_BOOLEAN	Specifies that UserVarPtr is passed an address of type *AIL_BOOL*.			e												b
M_TYPE_DOUBLE	Specifies that UserVarPtr is passed an address of type *AIL_DOUBLE*.			e												b
M_TYPE_INT64	Specifies that UserVarPtr is passed an address of type *AIL_INT64*.			e												b
M_TYPE_STRING +	Specifies that UserVarPtr is passed an address of type *AIL_TEXT_CHAR*. You must specify a combination value from the following table: • To specify the length of the array			e												b
M_TYPE_UINT8 +	Specifies that UserVarPtr is passed an address of type *AIL_UINT8*. You must specify a combination value from the following table: • To specify the length of the array			e												b

Combination value for M_TYPE_STRING; M_TYPE_UINT8.

You must add the following value to the above-mentioned values to specify the length of the array.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the length of the array																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FEATURE_USER_ARRAY_SIZE(AIL_INT Length)	Specifies the length of the array.			e												b
	Parameters															
	Length Sets the length, in bytes.															

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_TEXT_CHAR [optionally, in C++: a reference to an AIL_STRING]
- array of type AIL_UINT8 [optionally, in C++: a reference to a constant std::vector<AIL_UINT8>]
- AIL_BOOL
- AIL_DOUBLE
- AIL_INT
- AIL_INT64
- M_NULL

Specifies the address of the variable in which to write the value of the feature.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The table below lists possible values for the **ControlType** and **UserVarPtr** parameters.

- For specifying the type of information about the feature to set and the data type returned

To determine the data type of the FeatureName specified, use MsysInquireFeature() with M_FEATURE_VALUE.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of information about the feature to set and the data type returned																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr																
M_FEATURE_CHANGE_HOOK +		Sets whether to enable an event to occur when the value of the specified feature changes. Once enabled, use MsysHookFunction() with M_FEATURE_CHANGE to hook a specified function to the feature change event. Repeat for each feature that you want to enable a feature change event. Alternatively, to enable an event to occur when the value of any feature changes and at the same time hook a function to any feature change, use MsysHookFunction() with M_FEATURE_CHANGE + M_ALL. INQ															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e												b
	M_DISABLE	Specifies that no event is generated when the value of the specified feature changes. This is the default value.			e												b
	M_ENABLE	Specifies that an event is generated when the value of the specified feature changes.			e												b
M_FEATURE_EXECUTE +		Sets that the specified command feature must be executed.			e												b
	UserVarPtr extra info	Data type info: Data type: *Set this parameter to M_NULL*			e												b
M_FEATURE_MAX +		Specifies to set the feature to its maximum value.			e												b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *Set this parameter to M_NULL*			e												b
M_FEATURE_MIN +		Specifies to set the feature to its minimum value.			e												b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *address of an AIL_INT64*			e												b
M_FEATURE_VALUE +		Sets the current value of the feature.			e												b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_BOOL* **|** *address of an AIL_DOUBLE* **|** *address of an AIL_INT64* **|** *array of type AIL_UINT8 [optionally, in C++: a reference to a constant std::vector<AIL_UINT8>]* **|** *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e												b

Combination values for the values listed in For specifying the type of information about the feature to set and the data type

returned.

You can add one of the following values to the above-mentioned values to specify the GenTL configuration file (XML file) that is associated with the feature.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the configuration file associated with the feature																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GENTL_INTERFACE_NUMBER(AIL_INT IndexValue)	Specifies which instance of the GenTL interface configuration file is associated with the feature.			e												b
	Parameters															
	IndexValue Specifies the index of the GenTL interface configuration file.															
	0<= *Value*<=*MsysInquire*(M_GENTL_INTERFACE_COUNT) Specifies the index.			e												b
M_GENTL_SYSTEM	Specifies to display the GenTL system configuration information. This is the default value.			e												b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MsysFree

Synopsis

Free a system.

Syntax

```
void MsysFree(
    AIL_ID SysId //in
)
```

Description

This function deallocates a system previously allocated with MsysAlloc().

Prior to freeing a system, ensure that all buffers, displays, and digitizers allocated on the system are freed, unless M_UNIQUE_ID was specified during allocation.

Parameter

SysId

Specifies the identifier of the system to free.

Remark

- If you are creating a DLL that includes a call to **MsysFree**(), ensure that the call is not made from the `DllMain()` function, because **MsysFree**() might unload any DLL loaded with MsysAlloc() and you cannot unload a DLL from `DllMain()`. If necessary, call **MsysFree**() from a clean-up function in your DLL instead.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MsysGetHookInfo

Synopsis

Get information about a hook event.

Syntax

```
AIL_INT MsysGetHookInfo(
    AIL_ID SysId,          //in
    AIL_ID EventId,        //in
    AIL_INT64 InquireType, //in
    void *UserVarPtr       //out
)
```

Description

This function allows you to get information about the event that caused the hook function to be called. **MsysGetHookInfo**() should only be called within the scope of a system hook-handler function (see MsysHookFunction()).

Parameters

SysId

Specifies the identifier of the system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the system identifier																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e		g	a		i						n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c	d	e	f	g	a	h	i				l	m	n	b

EventId

Specifies the system event identifier received by the hook-handler function (see MsysHookFunction()).

InquireType

Specifies the type of information about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_TEXT_CHAR [optionally, in C++: a reference to an AIL_STRING]
- AIL_DOUBLE
- AIL_INT
- AIL_INT64
- AIL_UINT64

Specifies the address in which to write the requested information.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr**

parameter.

- For retrieving information about a camera-specific event
- For retrieving information about a timer start or timer end event
- For retrieving information about an I/O change event
- For retrieving information about a data latch value
- For retrieving information from a GenTL feature change event

The following allows you to retrieve information about a camera-specific event. The following information types are only available if **MsysGetHookInfo**() was called from a function hooked to a system event using MsysHookFunction() with M_CAMERA_PRESENT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information about a camera-specific event																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_CAMERA_PRESENT	Retrieves whether a camera has been added or re-connected to the system.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c			f	g	a		i				l	m	n	
M_FALSE	Specifies that a camera has been removed from the system.	c			f	g	a		i				l	m	n	
M_TRUE	Specifies that a camera has been added or re-connected to the system.	c			f	g	a		i				l	m	n	
M_GC_REMOTE_IP_ADDRESS_STRING +	Retrieves the IP address (IPv4) of the camera that generated the M_CAMERA_PRESENT event.				f	g										
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*				f	g										
AIL_TEXT("*nnn.nnn.nnn.nnn*") 1	Specifies the IP address of the camera as a string. The address string is expressed in dotted decimal notation, where each dotted decimal value (*nnn*) is a number between 0 and 255.				f	g										
M_GC_REMOTE_MAC_ADDRESS_STRING +	Retrieves the MAC address of the camera that generated the M_CAMERA_PRESENT event.				f	g										
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*				f	g										
AIL_TEXT("*nn-nn-nn-nn-nn-nn*") 1	Specifies the MAC address of the camera, as a string. The address string is expressed in hexadecimal pairs, where each pair (*nn*) is a hexadecimal number between 00 and FF.				f	g										
M_GC_UNIQUE_ID_STRING +	Retrieves the unique identifier for your camera.				f	g	a		i						n	
	System specific															
	This is the camera's MAC address.				f	g										

	This is the camera's global unique identifier (GUID).														n	
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*				f	g	a		i						n	
AIL_TEXT("*nnnnnnnnnnnn*") 1	Specifies the unique identifier, as a string. The string is expressed in hexadecimal.				f	g	a		i						n	
M_GC_USER_NAME	Retrieves the user-defined name of the camera that generated the M_CAMERA_PRESENT event.				f	g	a		i						n	
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*				f	g	a		i						n	
AIL_TEXT("*CameraName*") 1	Specifies the name of the camera.				f	g	a		i						n	
M_NUMBER	Retrieves the device number of the digitizer to which a camera has been connected or disconnected from, causing an M_CAMERA_PRESENT event. SET	c		e	f	g	a		i				l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i				l	m	n	b
M_DEV*n* +	Specifies the device number (rank) of the board. INFO	c		e	f	g	a		i				l	m	n	b
	System specific															
	This value is combined with a value from the following table: • To identify the GenTL library for which an Aurora Imaging Library GenTL system was allocated			e												b
M_TIME_STAMP	Retrieves the operating system's time stamp at the point which the event generated a service request. The time stamp is generated by the operating system's performance counter. Time stamps can be used to see if a frame has been missed since the last time stamp.		d	e	f	g	a	h	i						n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*		d	e	f	g	a	h	i						n	b
Value	Specifies operating system's time stamp, in sec.		d	e	f	g	a	h	i						n	b

[1] If the value was passed in a variable, it won't be returned enclosed in AIL_TEXT().

Combination value for [when InquireType=M_NUMBER: M_DEVn].

The following value is combined with the above-mentioned value to identifythe GenTL library for which an Aurora Imaging Library GenTL system was allocated.

All third-party proprietary GenTL libraries installed on your computer are sorted and indexed by Aurora Imaging Library. Use the GenTL Producer index number to identify the library that your Aurora Imaging Library system should use to communicate with the hardware devices on the specified transportation layer. These libraries are indexed and sorted for reference by Aurora Imaging Library (the GenTL Consumer). In all cases, the same number and type of libraries should be installed on every computer that will run your application; the order in which they are installed, however, is not important.

To determine the number of GenTL libraries installed on your computer, use MappInquire() with M_GENTL_PRODUCER_COUNT or use the **General Default Values** page of the Aurora Imaging Configurator utility.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For identifying the instance of the GenTL library to use																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GENTL_PRODUCER(AIL_INT IndexValue)	Specifies the GenTL Producer (library) for which this Aurora Imaging Library GenTL system was allocated.			e												b
	Parameters															
	IndexValue Specifies the index of the GenTL Producer.			e												b
	0 <= *Value* <= 127 Specifies the index.			e												b

The following allows you to retrieve information about a timer event. The following information types are only available if **MsysGetHookInfo**() was called from a function hooked to an event using M_TIMER_START or M_TIMER_END.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information about a timer start or timer end event																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_TIMER_INDEX	Retrieves the index value of the timer that generated an M_TIMER_END or an M_TIMER_START event.															
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
Value	Specifies the value of the index of the timer.		d				a	h	i							
M_TIMER_VALUE +	Retrieves the value of the specified timer. This is typically used with a timer end event (set using MsysHookFunction() with M_TIMER_END). Note that, if used with a timer start event (M_TIMER_START), it will return 0. To inquire the value of the timer that caused the event, use M_TIMER_INDEX to learn which timer to inquire. You must specify a combination value from the following table: • To specify the timer to inquire about		d				a	h	i							
	System specific															

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*		d				a	h	i							
Value	Specifies the current value of the timer, in nsec.		d				a	h	i							

Combination value for M_TIMER_VALUE.

You must add the following value to the above-mentioned value to specify the timer to inquire about.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the timer to inquire about																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_TIMER*n*	Specifies the timer to inquire about.		d				a	h	i							

The following allows you to retrieve information about an I/O change event. The following information types are only available if **MsysGetHookInfo**() was called from a function hooked to an event using M_IO_CHANGE.

If more than one input signal triggers an interrupt at the same time, the hook-handler function (or chain of hook-handler functions) is called for each input signal that generated an interrupt. As a result, any and all user-specified function(s) hooked using MsysHookFunction() with M_IO_CHANGE will be executed for each interrupt.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information about an I/O change event																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																	
M_IO_INTERRUPT_SOURCE	Retrieves the input signal that triggered the hook function. See the Aurora Imaging Library Hardware-specific Notes for a listing of available auxiliary signals and their corresponding numbers in Aurora Imaging Library.																
	System specific																
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_AUX_IO*n*	Specifies that auxiliary signal *n* triggered the hook function, where *n* is the number of the auxiliary signal.		d				a	h	i							
M_IO_STATUS	Retrieves the state of the input signal that generated the interrupt. The returned value reflects the state of the input signal at the time of the interrupt.		d				a	h	i							
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							

	M_OFF	Specifies that the I/O signal is off.		d				a	h	i							
	M_ON	Specifies that the I/O signal is on.		d				a	h	i							

The following allows you to retrieve information about a data latch event. The following information types are only available if using a latch that has previously saved the time, I/O command list counter, or rotary decoder position counter value upon a hardware event.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information about a data latch value																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_REFERENCE_LATCH_VALUE +	Retrieves the last timestamp or counter value stored by the specified latch, associated with the specified I/O command list. This setting is available in functions hooked to any type of event. You must specify a combination value from the following table: • To specify which of the I/O command lists to use															
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*		d				a	h	i							
Value	Specifies the last timestamp or counter value stored by the specified latch, associated with the specified command list.		d				a	h	i							
M_SYS_DATA_LATCH_VALUE +	Retrieves the last rotary decoder position counter value stored by the specified latch, associated with the rotary decoders of the system. This setting is only available if **MsysGetHookInfo**() was called from a function hooked to an event using M_TIMER_START, M_TIMER_END, or M_IO_CHANGE. [Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. You must specify a combination value from the following table: • To specify which latch to inquire about						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_UINT64*						a									

	Value	Specifies the last rotary decoder position counter value stored by the specified latch, associated with the rotary decoders of the system.						a										

Combination value for M_REFERENCE_LATCH_VALUE.

You must add the following value to the above-mentioned value to specify which of the I/O command lists to use.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the I/O command list to use																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_IO_COMMAND_LIST*n* +	Specifies to inquire about I/O command list *n*. You must specify a combination value from the following table: • To specify which latch to inquire about		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* can be either 1 or 2.						a									
	For Zebra Indio and Zebra Concord PoE *n* is a number from 1 to 2.		d					h								
	For Zebra Iris GTX, *n* must be 1.								i							

Combination value for M_SYS_DATA_LATCH_VALUE; M_IO_COMMAND_LISTn.

You must add the following value to the above-mentioned values to specify which latch to inquire about.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the latch to inquire																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_LATCH*n*	Specifies to inquire about latch *n*.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] For Zebra 4Sight EV6/EV7, *n* is a number from 1 to 4, unless used with M_SYS_DATA_LATCH_VALUE (in which case, *n* is a number from 1 to 2).						a									
	For Zebra Indio, *n* is a number from 1 to 4.							h								
	For Zebra Iris GTX, *n* is a number from 1 to 2.								i							

The following allows you to retrieve information from a GenTL feature change event. These information types are only available if **MsysGetHookInfo**() was called from a function hooked to GenICam feature change event using MsysHookFunction() with M_FEATURE_CHANGE.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information from a GenTL feature change event																		
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_GC_FEATURE_CHANGE_NAME +	Retrieves the name of the GenTL feature that changed. This information type is only available if **MsysGetHookInfo**() is used from a function hooked to GenTL feature change events using MsysHookFunction() with M_FEATURE_CHANGE.															
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e												b
Value	Specifies the feature name.			e												b

Combination value for M_GC_REMOTE_IP_ADDRESS_STRING; M_GC_REMOTE_MAC_ADDRESS_STRING; M_GC_UNIQUE_ID_STRING; M_GC_FEATURE_CHANGE_NAME.

You can add the following value to the above-mentioned values to get the string's length.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For getting the string size																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_STRING_SIZE	Retrieves the length of the string, including the terminating null character ("\0").															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e		g	a		i						n	b

Return value

The returned value is **M_NULL** if successful. If the operation fails, a non-null (!**M_NULL**) value is returned.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MsysHookFunction

Synopsis

Hook a function to a system event.

Syntax

```
void MsysHookFunction(
    AIL_ID SysId,                           //in
    AIL_INT HookType,                       //in
    AIL_SYS_HOOK_FUNCTION_PTR HookHandlerPtr, //in
    void *UserDataPtr                       //in-out
)
```

Description

This function allows you to attach or detach a user-defined function to a specified system event. Once a hook-handler function is defined and hooked to an event, it is automatically called when the event occurs.

You can hook more than one function to an event by making separate calls to **MsysHookFunction**() for each function that you want to hook. Aurora Imaging Library automatically chains and keeps an internal list of all these hooked functions. When a function is hooked, this new function is added to the end of the list. When the event happens, all user-defined functions in the list will be executed in the same order that they were hooked to the event. You can also remove any function from the list; in this case, Aurora Imaging Library preserves the order of the remaining functions in the list.

You can obtain more information about the event from within the hook-handler function using MsysGetHookInfo().

Note that functions hooked to an event execute on a distinct thread. This permits the functions to run concurrently from the operation that fired the event and from functions hooked to other events. Although there is a small queue to permit a certain amount of overlap, hooked functions should not take longer to execute than the period in which two of their associated events can occur. You cannot determine the instance of the event that fired the function.

Parameters

SysId

Specifies the identifier of the system on which to hook a function.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system identifier																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e		g	a		i						n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c	d	e	f	g	a	h	i		j	k	l	m	n	b

HookType

Specifies the system event to which to hook the function. This parameter can be set to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system event to hook																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_CAMERA_PRESENT +	Hooks the function to the presence of the camera.	c			f	g							l	m	n	

	System specific															
	Note that this hook requires the use of the GigE Vision Device Discovery Service. If the service is not started, an Aurora Imaging Library error will be generated when registering the M_CAMERA_PRESENT hook function.				f	g										
M_FEATURE_CHANGE +	Hooks the function to the event that occurs when the value of a GenICam feature changes. To enable an event to occur when the value of a specific feature changes, use MsysControlFeature() with M_FEATURE_CHANGE_HOOK set to M_ENABLE. Repeat for each feature for which you want to enable a feature change event. To enable an event to occur when the value of any feature changes, use M_FEATURE_CHANGE + M_ALL. This setting is only available for hooking to an image buffer. To retrieve the name of the feature that caused the event, use MsysGetHookInfo() with M_GC_FEATURE_CHANGE_NAME.			e												b
M_IO_CHANGE +	Hooks the function to the event that occurs when an I/O signal changes in accordance with its specified interrupt mode, set using MsysControl() with M_IO_INTERRUPT_ACTIVATION. Within the scope of the hook-handler function, it is prudent to verify that the appropriate I/O signal triggered the hook-handler function, using MsysGetHookInfo() with M_IO_INTERRUPT_SOURCE. If more than one input signal triggers an interrupt at the same time, the hook-handler function (or chain of hook-handler functions) is called for each input signal that generated an interrupt. As a result, any and all user-specified function(s) hooked using this hook type will be executed for each interrupt.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.		d				a	h								
M_TIMER_END +	Hooks the function to the event that occurs when the specified timer completes. You must specify a combination value from the following table: • To specify the timer to which to hook the function		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_TIMER_START +	Hooks the function to the event that occurs when the specified timer starts its active period (after the timer delay). You must specify a combination value from the following table: • To specify the timer to which to hook the function		d				a	h	i							
	System specific															

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_UART_DATA_RECEIVED +	Hooks the function to the event that occurs when the specified UART receives data. Use MsysInquire() with M_UART_PRESENT to return the number of UARTs available. Note that this value can be used in combination; see below.										j	k				

Combination values for M_FEATURE_CHANGE.

You can add one of the following values to the above-mentioned value to specify the GenTL configuration file (XML file) associated with the feature.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying to hook on a specific feature change event associated with a GenTL configuration file																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GENTL_INTERFACE_NUMBER(AIL_INT IndexValue)	Specifies that the hook will occur on a feature change event for a feature associated with a specific instance of the GenTL interface configuration file (XML file).			e												b
	Parameters															
	IndexValue Specifies the index of the GenTL interface configuration file.			e												b
	0 <= *Value* <=M_GENTL_INTERFACE_COUNT — Specifies the index.			e												b
M_ALL	Specifies that the hook will occur on all feature change events.			e												b
M_GENTL_SYSTEM	Specifies that the hook will occur on a feature change event of a feature associated with the GenTL system configuration file (XML file).			e												b

Combination value for M_UART_DATA_RECEIVED.

You can add the following value to the above-mentioned value to set the UART upon which the system event occurs.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the UART upon which the system event occurs																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_UART_NB(AIL_INT DeviceToHook)	Specifies the UART upon which the system event occurs. Use MsysInquire() with M_UART_PRESENT to determine the number of UARTS on the system.										j	k				
	Parameters															

	DeviceToHook Specifies a device (UART) on which the system event occurs.										j	k					
	M_DEV*n*	Specifies the device upon which the system event occurs, where *n* can be from 0 and 15.									j	k					

Combination value for M_TIMER_END; M_TIMER_START.

You must add the following value to the above-mentioned values to specify the timer to which to hook the function.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the timer to which to hook the function																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_TIMER*n*	Specifies the timer to hook.		d				a	h	i							

Combination value for the values listed in For specifying the system event to hook.

You can add the following value to the above-mentioned values to specify that the function should be unhooked.

For specifying that the function should be unhooked	
Combination value	Description
M_UNHOOK	Unhooks the specified function if hooked to the specified event. When you use M_UNHOOK, you must provide the same values for all the parameters as when you originally hooked the function.

HookHandlerPtr

Specifies the address of the function that should be called when the specified event occurs. The hook-handler function must be declared as follows:

```
AIL_INT MFTYPE HookHandler(
    AIL_INT HookType,
    AIL_ID EventId,
    void *UserDataPtr
)
```

Parameters

HookType

Type of system event hooked.

EventId

Event identifier to pass to MsysGetHookInfo() when inquiring about the hooked event.

UserDataPtr

Specifies the user data pointer passed to **MsysHookFunction**().

Upon successful completion, the hook-handler function should return **M_NULL**. Note, *AIL_SYS_HOOK_FUNCTION_PTR* and *MFTYPE* are reserved Aurora Imaging Library predefined types for functions and data pointers.

UserDataPtr

Specifies the address of the user data that you want to make available to the hook-handler function. This address is passed to the hook-handler function, through its UserDataPtr parameter, when the specified event occurs. Set this parameter to **M_NULL** if not used.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MsysInquire

Synopsis

Inquire about a system setting.

Syntax

```
AIL_INT MsysInquire(
    AIL_ID SysId,          //in
    AIL_INT64 InquireType, //in
    void *UserVarPtr       //out
)
```

Description

This function inquires about the specified system setting.

Note that you can use MsysControl() to control specific system settings.

You can also interactively inquire most of the system settings in real-time, using Aurora Imaging Intellicam's Feature Browser.

Parameters

SysId

Specifies the system identifier. This parameter should be set to one of the following values:

For the system identifier	
Value	Description
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.
System identifier	Specifies a valid system identifier, which you have allocated using the MsysAlloc() function.

InquireType

Specifies the type of system setting about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_TEXT_CHAR [optionally, in C++: a reference to an AIL_STRING]
- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT64

Specifies the address in which to write the requested information. Since the **MsysInquire**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr** parameter.

- For general system settings

- For inquiring CXP connection errors and settings
- For discovering connected devices
- For inquiring I/O signals and their mode
- For inquiring the state of specified user-bits in a static-user-output register
- For inquiring the settings of a timer
- For inquiring the settings of a rotary decoder
- For controlling the settings of a data latch associated with rotary encoders
- For UART settings
- For inquiring about an action command
- For inquiring a Trigger-over-Ethernet packet (action command or GigE Vision software trigger) for transmission using a ToE module

The following inquire types allow you to inquire general system settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For general system settings																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_ACCELERATOR_PRESENT	Inquires whether the processing accelerator is present.															
	System specific															
	This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d		f						j	k	l	m		
M_NO	Specifies that the processing accelerator is not present.		d		f						j	k	l	m		
M_YES	Specifies that the processing accelerator is present.		d		f						j	k	l	m		
M_ALLOCATION_OVERSCAN	Inquires whether image buffers, allocated on the system, will be allocated with an overscan region. SET	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DISABLE	Specifies that image buffers allocated on the system will have no overscan region.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_ENABLE	Specifies that image buffers are allocated on the system with an overscan region.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_ALLOCATION_OVERSCAN_SIZE	Inquires the size of the overscan region, added around all subsequently allocated image buffers (MbufAlloc...()). SET	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b

Value	Specifies the size of the overscan region, in pixels. INFO	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_ASYNCHRONOUS_CALL_SUPPORT	Inquires whether the system supports asynchronous function execution or not.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_NO	Specifies that the system does not support asynchronous execution.		d	e		g	a	h							n	b
M_YES	Specifies that the system supports asynchronous execution.	c			f				i		j	k	l	m		
M_BOARD_REVISION	Inquires the board revision number.	c			f		a	h	i		j	k	l	m		
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c			f		a	h	i		j	k	l	m		
Value	Specifies the board revision number.	c			f		a	h	i		j	k	l	m		
M_BOARD_SUB_REVISION	Inquires the board sub-revision number.		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
Value	Specifies the board sub-revision number.		d				a	h	i							
M_BOARD_TYPE	Inquires the type of system board. To return only the main board type and not the sub-board types (for example, to get M_RAPIXO without M_CL, M_PF, or M_SFCL), mask the return value with **M_BOARD_TYPE_MASK**. For an example of how to use this mask, refer to the examples below.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															

		This control type is also available for the Zebra Concord PoE base model.		d													
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	M_CLARITY_UHD	Specifies a Zebra Clarity UHD board.	c														
	M_CLARITY_UHD + M_H264	Specifies a Zebra Clarity UHD board with the optional on-board H.264 encoding module.	c														
	M_CONCORD_POE + M_DCH	Specifies a Zebra Concord PoE base model Dual-Port board.		d													
	M_CONCORD_POE + M_DCH + M_TOE	Specifies a Zebra Concord PoE with ToE Dual-Port board.		d													
	M_CONCORD_POE + M_QCH	Specifies a Zebra Concord PoE base model Quad-Port board.		d													
	M_CONCORD_POE + M_QCH + M_TOE	Specifies a Zebra Concord PoE with ToE Quad-Port board.		d													
	M_GENTL	Specifies a camera that uses the Zebra driver for GenTL.			e												
	M_GEVIQ + M_D25G	Specifies a Zebra GevIQ Dual 25 Gbps Ethernet board.				f											
	M_GIGE_VISION	Specifies a camera that uses the Zebra GigE Vision driver.					g										
	M_HOST	[This is only applicable to Non-Zebra computer] Specifies a Host system.						a									
	M_HOST+ M_4SIGHT + M_4SIGHT_EV6	[This is only applicable to Zebra 4Sight] Specifies a Host system allocated on Zebra 4Sight EV6.						a									
	M_HOST+ M_4SIGHT + M_4SIGHT_EV7	[This is only applicable to Zebra 4Sight] Specifies a Host system allocated on Zebra 4Sight EV7.						a									
	M_INDIO	Specifies a Zebra Indio board.							h								
	M_IRIS_GTX	Specifies a Zebra Iris GTX smart camera.								i							
	M_RADIENT + M_CL + M_DBCL	Specifies a Zebra Radient eV-CL dual-Base Camera Link board.										j					

	M_RADIENT + M_CL + M_DFCL	Specifies a Zebra Radient eV-CL dual-Full Camera Link board.									j						
	M_RADIENT + M_CL + M_QBCL	Specifies a Zebra Radient eV-CL quad-Base Camera Link board.									j						
	M_RADIENT + M_CL + M_SBCL	Specifies a Zebra Radient eV-CL single-Base Camera Link board.									j						
	M_RADIENT + M_CL + M_SFCL	Specifies a Zebra Radient eV-CL single-Full Camera Link board.									j						
	M_RAPIXO + M_CL + M_DBCL	Specifies a Zebra Rapixo DB-CL (dual-Base Camera Link) board.										k					
	M_RAPIXO + M_CL + M_DBCL + M_PF	Specifies a Zebra Rapixo Pro DB-CL (dual-Base Camera Link) with a Processing FPGA.										k					
	M_RAPIXO + M_CL + M_DFCL + M_PF	Specifies a Zebra Rapixo Pro DF-CL (dual-Full Camera Link) with a Processing FPGA.										k					
	M_RAPIXO + M_CL + M_QBCL + M_PF	Specifies a Zebra Rapixo Pro QB-CL (squad-Base Camera Link) with a Processing FPGA.										k					
	M_RAPIXO + M_CL + M_SFCL	Specifies a Zebra Rapixo SF-CL (single-Full Camera Link) board.										k					
	M_RAPIXO + M_CL + M_SFCL + M_PF	Specifies a Zebra Rapixo Pro SF-CL (single-Full Camera Link) with a Processing FPGA.										k					
	M_RAPIXO + M_COF + M_Q10G	Specifies a Zebra Rapixo CoF Quad 10 Gbps board.											l				
	M_RAPIXO + M_COF + M_Q25G + M_PF	Specifies a Zebra Rapixo CoF Quad 25 Gbps board with a Processing FPGA.											l				
	M_RAPIXO + M_CXP + M_D12G	Specifies a Zebra Rapixo CXP Dual 12G board.												m			
	M_RAPIXO + M_CXP + M_Q6G	Specifies a Zebra Rapixo CXP (Quad CXP-6) board.												m			
	M_RAPIXO + M_CXP + M_Q6G + M_DFWD	Specifies a Zebra Rapixo CXP DF (Quad CXP-6 with data forwarding) board.												m			
	M_RAPIXO + M_CXP + M_Q12G	Specifies a Zebra Rapixo CXP (Quad CXP-12) board.												m			

M_RAPIXO + M_CXP + M_Q12G + M_DFWD	Specifies a Zebra Rapixo CXP DF (Quad CXP-12 with data forwarding) board.													m		
M_RAPIXO + M_CXP + M_Q12G + M_PF	Specifies a Zebra Rapixo CXP Pro (Quad CXP-12 with a Processing FPGA) board.													m		
M_RAPIXO + M_CXP + M_S12G	Specifies a Zebra Rapixo CXP Single 12G board.													m		
M_USB3_VISION	Specifies a camera that uses the Zebra USB3 Vision driver.														n	
M_V4L2	Specifies a camera that uses the Video4Linux2 driver.															b
M_COM_SUPPORTED	Inquires whether the system can be used with the Aurora Imaging Library Industrial Communication module (Mcom).	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_NO	Specifies that the system cannot be used.	c	d	e	f	g					j	k	l	m	n	b
M_YES	Specifies that the system can be used.						a	h	i							
M_CURRENT_THREAD_ID	Inquires the identifier of the current system thread. This identifier can be used with the thread module.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															
	This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
System thread identifier	Specifies the identifier of the current system thread.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DCF_SUPPORTED	Inquires whether the system supports downloadable digitizer configuration format (*DCF*) files.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_FALSE	Specifies that DCF files are not supported.		d				a	h	i							
M_TRUE	Specifies that DCF files are supported.	c		e	f	g					j	k	l	m	n	b

M_DEFAULT_PITCH_BYTE_MULTIPLE	Inquires the pitch (or stride) multiple (in bytes) for the buffers allocated on the system. SET	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
Value	Specifies the pitch multiple, in bytes.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DEVICE_NAME +	Inquires the user-defined name for the board. This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*		d													
Value	Specifies the device name that was set using M_DEVICE_NAME.		d													
M_DEVICE_NAME_MAX_SIZE	Inquires the maximum length that the device name can have of the string returned by M_DEVICE_NAME. This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d													
Value	Specifies the maximum length of the string returned by M_DEVICE_NAME.		d													
M_DIGITIZER_NUM	Inquires the total number of possible independent acquisition paths on the system. Note that this is not the same as the total number of allocated acquisition paths on the system.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															
	Inquires the total number of Video4Linux2 cameras discovered.					g										b
	Inquires the total number of GigE Vision cameras discovered.					g										
	Inquires the total number of USB cameras discovered.														n	
	Inquires the total number of GenICam complaint cameras discovered.			e												
	This inquire type is also available for the Zebra Concord PoE base model.		d													

	Inquires the total number of GigE Vision cameras connected directly to Zebra GevIQ or behind a switch.				f											
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
Value >= 0	Specifies the number of independent acquisition paths available.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DIGITIZER_TYPE +	Inquires the type of frame grabber(s) available to allocate a digitizer on the system.								i							
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*								i							
M_GTX2000	Specifies a Zebra Iris GTX 2000.								i							
M_GTX2000C	Specifies a Zebra Iris GTX 2000C.								i							
M_GTX3000	Specifies a Zebra Iris GTX 3000.								i							
M_GTX3000C	Specifies a Zebra Iris GTX 3000C.								i							
M_GTX5000	Specifies a Zebra Iris GTX 5000.								i							
M_GTX5000C	Specifies a Zebra Iris GTX 5000C.								i							
M_GTX8000	Specifies a Zebra Iris GTX 8000.								i							
M_GTX8000C	Specifies a Zebra Iris GTX 8000C.								i							
M_GTX12000	Specifies a Zebra Iris GTX 12000.								i							
M_GTX12000C	Specifies a Zebra Iris GTX 12000C.								i							
M_GTX16000	Specifies a Zebra Iris GTX 16000.								i							
M_GTX16000C	Specifies a Zebra Iris GTX 16000C.								i							
M_DISPLAY_OUTPUT_NUM	Inquires the number of video output controllers available on your Zebra imaging board.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
Value >= 0	Specifies the number of video output controllers.	c	d	e	f	g	a	h	i		j	k	l	m	n	b

M_DISTRIBUTED_AIL_PROTOCOL	Inquires the protocol used for DAIL.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DAIL_NOT_USED	Specifies that DAIL is not used.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DAIL_SHM	Specifies that DAIL is using the SHM protocol.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DAIL_TCPIP	Specifies that DAIL is using the TCP/IP protocol.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DISTRIBUTED_AIL_REMOTE_COMPUTER_NAME +	Inquires the name of the remote computer.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
Value	Specifies the computer name that was set in MsysAlloc().	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DISTRIBUTED_AIL_TYPE	Inquires how the system is allocated via DAIL.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DAIL_LOCAL_HOST	Specifies that the system is allocated on the local computer with DAIL.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DAIL_NOT_USED	Specifies that the system is not allocated with DAIL.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DAIL_REMOTE	Specifies that the system is allocated on a remote computer with DAIL.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_EXTENDED_INIT_FLAG	Inquires the system initialization flag. SET	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. INFO	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_CL	Specifies to initialize Camera Link transport layer technology.			e												
M_COMPLETE +	Specifies to initialize the system completely; the system is initialized to its default state and any required resident software is downloaded. INFO	c	d		f	g	a	h	i		j	k	l	m	n	

M_CXP	Specifies to initialize CoaXPress transport layer technology.			e												
M_GEV	Specifies to initialize Ethernet transport layer technology.			e												
M_MIXED	Specifies to initialize the transport layer technology specified by the GenTL Producer (library).			e												
M_U3V	Specifies to initialize USB transport layer technology.			e												
M_FIRMWARE_BUILDDATE	Inquires the date when the grab firmware was built.	c	d		f		a	h	i		j	k	l	m		
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d		f		a	h	i		j	k	l	m		
Value	Specifies the grab firmware build date.	c	d		f		a	h	i		j	k	l	m		
M_FIRMWARE_REVISION	Inquires the revision number of the grab firmware.	c	d		f		a	h	i		j	k	l	m		
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d		f		a	h	i		j	k	l	m		
Value	Specifies the grab firmware revision number.	c	d		f		a	h	i		j	k	l	m		
M_GC_FEATURE_EXECUTE_POLLING_MODE	Inquires whether to automatically perform execution-complete polling on executable camera features. SET			e	f	g							l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e	f	g							l	m		

	M_AUTOMATIC	Specifies that the specific executable camera feature is executed synchronously. INFO			e	f	g							l	m		
	M_MANUAL	Specifies that the executable camera feature is executed asynchronously. INFO			e	f	g							l	m		
M_GC_LOCAL_MAC_ADDRESS + *n*		Inquires the MAC address at port *n*, where *n* is a number between 0 and the number of Ethernet ports (M_GC_NIC_PORT_COUNT). This inquire type is also available for the Zebra Concord PoE base model.		d													
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*		d													
	Value	Specifies the MAC address at the specified port. The MAC address is stored as a 48-bit integer in an *AIL_INT64*. Typically, MAC addresses are written as 6 separate 8-bit numbers (hexadecimal number pairs). To retrieve these numbers, map the 48-bit MAC address to 6 *AIL_UINT8* variables. For example, the stored representation of the MAC address `84:20:fc:33:0e:e9` is `0x0000e90e33fc2084`. The least-significant byte (`84`) is the first number of the MAC address, the second least-significant byte (`0e`) is the second number of the MAC address and so on. The first 4 hexadecimal digits (2 most significant bytes) are not part of the MAC address.		d													
M_GC_NIC_PORT_COUNT		Inquires the number of Ethernet ports. This inquire type is also available for the Zebra Concord PoE base model.		d													
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d													
	2	Specifies that the system has 2 Ethernet ports.		d													
	4	Specifies that the system has 4 Ethernet ports.		d													
M_GENICAM_AVAILABLE		Inquires whether the system supports a GenICam-compliant device (such as a GigE Vision device).	c	d	e	f	g	a	h	i		j	k	l	m	n	b
		System specific															
		This inquire type is also available for the Zebra Concord PoE base model.		d													

UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_FALSE	Specifies that GenICam is not available.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_TRUE	Specifies that GenICam is available.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_GENTL_DEVICE_COUNT +	Inquires the number of GenTL devices associated with the specified interface.			e												
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e												
Value	Specifies the number of GenTL devices.			e												
M_GENTL_INTERFACE_COUNT	Inquires the number of GenTL interfaces available to your Aurora Imaging Library system. This is useful when the GenTL Producer (library) is of a mixed type.			e												
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e												
0 <= *Value* <= 33	Specifies the number of GenTL interfaces.			e												
M_GRAB_FPGA_FAN_RPM	Inquires the revolutions per minute (RPM) of the grab firmware's fan.	c			f								l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c			f								l	m		
Value	Specifies the speed of the grab firmware's fan, in RPM.	c			f								l	m		
M_INSTALLED_SYSTEM_DEVICE_COUNT	Inquires the number of installed devices of the specified system type.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_UNKNOWN	Specifies that the number of devices is unknown.	c	d	e				h	i						n	
Value	Specifies the number of installed devices of the specified system type.	c	d		f	g	a	h	i		j	k	l	m	n	b
	System specific															
	Note that for this system type, this inquire type will return 1.					g	a									b
M_LED_USER	Inquires the color of the user LED on your Zebra Iris GTX. SET								i							
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*								i							

M_GREEN	Specifies to turn the user LED green.								i							
M_OFF	Specifies to turn the user LED off.								i							
M_ORANGE	Specifies to turn the user LED orange.								i							
M_RED	Specifies to turn the user LED red.								i							
M_LOCATION	Inquires the location of the specified system.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_LOCAL	Specifies that the system is on the local computer.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_REMOTE	Specifies that the system is on a remote computer or was allocated via Distributed Aurora Imaging Library.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_MEMORY_FREE	Inquires the total amount of free on-board memory. Note that this memory might not be contiguous.				f						j	k	l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*				f						j	k	l	m		
Value	Specifies the amount of free on-board memory, in bytes.				f						j	k	l	m		
M_MEMORY_FREE_BANK_0	Inquires the total amount of free on-board memory the first SDRAM bank. Note that this memory might not be contiguous.				f						j	k	l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*				f						j	k	l	m		
Value	Specifies the amount of free on-board memory in the first SDRAM bank, in bytes.				f						j	k	l	m		
M_MEMORY_FREE_BANK_1	Inquires the total amount of free on-board memory second SDRAM bank. Note that this memory might not be contiguous.										j	k				
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*										j	k				
Value	Specifies the amount of free on-board memory second SDRAM bank, in bytes.										j	k				
M_MEMORY_SIZE	Inquires the amount of on-board memory.	c			f						j	k	l	m		

UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c			f						j	k	l	m		
Value	Specifies the total amount of on-board memory, in Mbytes.	c			f						j	k	l	m		
M_MEMORY_SIZE_BANK_0	Inquires the total amount of on-board shared memory in the first SDRAM bank.				f						j	k	l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*				f						j	k	l	m		
Value	Specifies the total amount of on-board shared memory in the first SDRAM bank, in Mbytes.				f						j	k	l	m		
M_MODIFIED_BUFFER_HOOK_MODE	Inquires whether to run user-defined functions hooked to a buffer modification on separate threads, up to the number of CPU cores present in the computer. SET	c		e	f	g					j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g					j	k	l	m	n	b
M_MULTI_THREAD +	Specifies to run user-defined functions hooked to a buffer modification on separate threads. INFO	c		e	f	g					j	k	l	m	n	b
M_SINGLE_THREAD	Specifies that only one thread should be created and that all user-defined functions hooked to buffer modifications are run on the same thread.	c		e	f	g					j	k	l	m	n	b
M_NUM_CAMERA_PRESENT	Inquires the number of GigE Vision-compliant cameras connected to the allocated system.			e		g									n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e		g									n	b
Value	Specifies the number of GigE Vision-compliant cameras.			e		g									n	b
M_NUMBER	Inquires the board number of the system. SET	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															
	This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_DEV*n* +	Specifies the device number (rank) of the board. INFO	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															

	This value is combined with a value from the following table: • To specify the GenTL Producer (library) available to this Aurora Imaging Library GenTL system			e												
"BoardIdentifierString"	Specifies the user-defined name of the board. INFO		d													
M_OWNER_APPLICATION	Inquires the Aurora Imaging Library identifier of the application context in which the system has been allocated.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
Application identifier	Specifies the Aurora Imaging Library application context identifier.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_PCIE_NUMBER_OF_LANES	Inquires the number of PCIe lanes that are currently active.	c			f						j	k	l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c			f						j	k	l	m		
1; 2; 4; 8; 16	Specifies the number of active lanes.	c			f						j	k	l	m		
M_PCIE_NUMBER_OF_LANES_MAX	Inquires the total number of PCIe lanes that your board can use.				f						j	k	l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*				f						j	k	l	m		
1; 2; 4; 8; 16	Specifies the total number of lanes.				f						j	k	l	m		
M_PCIE_SPEED	Inquires the negotiated generation of the PCIe standard currently being used by the active PCIe lanes. For example, if your board is a PCIe 3.0 device, but it is connected to a PCIe 2.0 device, this inquire type returns M_GEN2. To establish the current PCIe speed possible for your board, use the following equation: M_PCIE_NUMBER_OF_LANES x 2 x *speed of PCIe generation*	c			f						j	k	l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c			f						j	k	l	m		
M_GEN*n*	Specifies the PCIe generation, where *n* is a value from 1 to 3.	c			f						j	k	l	m		
M_INVALID	Specifies that the PCIe generation cannot be returned.	c			f						j	k	l	m		

M_PCIE_SPEED_MAX	Inquires the generation of the PCIe standard of your board. For example, if your board is a PCIe 3.0 device, this inquire type returns M_GEN3. To establish the maximum PCIe speed possible for your board, use the following equation: M_PCIE_NUMBER_OF_LANES_MAX x 2 x *maximum speed of PCIe generation*				f						j	k	l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*				f						j	k	l	m		
M_GEN*n*	Specifies the PCIe generation, where *n* is a value from 1 to 3.				f						j	k	l	m		
M_INVALID	Specifies that the PCIe generation cannot be returned.				f						j	k	l	m		
M_POWER_OVER_CABLE +	Inquires whether the board provides power to connected devices. SET		d				a							m		
	System specific															
	You must specify a combination value from the following table: • To specify which connector or port to inquire						a							m		
	For a Host system, this inquires whether PoE (Power over Ethernet) is enabled on the specified port. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
	For Zebra Concord PoE, this inquires whether PoE (Power over Ethernet) is enabled on all ports. This control type is also available for the Zebra Concord PoE base model.		d													
	For Zebra Rapixo CXP, this inquires whether PoCXP (Power over CXP) is automatically enabled when a PoCXP-compliant camera is connected to the specified connector. M_ON and M_OFF are only returned when PoCXP has been manually enabled or disabled using MsysControl() with M_POWER_OVER_CABLE. When automatic detection is enabled (M_AUTOMATIC), use M_POWER_OVER_CABLE_STATUS to learn whether PoCXP is currently enabled or disabled.													m		

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a						m		
	M_AUTOMATIC	Specifies to automatically enable or disable PoCXP on this CXP connector, depending on detected device support.												m		
	M_OFF	Specifies not to provide power to connected devices. INFO		d				a						m		
	M_ON	Specifies to provide power to connected devices. INFO		d				a						m		
M_POWER_OVER_CABLE_STATUS +		Inquires whether PoCXP (power over CXP) is enabled on a CXP connector, and whether an over- or under-current condition has been detected. Once detected, an over- or under-current condition persists until the PoCXP status is reset for the connector (using MsysControl() with M_POWER_OVER_CABLE and M_RESET). Typically, to prevent the over- or under-current condition from immediately recurring, you should first restart PoCXP for this connector using MsysControl() with M_POWER_OVER_CABLE and M_OFF and then returning it to its previous setting (M_AUTOMATIC is recommended). If the condition persists, discontinue using PoCXP with your camera by leaving M_POWER_OVER_CABLE set to M_OFF. SET You must specify a combination value from the following table: • To specify which connector or port to inquire												m		
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*												m		
	M_OFF	Specifies that PoCXP is disabled on the CXP connector, either because a non-PoCXP-compatible camera is connected, or because PoCXP was manually disabled (using MsysControl() with M_POWER_OVER_CABLE and M_OFF).												m		
	M_ON	Specifies that PoCXP is enabled on the CXP connector, either because a PoCXP-compatible camera is connected, or because PoCXP was manually enabled (using MsysControl() with M_POWER_OVER_CABLE and M_ON).												m		
	M_OVER_CURRENT	Specifies that PoCXP is disabled on the connector because an over-current condition was detected and has not been reset.												m		

M_SENSE	Specifies that PoCXP will be enabled or disabled automatically on the connector when a camera is connected (depending on detected device support).													m		
M_UNDER_CURRENT	Specifies that PoCXP is disabled on the connector because an under-current condition was detected and has not been reset.													m		
M_PROCESSOR_NUM	Inquires the number of processors (CPUs) available on the allocated Zebra imaging board.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															
	This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
Value >= 0	Specifies the number of processors available.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_PROFINET_HARDWARE_SUPPORTED	Inquires whether the system has access to a PROFINET Engine.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_NO	Specifies that the system does not have access.	c	d	e	f	g	a				j	k	l	m	n	b
M_YES	Specifies that the system has access.						a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_SERIAL_NUMBER +	Inquires the serial number of the Zebra Imaging board, as a string.	c	d	e	f	g	a	h	i		j	k	l	m		b
	System specific															
	[This is only applicable to Zebra 4Sight] M_SERIAL_NUMBER only applies to Zebra products. When a host system is allocated on a non-Zebra computer, no serial number is returned.						a									

	This inquire type is also available for the Zebra Concord PoE base model.		d													
	When a Video4Linux2 system is allocated, no serial number is returned.															b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*	c	d	e	f	g	a	h	i		j	k	l	m		b
M_SHARED_MEMORY_FREE	Inquires the total amount of free on-board shared memory.				f						j	k	l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*				f						j	k	l	m		
Value	Specifies the total amount of free on-board shared memory, in bytes.				f						j	k	l	m		
M_SHARED_MEMORY_SIZE	Inquires the total amount of on-board shared memory.				f						j	k	l	m		
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*				f						j	k	l	m		
Value	Specifies the total amount of on-board shared memory, in Mbytes.				f						j	k	l	m		
M_SYSTEM_DESCRIPTOR +	Inquires the system descriptor. SET	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															
	This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_SYSTEM_TYPE	Inquires the type of system allocated. If a DAIL remote system has been registered, you can use the following macros to further inquire about the system: SET • To determine whether the system is a DAIL remote system, use the **M_IS_SYSTEM_DISTRIBUTED** macro. This macro returns a non-zero value if the system is a DAIL remote system and zero if the system is local. • To remove the bit flag for DAIL remote systems, use the **M_REAL_SYSTEM_TYPE** macro. This macro returns only the system type and not the system type and DAIL remote system bit flag.	c	d	e	f	g	a	h	i		j	k	l	m	n	b

			c	d	e	f	g	a	h	i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*															
	M_SYSTEM_CLARITY_UHD_TYPE +	Specifies an Aurora Imaging Library Clarity UHD system.	c														
	M_SYSTEM_CONCORD_POE_TYPE +	Specifies an Aurora Imaging Library Concord POE system. Note that to use Zebra Concord PoE for acquisition, you must allocate and use an Aurora Imaging Library GigE Vision system (using MsysAlloc() with M_SYSTEM_GIGE_VISION) instead; refer to information denoted for a GigE Vision system. You only need to allocate and use an Aurora Imaging Library Concord PoE system (using MsysAlloc() with M_SYSTEM_CONCORD_POE) to use the other functionality on the board and to inquire about the board itself. So information in this reference, for use with a Concord PoE system, is denoted Concord PoE with ToE since it is typically only applicable to this model of the board. If the information is applicable to the Zebra Concord PoE base model it will be explicitly specified. For more information, see the Using an Aurora Imaging Library Concord PoE system with the Zebra Concord PoE base model section of the Zebra Concord PoE chapter in the Hardware-specific Notes. This inquire type is also available for the Zebra Concord PoE base model.		d												
	M_SYSTEM_GENTL_TYPE +	Specifies an Aurora Imaging Library GenTL system.			e												
	M_SYSTEM_GEVIQ_TYPE +	Specifies an Aurora Imaging Library GevIQ system.				f											
	M_SYSTEM_GIGE_VISION_TYPE +	Specifies an Aurora Imaging Library GigE Vision system.					g										
	M_SYSTEM_HOST_TYPE +	Specifies the Host.		d				a	h								
	M_SYSTEM_INDIO_TYPE +	Specifies an Aurora Imaging Library Indio system.							h								
	M_SYSTEM_IRIS_GTX_TYPE +	Specifies an Aurora Imaging Library Iris GTX system.								i							
	M_SYSTEM_RADIENTEVCL_TYPE +	Specifies an Aurora Imaging Library Radient eV-CL system.										j					

M_SYSTEM_RAPIXOCL_TYPE +	Specifies an Aurora Imaging Library Rapixo Pro CL system.											k				
M_SYSTEM_RAPIXOCOF_TYPE +	Specifies an Aurora Imaging Library Rapixo CoF system.												l			
M_SYSTEM_RAPIXOCXP_TYPE +	Specifies an Aurora Imaging Library Rapixo CXP system.													m		
M_SYSTEM_USB3_VISION_TYPE +	Specifies an Aurora Imaging Library USB3 Vision system.														n	
M_SYSTEM_V4L2_TYPE +	Specifies an Aurora Imaging Library Video4Linux2 system.															b
M_TEMPERATURE	Inquires the current temperature of the sensor.	c						i								
	System specific															
	On Zebra Clarity UHD, this value returns the temperature of the Host interface.	c														
	On Zebra Iris GTX, this value returns the temperature of the daughter board (inbox).							i								
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c						i								
Value	Specifies the current temperature, in degrees Celsius.	c						i								
M_TEMPERATURE_CPU	Inquires the current temperature of the CPU.							i								
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*							i								
Value	Specifies the current temperature, in degrees Celsius.							i								
M_TEMPERATURE_FPGA	Inquires the current temperature of the grab firmware.	c			f			i		j	k	l	m			
	System specific															
	Note that, on your Zebra Clarity UHD, this value inquires the current temperature of the Acquisition controller.	c														
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c			f			i		j	k	l	m			

Value	Specifies the current temperature, in degrees Celsius.	c			f				i		j	k	l	m		
M_TEMPERATURE_FPGA_MAX_MEASURED	Inquires the maximum temperature of the grab firmware measured.								i							
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*								i							
Value	Specifies the maximum measured temperature, in degrees Celsius.								i							
M_TEMPERATURE_IMAGE_SENSOR +	Inquires the current temperature of the image sensor.								i							
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*								i							
Value	Specifies the current temperature, in degrees Celsius.								i							
M_THREAD_MODE	Inquires whether threads allocated on the system can execute in asynchronous mode. SET	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															
	This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_ASYNCHRONOUS	Specifies that threads allocated on the system can execute in asynchronous mode. INFO	c		e	f	g			i		j	k	l	m	n	b
M_SYNCHRONOUS	Specifies that threads allocated on the system can only execute in synchronous mode. INFO	c	d	e	f	g	a	h	i		j	k	l	m	n	b
M_TIMEOUT	Inquires the maximum amount of time for the Host to wait for a synchronous function to return before generating a time-out error, in sec. SET	c	d	e	f	g			i		j	k	l	m	n	b
	System specific															
	This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g			i		j	k	l	m	n	b
M_INFINITE	Waits indefinitely.								i							
Value	Specifies the time to wait, in sec.	c	d	e	f	g			i		j	k	l	m	n	b

M_UART_PRESENT	Inquires the number of UARTs on the system.	c	d	e	f	g	a	h	i		j	k	l	m	n	b
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
	This inquire type is also available for the Zebra Concord PoE base model.		d													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	d	e	f	g	a	h	i		j	k	l	m	n	b
Value	Specifies the number of UARTs on the system.	c	d	e	f	g	a	h	i		j	k	l	m	n	b

Combination value for [when InquireType=M_SYSTEM_TYPE: M_SYSTEM_CLARITY_UHD_TYPE; M_SYSTEM_CONCORD_POE_TYPE; M_SYSTEM_GENTL_TYPE; M_SYSTEM_GEVIQ_TYPE; M_SYSTEM_GIGE_VISION_TYPE; M_SYSTEM_HOST_TYPE; M_SYSTEM_INDIO_TYPE; M_SYSTEM_IRIS_GTX_TYPE; M_SYSTEM_RADIENTEVCL_TYPE; M_SYSTEM_RAPIXOCL_TYPE; M_SYSTEM_RAPIXOCOF_TYPE; M_SYSTEM_RAPIXOCXP_TYPE; M_SYSTEM_USB3_VISION_TYPE; M_SYSTEM_V4L2_TYPE].

The following value might be combined with the above-mentioned values to determine if the inquired system type is remote.

By default, an installed systems that is remote will have a distributed flag appended to the returned system type.

For inquiring if a system is remote	
Combination value	Description
M_SYSTEM_DISTRIBUTED_FLAG	Specifies that the system is a DAIL remote system.

The following inquire types and inquire values relate to (if available) the connection.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring CXP connection errors and settings																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_CONNECTION_STATE +	Inquires the state of the connection for a Zebra Rapixo CXP or Zebra Rapixo CoF input connector. Note that, connection errors are only meaningful if the specified connection is locked. You must specify a combination value from the following table: • To specify the CXP input connector															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*												l	m		

	M_INVALID	Specifies that the state of the connection is invalid. > For example, if you have a Zebra Rapixo CXP Dual board, you can only have connections at input connectors 0 and 1. Therefore, input connectors 2 and 3 would return M_INVALID.											I	m		
	M_LOCK	Specifies that the state of the connection is locked.											I	m		
	M_UNLOCK	Specifies that the state of the connection is unlocked.											I	m		
M_TL_ERROR_CORRECTED_COUNT +		Inquires the count of corrected duplicate character errors in the CXP control words. SET You must specify a combination value from the following table: • To specify the CXP input connector											I	m		
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*											I	m		
	Value	Specifies the current count of corrected duplicate character errors in the CXP control words.											I	m		
M_TL_ERROR_COUNT +		Inquires the count of all errors encountered. SET You must specify a combination value from the following table: • To specify the CXP input connector											I	m		
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*											I	m		
	Value	Specifies the current count of all errors encountered.											I	m		
M_TL_ERROR_CTRL_CRC_COUNT +		Inquires the count of CRC errors detected in a control packet. SET You must specify a combination value from the following table: • To specify the CXP input connector											I	m		
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*											I	m		
	Value	Specifies the current count of CRC errors detected in a control packet.											I	m		
M_TL_ERROR_DATA_CRC_COUNT +		Inquires the count of CRC errors detected in a data packet. SET You must specify a combination value from the following table: • To specify the CXP input connector											I	m		
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*											I	m		

	Value	Specifies the current count of CRC errors detected in a data packet.												I	m		
M_TL_ERROR_ENCODING_COUNT +		Inquires the count of protocol encoding errors detected. SET You must specify a combination value from the following table: • To specify the CXP input connector												I	m		
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*												I	m		
	Value	Specifies the current count of protocol encoding errors detected.												I	m		
M_TL_ERROR_EVENT_CRC_COUNT +		Inquires the count of CRC errors detected in an event packet. SET You must specify a combination value from the following table: • To specify the CXP input connector												I	m		
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*												I	m		
	Value	Specifies the current count of CRC errors detected in an event packet.												I	m		
M_TL_ERROR_LOCK_LOSS_COUNT +		Inquires the count of lock losses encountered. SET You must specify a combination value from the following table: • To specify the CXP input connector												I	m		
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*												I	m		
	Value	Specifies the current count of lock losses encountered.												I	m		
M_TL_ERROR_UNCORRECTED_COUNT +		Inquires the count of uncorrected duplicate character errors in the CXP control words. SET You must specify a combination value from the following table: • To specify the CXP input connector												I	m		
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*												I	m		
	Value	Specifies the current count of uncorrected duplicate character errors in the CXP control words.												I	m		

Combination value for the values listed in For inquiring CXP connection errors and settings.

You must add the following value to the above-mentioned values to specify the CXP input connector.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the CXP input connector																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_CONNECTION*n*	Specifies the connection made at CXP input connector *n*, where *n* is the CXP input connector number from 0 to 3.												l	m		

Combination value for [when InquireType=M_NUMBER: M_DEVn].

The following value is combined with the above-mentioned value to specify the GenTL Producer (library) available to this Aurora Imaging Library GenTL system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For identifying the instance of the GenTL Producer library identifier to use																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GENTL_PRODUCER(AIL_INT IndexValue)	Specifies the GenTL Producer (library) available to this Aurora Imaging Library GenTL system.			e												
	Parameters															
	IndexValue Specifies the index of the GenTL Producer.			e												
	0 <= *Value* <= 127 — Specifies the index.			e												

Combination value for M_GENTL_DEVICE_COUNT.

You can add the following value to the above-mentioned value to specify about which interface type to inquire.

The GenTL interface refers to the type of physical connection (port) between the Host and the device (such as Camera Link or Ethernet connections). A mixed-type GenTL Producer (library) supports multiple interface types and M_GENTL_INTERFACEn can be used with any GenTL Producer provided it reports more than one type of interface. You can use the value below to specify about which interface type you would like to inquire.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which configuration information file to access																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GENTL_INTERFACE*n*	Specifies the GenTL interface to use.			e												

Combination value for [when InquireType=M_MODIFIED_BUFFER_HOOK_MODE: M_MULTI_THREAD].

The following value might be combined with the above-mentioned value to determine the number of hook threads.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

Returns the number of hook threads that were allocated																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Value >= 1	Specifies the number of hook threads that were allocated . INFO			e		g					j	k	l	m	n	b

Combination value for M_DIGITIZER_TYPE; M_TEMPERATURE_IMAGE_SENSOR.

You can add the following value to the above-mentioned values to specify the acquisition path to inquire.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the acquisition path to inquire																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEV0	Specifies the number of the acquisition path to inquire.								i							

Combination value for [when InquireType=M_EXTENDED_INIT_FLAG: M_COMPLETE].

The following value might be combined with the above-mentioned value to set that the board will have a user-defined board name..

You can use the following value on its own, or add it to the above-mentioned value, to specify that the target board is identified using its user-defined name.

> This InitFlag is also available for the Zebra Concord PoE base model.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying that the target board is identified using its user-defined name																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEVICE_NAME	Specifies that the SystemNum parameter identifies the target board using its user-defined name.		d													

Combination value for M_POWER_OVER_CABLE; M_POWER_OVER_CABLE_STATUS.

You must add the following value to the above-mentioned values to specify which connector or port to inquire.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which connector or port to inquire																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_CONNECTION*n*	Specifies to inquire connector *n*, where *n* corresponds to a physical connector or port on your board or industrial computer. INFO INFO						a							m		

The following inquire types allow you to inquire information about discovered devices accessible by the specified system.

> Before using these inquire types, you must refresh the list of discovered devices for the specified system using MsysControl() with M_DISCOVER_DEVICE. This list is cached; if devices are connected or disconnected, you must refresh the list (until you do, using these inquire types will return outdated information).

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For discovering connected devices																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																

M_DISCOVER_DEVICE_ADDRESS + *n*	Inquires the IP address (IPv4) or the connector of device *n* as a string, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT).															
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*				f	g						l	m		
AIL_TEXT("*BoardString*") 1	Specifies the string containing information about the board type, board device number (previously allocated using MsysAlloc() with M_DEVn), and the acquisition path (previously allocated using MdigAlloc() with M_DEVn).											l	m		
AIL_TEXT("*nnn.nnn.nnn.nnn*") 1	Specifies the IP address of the GigE Vision-compliant grabberless interface camera, as a string. The address string is expressed in dotted decimal notation, where each dotted decimal (*nnn*) is a number between 000 and 255.				f	g									
M_DISCOVER_DEVICE_ALLOCATION_STATUS + *n*	Inquires whether device *n* has been allocated, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT).			e	f	g						l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e	f	g						l	m	n	b
M_DEVICE_ALLOCATED	Specifies that the device is allocated (either by this application, or by another application).			e	f	g						l	m	n	b
	System specific														
	The device might be allocated by an application running on another computer on the network.					g									
M_DEVICE_FREE	Specifies that the device is not allocated. You can allocate the device using MdigAlloc().			e	f	g						l	m	n	b
M_DEVICE_UNREACHABLE	Specifies that the device is unreachable.				f	g								n	
	System specific														
	The device is using an IP address that is outside the range of valid addresses on the network.				f	g									
	The device is controlled by a third-party library. Install the Zebra USB3 Vision driver to resolve the issue.													n	

M_DEVICE_UNREACHABLE_ON_DHCP_NETWORK	Specifies that the device is unreachable on the DHCP network. The device has DHCP disabled and is using a static IP or link-local address (LLA). Connect the device peer-to-peer to resolve the issue. > Note that to connect the device peer-to-peer, you must enable DHCP and disable static IP on the device.				f	g										
M_DISCOVER_DEVICE_COUNT	Inquires the number of devices that are accessible by the specified system.			e	f	g						l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e	f	g						l	m	n	b
Value	Specifies the number of devices that are accessible by the specified system.			e	f	g						l	m	n	b
M_DISCOVER_DEVICE_DIGITIZER_NUMBER + *n*	Inquires the digitizer number that you can pass to MdigAlloc() to allocate device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT).			e	f	g						l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e	f	g						l	m	n	b
M_DEV*n*	Specifies the digitizer number.			e	f	g						l	m	n	b
M_DISCOVER_DEVICE_INTERFACE_NAME + *n*	Inquires the name of the host interface used by device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT).			e	f	g						l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e	f	g						l	m	n	b
M_DISCOVER_DEVICE_INTERFACE_TYPE + *n*	Inquires the type of interface used by device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT).			e	f	g						l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e	f	g						l	m	n	b
M_CLHS	Specifies that the device is connected using Camera Link HS.			e											
M_COF	Specifies that the device is connected using CoaXPress-over-Fiber.			e								l			

	M_CUSTOM	Specifies that the device is connected using a custom/non-standard interface type.			e											
	M_CXP	Specifies that the device is connected using CoaXPress.			e									m		
	M_GIGE_VISION	Specifies that the device is connected using GigE Vision.			e	f	g									
	M_USB3_VISION	Specifies that the device is connected using USB3 Vision.			e										n	
	M_V4L2	Specifies that the device is connected using Video4Linux2.														b
M_DISCOVER_DEVICE_MANUFACTURER_INFO + *n* +		Inquires the manufacturer-specific information string reported by device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT). If the camera does not provide this information, an empty string "\0" is written.			e	f	g						l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e	f	g						l	m	n	b
M_DISCOVER_DEVICE_MANUFACTURER_NAME + *n* +		Inquires the manufacturer name string reported by device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT). If the camera does not provide this information, an empty string "\0" is written.			e	f	g						l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e	f	g						l	m	n	b
M_DISCOVER_DEVICE_MODEL_NAME + *n* +		Inquires the model name string reported by device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT). If the camera does not provide this information, an empty string "\0" is written.			e	f	g						l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e	f	g						l	m	n	b

M_DISCOVER_DEVICE_SERIAL_NUMBER + *n*	Inquires the serial number of device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT).			e	f	g							l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e	f	g							l	m	n	b
M_DISCOVER_DEVICE_UNIQUE_IDENTIFIER + *n*	Inquires the unique identifier reported by device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT).			e	f	g							l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e	f	g							l	m	n	b
M_DISCOVER_DEVICE_USER_NAME + *n* +	Inquires the user-defined name string reported by device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT). Refer to your camera manual to learn how to set this name string. If the camera does not provide this information, an empty string "\0" is written.			e	f	g							l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e	f	g							l	m	n	b
M_DISCOVER_DEVICE_VERSION + *n* +	Inquires the version number string reported by device *n*, where *n* is a number between 0 and the number of discovered devices (M_DISCOVER_DEVICE_COUNT). If the camera does not provide this information, an empty string "\0" is written.			e	f	g							l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e	f	g							l	m	n	b

[1] If the value was passed in a variable, it won't be returned enclosed in AIL_TEXT().

The following inquire types and inquire values allow you to inquire the number, mode, and the purpose of I/O signals (such as, auxiliary signal). Once the format, routing, and mode are determined for an I/O signal, you can further inquire the I/O signal using the inquire types described in the For inquiring the state of specified user-bits in a static-user-output register table.

> Note that for other Zebra imaging boards that have auxiliary I/O signals, but are not supported with the constants below, see MdigInquire().

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring I/O signals and their mode

InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_AUX_IO_COUNT	Inquires the total number of auxiliary signals.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
Value	Specifies the total number of auxiliary signals.		d				a	h	i							
M_AUX_IO_COUNT_IN	Inquires the total number of auxiliary input and I/O signals.		d				a	h	i							
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
Value	Specifies the total number of auxiliary input and I/O signals.		d				a	h	i							
M_AUX_IO_COUNT_OUT	Inquires the total number of auxiliary output and I/O signals.		d				a	h	i							
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
Value	Specifies the total number of auxiliary output and I/O signals.		d				a	h	i							
M_IO_DEBOUNCE_TIME +	Inquires the amount of time that the specified auxiliary input signal is debounced. SET You must specify a combination value from the following table: • To specify the type and number of the I/O signal		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
Value >= 0	Specifies the minimum amount of time to ignore any additional signal transitions after accepting a signal transition, in nsec. INFO		d				a	h	i							
M_IO_FORMAT +	Inquires the format for an I/O signal. For more information, refer to the Technical information appendix in the Installation and Hardware Reference manual for your Zebra imaging board. You must specify a combination value from the following table: • To specify the type and number of the I/O signal		d				a	h								
	System specific															

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h								
M_CURRENT_SOURCE	Specifies that the specified I/O is an current-controlled output.		d				a	h								
M_OPEN_DRAIN	Specifies that the specified I/O is an open collector (open drain) I/O signal.		d				a	h								
M_OPTO	Specifies that the specified I/O is an opto-coupled I/O signal.		d				a	h								
M_IO_GLITCH_FILTER_STATE +	Inquires whether the glitch filter is enabled. SET You must specify a combination value from the following table: • To specify the type and number of the I/O signal		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_DISABLE	Specifies not to use a glitch filter.		d				a	h	i							
M_ENABLE	Specifies to use a glitch filter.		d				a	h	i							
M_IO_INTERRUPT_ACTIVATION +	Inquires the signal transition upon which to generate an interrupt, if interrupt generation has been enabled for the specified I/O signal. SET You must specify a combination value from the following table: • To specify the type and number of the I/O signal		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_ANY_EDGE	Specifies to generate an interrupt upon both a low-to-high and a high-to-low signal transition. INFO						a	h	i							

	M_EDGE_FALLING	Specifies that an interrupt will be generated upon a high-to-low signal transition.		d				a	h	i						
	M_EDGE_RISING	Specifies that an interrupt will be generated upon a low-to-high signal transition.		d				a	h	i						
M_IO_INTERRUPT_STATE +		Inquires whether to generate an interrupt upon the specified transition of the I/O signal. SET You must specify a combination value from the following table: • To specify the type and number of the I/O signal		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
	M_DISABLE	Specifies not to generate an interrupt.		d				a	h	i						
	M_ENABLE	Specifies to generate an interrupt.		d				a	h	i						
M_IO_INVERTER +		Inquires whether the output of the specified I/O signal should be inverted. SET You must specify a combination value from the following table: • To specify the type and number of the I/O signal		d				a	h	i						
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
	M_DEFAULT	Same as M_DISABLE.		d				a	h	i						
	M_DISABLE	Specifies not to invert the specified I/O signal.		d				a	h	i						
	M_ENABLE	Specifies to invert the specified I/O signal.		d				a	h	i						
M_IO_MODE +		Inquires mode of the specified I/O signal. You must specify a combination value from the following table: • To specify the type and number of the I/O signal						a	h	i						
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*						a	h	i						
	M_INPUT	Specifies that the signal is for input.						a	h	i						
	M_OUTPUT	Specifies that the signal is for output.						a	h	i						

M_IO_SOURCE +	Inquires the type of signal to route to an output signal, or an I/O signal set to output mode. SET You must specify a combination value from the following table: • To specify the type and number of the I/O signal		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_EXPOSURE	Specifies to route the exposure signal of the camera.								i							
M_GRAB_TRIGGER_READY	Specifies to route the internal grab trigger ready signal.								i							
M_IO_COMMAND_LIST*n* +	Specifies to route a bit of the I/O command register of I/O command list *n*, where *n* is the number of the I/O command list. INFO This value is combined with a value from the following table: • To specify which I/O command register bit was used		d				a	h	i							
M_ROTARY_ENCODER*n*	Specifies to route the output of rotary encoder *n*, where *n* is the number of rotary encoders available. INFO		d				a	h	i							
M_TIMER_STROBE	Specifies to route the internal timer strobe signal.								i							
M_TIMER*n*	Specifies to route the output of timer *n*, where *n* is the number of timers available. INFO		d				a	h	i							
M_USER_BIT*n*	Specifies to route the state of bit n of the main static-user-output register, where *n* is the bit number. INFO		d				a	h	i							
M_IO_STATUS +	Inquires the status of the specified I/O signal. You must specify a combination value from the following table: • To specify the type and number of the I/O signal		d				a	h	i							
	System specific															
	Inquiring the status of signals dedicated for trigger inputs will generate an error.								i							
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									

UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i								
M_INVALID	Specifies that the I/O signal is disabled.						a										
M_OFF	Specifies that the I/O signal is off.		d				a	h	i								
M_ON	Specifies that the I/O signal is on.		d				a	h	i								
M_UNKNOWN	Specifies that the I/O signal cannot be inquired with its current configuration.		d				a	h	i								
M_IO_STATUS_ALL +	Inquires the status of all available I/O signals. Note that if there are I/O signals that cannot be inquired, the bits representing those signals, in the bit-encoded value returned, are not necessarily valid; these bits should be ignored.		d				a	h	i								
	System specific																
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a										
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*		d				a	h	i								
Value	Specifies the bit-encoded value representing the status of all available and inquirable I/O signals.		d				a	h	i								
	System specific																
	The status of signals dedicated for trigger inputs are indeterminable. The bit(s) representing dedicated trigger signal(s) are to be ignored.								i								

Combination value for M_IO_DEBOUNCE_TIME; M_IO_FORMAT; M_IO_GLITCH_FILTER_STATE; M_IO_INTERRUPT_ACTIVATION; M_IO_INTERRUPT_STATE; M_IO_INVERTER; M_IO_MODE; M_IO_SOURCE; M_IO_STATUS.

You must add the following value to the above-mentioned values to specify the type and number of the I/O signal.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the type and number of the I/O signal																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_AUX_IO*n*	Specifies to affect auxiliary signal n, where *n* is the signal number. INFO		d				a	h	i							

Combination value for M_IO_STATUS_ALL.

You can add the following value to the above-mentioned value to specify the type of I/O signal to inquire.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of I/O signal to inquire																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_AUX_IO	Specifies to inquire all auxiliary signals.		d				a	h	i								

The following inquire types and inquire values specify the settings for specified bits in a static-user-output register. The user-bits are the bits associated with output signals or I/O signals set to output. To establish which user-bits can be routed to a specific signal, see the connectors and signal names section of the Aurora Imaging Library Hardware-specific Notes chapter for your Zebra imaging board.

> Note that for other Zebra imaging boards that have user-bits, but are not supported with the constants below, see MdigInquire().

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the state of specified user-bits in a static-user-output register																	
InquireType UserVarPtr - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_USER_BIT_COUNT	Inquires the total number of bits of the main static-user-output register.															
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
Value	Specifies the number of bits in the main static-user-output register.		d				a	h	i							
M_USER_BIT_STATE +	Inquires the state of the specified bit in a static-user-output register. SET You must specify a combination value from the following table: • To specify the bit in the static-user-output register		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_OFF	Specifies that the specified bit is set to off.		d				a	h	i							
M_ON	Specifies that the specified bit is set to on.		d				a	h	i							
M_USER_BIT_STATE_ALL	Inquires the state of all the bits in the main static-user-output register. SET		d				a	h	i							
	System specific															

		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*		d				a	h	i							
	Value	Specifies a bit-encoded value that establishes the value of all the bits of the specified static-user-output register. INFO		d				a	h	i							

Combination value for M_USER_BIT_STATE.

You must add the following value to the above-mentioned value to specify the bit in the static-user-output register.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the bit in the static-user-output register																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)	
M_USER_BIT*n*	Specifies to affect bit n of the main static-user-output register.		d				a	h	i								

The following inquire types and inquire values allow you to inquire about timers and the signals generated from a timer (timer output signals). For more information, see the Timers and coordinating events section of Chapter 56: I/O signals and communicating with external devices.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the settings of a timer																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)	
UserVarPtr - Possible values returned																	
M_TIMER_ARM +	Inquires whether to enable the timer arming mechanism. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire																
	System specific																
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a										
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i								
M_DISABLE	Specifies that timer arming is disabled.		d				a	h	i								
M_ENABLE	Specifies that timer arming is enabled.		d				a	h	i								

M_TIMER_ARM_ACTIVATION +	Inquires the signal transition upon which to arm the timer. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_ANY_EDGE	Specifies that the timer will be armed by both a high-to-low and a low-to-high signal transition.		d				a	h	i							
M_EDGE_FALLING	Specifies that the timer will be armed upon a high-to-low signal transition.		d				a	h	i							
M_EDGE_RISING	Specifies that the timer will be armed upon a low-to-high signal transition.		d				a	h	i							
M_LEVEL_HIGH	Specifies that a timer is continuously armed during a high signal polarity.		d				a	h	i							
M_LEVEL_LOW	Specifies that a timer is continuously armed during a low signal polarity.		d				a	h	i							
M_TIMER_ARM_SOURCE +	Inquires which input signal will arm the timer. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the source signal used to arm the specified timer, where *n* is the number of the auxiliary signal. INFO		d				a	h	i							
M_EXPOSURE	Specifies to use the exposure signal as the trigger source.								i							

	M_GRAB_TRIGGER_READY	Specifies to route the internal grab trigger ready signal to the specified signal.								i						
	M_SOFTWARE	Specifies to use software to arm the specified timer.		d				a	h	i						
	M_TIMER_STROBE	Specifies to route the strobe's timer signal to the specified signal.								i						
	M_TIMER*n*	Specifies to use the output signal of the specified timer as the source signal to arm the timer, where *n* is the timer number. INFO		d				a	h	i						
M_TIMER_CLOCK_ACTIVATION +		Inquires the edge of the signal that will increment the clock used to control the active portion of the timer's output signal. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
	M_ANY_EDGE	Specifies that the clock will be incremented by both a high-to-low and a low-to-high signal transition.		d				a	h	i						
	M_EDGE_FALLING	Specifies that the clock will be incremented by a high-to-low signal transition.		d				a	h	i						
	M_EDGE_RISING	Specifies that the clock will be incremented by a low-to-high signal transition.		d				a	h	i						
M_TIMER_CLOCK_FREQUENCY +		Inquires the frequency of the clock source signal for the active portion of the timer's output signal (M_TIMER_CLOCK_SOURCE). SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*		d				a	h	i						

	M_UNKNOWN	Specifies that the signal is not periodic or the frequency is unknown.		d				a	h	i						
	0 < *Value* <= *Frequency of* M_SYSCLK	Specifies the frequency, in Hz.		d				a	h	i						
M_TIMER_CLOCK_SOURCE +		Inquires the source of the clock that drives the active portion of the specified timer's output signal. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
	M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the clock source, where *n* is the number of the auxiliary signal. INFO		d				a	h	i						
	M_ROTARY_ENCODER*n*	Specifies to use rotary decoder *n* as the clock source, where *n* is the number of the rotary decoder. INFO		d				a	h	i						
	M_SYSCLK	Specifies to use the allocated system's clock source.		d				a	h	i						
M_TIMER_DELAY +		Inquires the delay between the timer trigger and the active portion of the timer's output signal. Note, an error is generated if the specified delay cannot be respected. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*		d				a	h	i						
	0 <= *Value* <= *Max. value*	Specifies the delay. INFO		d				a	h	i						

M_TIMER_DELAY_CLOCK_ACTIVATION +	Inquires the signal transition that will increment the clock used to control the delay between the timer's trigger and the active portion of the timer's output signal. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_ANY_EDGE	Specifies that the clock will be incremented by both a high-to-low and a low-to-high signal transition.		d				a	h	i							
M_EDGE_FALLING	Specifies that the clock will be incremented by a high-to-low signal transition.		d				a	h	i							
M_EDGE_RISING	Specifies that the clock will be incremented by a low-to-high signal transition.		d				a	h	i							
M_TIMER_DELAY_CLOCK_FREQUENCY +	Inquires the frequency of the clock source signal for the delay between the timer's trigger and the active portion of the timer's output signal (M_TIMER_DELAY_CLOCK_SOURCE). SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*		d				a	h	i							
M_UNKNOWN	Specifies that the signal is not periodic or the frequency is unknown.		d				a	h	i							
0 < *Value* <= *Frequency of* M_SYSCLK	Specifies the frequency, in Hz.		d				a	h	i							

M_TIMER_DELAY_CLOCK_SOURCE +	Inquires the source of the clock that drives the delay between the timer's trigger and the active portion of the specified timer's output signal. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the clock source, where *n* is the number of the auxiliary signal. INFO		d				a	h	i							
M_FOLLOW_TIMER_CLOCK	Specifies to use the clock source specified by M_TIMER_CLOCK_SOURCE. INFO		d				a	h	i							
M_ROTARY_ENCODER*n*	Specifies to use rotary decoder *n* as the clock source, where *n* is the number of the rotary decoder. INFO		d				a	h	i							
M_SYSCLK	Specifies to use the allocated system's clock source.		d				a	h	i							
M_TIMER_DURATION +	Inquires the duration for the active portion of the timer's output signal. Note, an error is generated if the specified duration cannot be respected. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*		d				a	h	i							
0 <= *Value* <= *Max. value*	Specifies the duration of the active portion of the timer output signal. INFO		d				a	h	i							

M_TIMER_OUTPUT_INVERTER +	Inquires whether the output of the timer should be inverted. This causes the low portion of the signal (the delay period) to be high and the high portion of the signal (the active portion) to be low. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_DISABLE	Specifies not to invert the output of the timer.		d				a	h	i							
M_ENABLE	Specifies to invert the output of the timer.		d				a	h	i							
M_TIMER_STATE +	Inquires the state of the specified timer. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_DISABLE	Specifies that the timer is disabled.		d				a	h	i							
M_ENABLE	Specifies that the timer is enabled.		d				a	h	i							
M_TIMER_TRIGGER_ACTIVATION +	Inquires the signal variation upon which to generate a timer trigger, if the specified timer is enabled. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_ANY_EDGE	Specifies that a timer trigger will be generated both upon a high-to-low and a low-to-high signal transition.		d				a	h	i							
M_EDGE_FALLING	Specifies that a timer trigger will be generated upon a high-to-low signal transition.		d				a	h	i							
M_EDGE_RISING	Specifies that a timer trigger will be generated upon a low-to-high signal transition.		d				a	h	i							
M_LEVEL_HIGH	Specifies that a timer trigger is continuously issued during a high signal polarity.		d				a	h	i							
M_LEVEL_LOW	Specifies that a timer trigger is continuously issued during a low signal polarity.		d				a	h	i							
M_TIMER_TRIGGER_OVERLAP +	Inquires how to deal with a new trigger that occurs while the previously triggered timer (both its delay and duration) has not expired. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_DEFAULT	Same as M_OFF.		d				a	h	i							
M_LATCH	Specifies that a trigger received, while the associated timer has not expired, will be latched (stored). INFO		d				a	h	i							
M_OFF	Specifies that a new trigger is ignored.		d				a	h	i							
M_RESET	Specifies that a new trigger automatically resets the timer (regardless of whether it is in its delay or active period) and then restarts the timer. INFO		d				a	h	i							

M_TIMER_TRIGGER_SOURCE +	Inquires the trigger source for the specified timer. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i							
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the trigger source for the specified timer, where *n* is the number of the auxiliary signal. INFO		d				a	h	i							
M_CONTINUOUS	Specifies to run the specified timer in periodic mode; no actual trigger signal is used. INFO		d				a	h	i							
M_EXPOSURE	Specifies to use the exposure signal as the trigger source.								i							
M_GRAB_TRIGGER_READY	Specifies to route the internal grab trigger ready signal to the specified signal.								i							
M_IO_COMMAND_LIST*n* +	Specifies to use a bit of the I/O command register of I/O command list *n*, where *n* is the number of the I/O command list. INFO This value is combined with a value from the following table: • To specify which I/O command register bit was used		d				a	h	i							
M_ROTARY_ENCODER*n*	Specifies to use rotary decoder *n* as the trigger source, where *n* is the number of the rotary decoder. INFO		d				a	h	i							
M_SOFTWARE	Specifies to use a software trigger as the trigger source. INFO		d				a	h	i							
M_TIMER_STROBE	Specifies to route the strobe's timer signal to the specified signal.								i							
M_TIMER*n*	Specifies to use the output signal of the specified timer as the trigger source, where *n* is the timer number. INFO		d				a	h	i							

M_TIMER_USAGE +	Inquires how the timer should be used. SET You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i								
	System specific																
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a										
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i								
M_PULSE_GENERATION	Specifies the normal use of the timer, as described in the Timers and coordinating events section of Chapter 56: I/O signals and communicating with external devices.		d				a	h	i								
M_PULSE_MEASUREMENT	Specifies to use the timer to measure the duration of the pulse that occurs on the timer's trigger source. INFO		d				a	h	i								
M_TIMER_VALUE +	Inquires the current value of the timer's duration. You must specify a combination value from the following table: • To specify which on-board timer to inquire		d				a	h	i								
	System specific																
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a										
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*		d				a	h	i								
Value >= 0	Specifies current value of the timer's duration. If M_TIMER_DELAY_CLOCK_SOURCE is set to M_SYSCLK or a frequency is specified using M_TIMER_DELAY_CLOCK_FREQUENCY, this value is specified in nsec. Otherwise, it is specified in number of signal transitions on the source signal.		d				a	h	i								

Combination value for the values listed in For inquiring the settings of a timer.

You must add the following value to the above-mentioned values to specify which on-board timer to inquire.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which on-board timer to inquire																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_TIMER*n*	Specifies on-board timer *n*, where *n* is the number of the timer. INFO		d				a	h	i								

Combination values for M_TIMER_DELAY; M_TIMER_DURATION.

You can add one of the following values to the above-mentioned values to determine the maximum or minimum value for the setting.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the maximum or minimum value for the setting																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_MAX_VALUE	Specifies the maximum value for this setting.		d				a	h								
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
M_MIN_VALUE	Specifies the minimum value for this setting.		d				a	h								
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									

The following inquire types and inquire values allow you to inquire about the settings of a rotary decoder.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the settings of a rotary decoder																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_ROTARY_ENCODER_BIT0_SOURCE +	Inquires the auxiliary input signal on which to receive bit 0 of the 2-bit Gray code. SET You must specify a combination value from the following table: • To specify which rotary decoder to inquire															
	System specific															
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a									
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i							

	M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the signal on which to receive bit 0 of the 2-bit Gray code, where *n* is the number of the auxiliary signal. INFO		d				a	h	i						
M_ROTARY_ENCODER_BIT1_SOURCE +		Inquires the auxiliary input signal on which to receive bit 1 of the 2-bit Gray code. SET You must specify a combination value from the following table: • To specify which rotary decoder to inquire		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
	M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the signal on which to receive bit 1 of the 2-bit Gray code, where *n* is the number of the auxiliary signal. INFO		d				a	h	i						
M_ROTARY_ENCODER_OUTPUT_MODE +		Inquires the rotary decoder's counter value and/or the direction of movement upon which the rotary decoder should output a pulse. SET You must specify a combination value from the following table: • To specify which rotary decoder to inquire		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
	M_POSITION_TRIGGER	Specifies to output a pulse upon the trigger generated by M_ROTARY_ENCODER_POSITION_TRIGGER.		d				a	h	i						
	M_STEP_ANY	Specifies to output a pulse upon any change in the rotary decoder's counter value (position change in any direction).		d				a	h	i						
	M_STEP_BACKWARD	Specifies to output a pulse upon a rotary decoder counter decrement only.		d				a	h	i						

	M_STEP_FORWARD	Specifies to output a pulse upon a rotary decoder counter increment only.		d				a	h	i						
	M_STEP_FORWARD_NEW_POSITIVE	Specifies to output a pulse upon a rotary decoder counter increment of a new value that has not been reached before. INFO		d				a	h	i						
M_ROTARY_ENCODER_POSITION +		Inquires the current value of the rotary decoder's counter. You must specify a combination value from the following table: • To specify which rotary decoder to inquire		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
	0 <= *Value* <= 429497295	Specifies the current value of the counter.		d				a	h	i						
M_ROTARY_ENCODER_POSITION_TRIGGER +		Inquires the value of the rotary decoder's counter upon which a trigger is generated. SET You must specify a combination value from the following table: • To specify which rotary decoder to inquire		d				a	h	i						
		System specific														
		[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
	0 <= *Value* <= 0*xFFFFFFFF*	Specifies the value of the counter upon which a trigger is generated. INFO		d				a	h	i						
M_ROTARY_ENCODER_RESET_ACTIVATION +		Inquires the signal transition upon which to reset the rotary decoder's counter to 0. SET You must specify a combination value from the following table: • To specify which rotary decoder to inquire		d				a	h	i						
		System specific														

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
M_EDGE_FALLING	Specifies to reset the rotary decoder upon a high-to-low signal transition.		d				a	h	i						
M_EDGE_RISING	Specifies to reset the rotary decoder upon a low-to-high signal transition.		d				a	h	i						
M_ROTARY_ENCODER_RESET_SOURCE +	Inquires the signal source to use to reset the rotary decoder's counter to 0. SET You must specify a combination value from the following table: • To specify which rotary decoder to inquire		d				a	h	i						
	System specific														
	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a								
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i						
M_NULL	Specifies not to reset using a hardware signal source.		d				a	h	i						
M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the trigger source for the specified timer, where *n* is the number of the auxiliary signal. INFO		d				a	h	i						
M_POSITION_TRIGGER	Specifies to use the trigger signal generated by the rotary decoder when the counter reaches the value specified with M_ROTARY_ENCODER_POSITION_TRIGGER.		d				a	h	i						
M_ROTARY_ENCODER_STATE +	Inquires whether the specified rotary decoder is enabled. SET You must specify a combination value from the following table: • To specify which rotary decoder to inquire		d				a	h	i						
	System specific														

	[This is only applicable to Zebra 4Sight] This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7.						a										
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d				a	h	i								
M_DISABLE	Specifies that the rotary decoder is disabled.		d				a	h	i								
M_ENABLE	Specifies that the rotary decoder is enabled.		d				a	h	i								

Combination value for the values listed in For inquiring the settings of a rotary decoder.

You must add the following value to the above-mentioned values to specify which rotary decoder to inquire.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which rotary decoder to inquire about																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ROTARY_ENCODER*n*	Specifies rotary decoder *n*, where *n* is the number of the rotary decoder. INFO		d				a	h	i								

The following control types allow you to control the settings of a data latch associated with the rotary decoders of your system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling the settings of a data latch associated with rotary encoders																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																	
M_SYS_DATA_LATCH_STATE +	Inquires the state of the specified data latch. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. SET You must specify a combination value from the following table: • To specify which data latch to affect																
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*						a										
M_DISABLE	Specifies that the data latch is disabled.						a										
M_ENABLE	Specifies that the data latch is enabled. INFO						a										

M_SYS_DATA_LATCH_TRIGGER_ACTIVATION +		Inquires the trigger signal transition upon which to store the specified information to the specified data latch. To set the signal with which to trigger the data latch, use M_SYS_DATA_LATCH_TRIGGER_SOURCE. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. SET You must specify a combination value from the following table: • To specify which data latch to affect						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*						a								
	M_ANY_EDGE	Specifies that the specified information is stored in the data latch both upon a high-to-low and a low-to-high signal transition.						a								
	M_EDGE_FALLING	Specifies that the specified information is stored in the data latch upon a high-to-low signal transition.						a								
	M_EDGE_RISING	Specifies that the specified information is stored in the data latch upon a low-to-high signal transition.						a								
M_SYS_DATA_LATCH_TRIGGER_SOURCE +		Inquires what triggers storing the specified information to the specified data latch. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. SET You must specify a combination value from the following table: • To specify which data latch to affect						a								
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*						a								
	M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the trigger source, where *n* is the number of the auxiliary signal. INFO						a								
	M_TIMER*n*	Specifies to use the output signal of timer *n* as the trigger source, where *n* is the number of the timer. INFO						a								

M_SYS_DATA_LATCH_TYPE +		Inquires which rotary decoder the specified data latch will store the position counter value of when the data latch is triggered. This constant is only available on a Host system if the Host system was previously allocated on a Zebra 4Sight EV6/EV7. It is not available on Zebra 4Sight XV6/XV7. SET You must specify a combination value from the following table: • To specify which data latch to affect						a									
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*						a									
	M_ROTARY_ENCODER*n*	Specifies to store the value of the counter of rotary decoder *n*, where *n* is a number between 1 and 2. INFO						a									

Combination value for the values listed in For controlling the settings of a data latch associated with rotary encoders.

You must add the following value to the above-mentioned values to specify which data latch to affect.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which data latch to set																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_LATCH*n*	Specifies which data latch to inquire, where *n* is 1 or 2.						a									

The following inquire types allow you to inquire UART settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For UART settings																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_COM_PORT_NUMBER +	Inquires the Microsoft Windows COM port number associated with the specified UART. Note that the number of UARTs available differs, depending on the version of your board. You must specify which COM port to inquire. See below for combination values.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*										j	k				
Value	Specifies the Microsoft Windows COM port number associated with the specified UART.										j	k				
M_UART_BYTES_READ +	Inquires the number of bytes read from the UART, after waiting for the UART read operation (using MsysControl() with M_UART_READ_STRING) to complete.										j	k				

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	Value	Specifies the number of bytes read from the UART.									j	k				
M_UART_BYTES_WRITTEN +		Inquires the number of bytes written to the UART, after waiting for the UART write operation (using MsysControl() with M_UART_WRITE_STRING) to complete.									j	k				
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	Value	Specifies the number of bytes written to the UART, in bytes.									j	k				
M_UART_DATA_PENDING +		Inquires the number of pending data bytes that are currently available to read from the UART input buffer.									j	k				
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	Value	Specifies the number of pending data bytes.									j	k				
M_UART_DATA_SIZE +		Inquires the number of data bits per character that is sent or received by the UART. SET									j	k				
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	7	Specifies that the data length is 7 bits.									j	k				
	8	Specifies that the data length is 8 bits.									j	k				
M_UART_INTERFACE_TYPE +		Inquires the type of interface used by the UART.									j	k				
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	M_RS232	Specifies that an RS-232 interface is used.									j	k				
	M_RS485	Specifies that an RS-485 interface is used.									j	k				
M_UART_PARITY +		Inquires whether character data is sent or received with a parity bit and how the parity bit is set. SET									j	k				
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	M_DISABLE	Specifies that no extra bit is added (no parity).									j	k				

	M_EVEN	Specifies that the number of 1's will be even.									j	k				
	M_ODD	Specifies that the number of 1's will be odd.									j	k				
M_UART_READ_STRING_MAXIMUM_SIZE +		Inquires the maximum length of the string to be read using M_UART_READ_STRING. SET									j	k				
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	Value	Specifies the maximum length of the string, in bytes.									j	k				
M_UART_READ_STRING_SIZE +		Inquires the number of bytes to read when calling MsysControl() with M_UART_READ_STRING. If this length is specified by M_UART_STRING_DELIMITER, the returned result will be M_DEFAULT. SET									j	k				
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	M_DEFAULT	Specifies the use of M_UART_STRING_DELIMITER to delineate the end of the string to read.									j	k				
	Value	Specifies the string length, in bytes.									j	k				
M_UART_SPEED +		Inquires the baud rate of the UART. SET									j	k				
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	300;600;1200;1800;2400;4800;7200;9600;14400;19200;38400;57600;115200	Specifies the baud rate of the UART. INFO									j	k				
M_UART_STOP_BITS +		Inquires the number of extra data bit(s) that are added to each character to indicate the end of a character. SET									j	k				
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
	1	Specifies that there is 1 stop bit.									j	k				
	2	Specifies that there are 2 stop bits.									j	k				
M_UART_STRING_DELIMITER +		Inquires the character used to terminate strings of incoming or outgoing data. The delimiter is used but not sent when writing data with M_UART_WRITE_STRING; it is read for incoming data with M_UART_READ_STRING. SET									j	k				

UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
Value	Specifies the character used to terminate strings.									j	k				
M_UART_TIMEOUT +	Inquires the maximum time to wait between each byte when reading incoming data. SET									j	k				
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
M_INFINITE	Waits indefinitely.									j	k				
Value	Specifies the time to wait, in msec.									j	k				
M_UART_WRITE_STRING_SIZE +	Inquires the length of the string to be sent to the UART for transmission. SET									j	k				
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*									j	k				
Value	Specifies the string length, in bytes.									j	k				

Combination value for the values listed in For UART settings.

You can add the following value to the above-mentioned values to set which UART to inquire.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For COM Ports and UARTs																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_UART_NB(AIL_INT DeviceToInquire)	Specifies which UART should be inquired.										j	k				
	Parameters															
	DeviceToInquire Specifies a device (UART) to inquire.										j	k				
	M_DEV*n* — Specifies the device to inquire, where *n* can be from 0 to 15.										j	k				
	System specific															
	There are 2 UARTS available on Zebra Radient eV-CL DB and eV-CL DF, 4 on Zebra Radient eV-CL QB, and 1 on Zebra Radient eV-CL SB and SF. Note that, using the CLProtocol will occupy one COM port per allocated digitizer.										j					

		There are 2 UARTS available on Zebra Rapixo CL Pro DB and DF, 4 on Zebra Rapixo CL Pro QB, and 1 on Zebra Rapixo CL Pro SF. Note that, using the CLProtocol will occupy one COM port per allocated digitizer.										k					

The following inquire types and inquire values allow you to inquire the details of an action command. Action commands require both an Aurora Imaging Library-side and a camera-side configuration. The following inquire types and inquire values inquire the Aurora Imaging Library-side of the action command. To inquire the camera-side, use MdigInquireFeature() with the appropriate feature values. For more information, refer to the Triggering simultaneous actions in multiple GigE Vision cameras section of the Zebra GigE Vision driver chapter in the Hardware-specific Notes.

> To inquire about features of the camera, use MdigInquireFeature() with a digitizer allocated for the camera on an Aurora Imaging Library GigE system.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about an action command																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_GC_ACTION_ACKNOWLEDGE_NUMBER +	Inquires the number of acknowledgments that Aurora Imaging Library should expect to receive when the action command is issued. SET You must specify a combination value from the following table: • To set which action command to inquire															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*				f	g										
Value	Specifies the number of acknowledgments expected.				f	g										
M_GC_ACTION_DEVICE_KEY +	Inquires the action device key for the action command. The action device key identifies the cameras on which the action should be performed. SET You must specify a combination value from the following table: • To set which action command to inquire		d		f	g										
	System specific															
	To inquire the features of the camera, use MdigInquireFeature() with a digitizer allocated for the camera on an Aurora Imaging Library GigE system. If using an Aurora Imaging Library Concord PoE system, this inquire type is only available for Zebra Concord PoE with ToE. To inquire about an action command when using the Zebra Concord PoE base model, use an Aurora Imaging Library GigE system with this inquire type.		d													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d		f	g										

	Value	Specifies the action device key.		d		f	g									
M_GC_ACTION_GROUP_KEY +		Inquires the action group key for the action command. The action group key identifies which action you want to perform on the camera. SET You must specify a combination value from the following table: • To set which action command to inquire		d		f	g									
		System specific														
		To inquire the features of the camera, use MdigInquireFeature() with a digitizer allocated for the camera on an Aurora Imaging Library GigE system. If using an Aurora Imaging Library Concord PoE system, this inquire type is only available for Zebra Concord PoE with ToE. To inquire about an action command when using the Zebra Concord PoE base model, use an Aurora Imaging Library GigE system with this inquire type.		d												
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d		f	g									
	Value	Specifies the action group key.		d		f	g									
M_GC_ACTION_GROUP_MASK +		Inquires the action group mask for the action command. In the case where you need one (or more) cameras to temporarily ignore an action command, you can mask out the action command by changing the action group mask. SET You must specify a combination value from the following table: • To set which action command to inquire		d		f	g									
		System specific														
		To inquire the features of the camera, use MdigInquireFeature() with a digitizer allocated for the camera on an Aurora Imaging Library GigE system. If using an Aurora Imaging Library Concord PoE system, this inquire type is only available for Zebra Concord PoE with ToE. To inquire about an action command when using the Zebra Concord PoE base model, use an Aurora Imaging Library GigE system with this inquire type.		d												
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d		f	g									
	Value	Specifies the action group mask.		d		f	g									

M_GC_ACTION_TIME +		Inquires the time at which the action command should execute. SET You must specify a combination value from the following table: • To set which action command to inquire				f	g										
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*				f	g										
	Value	Specifies the time at which to execute the action command on your camera, relative to the time at which the action command was sent, in sec.				f	g										

The following inquire types and inquire values allow you to inquire the details of a Trigger-over-Ethernet packet for transmission using a ToE module. The Trigger-over-Ethernet packet can be sent as an action command or a GigE Vision software trigger. Action commands and GigE Vision software triggers require both an Aurora Imaging Library-side and a camera-side configuration. The following inquire types and inquire values inquire about the Aurora Imaging Library-side of the action command or GigE Vision software trigger. To configure the camera-side, you need to use a digitizer allocated for the camera on an Aurora Imaging Library GigE Vision system and use MdigControlFeature() with the appropriate feature values. For more information, refer to the Triggering simultaneous actions in multiple GigE Vision cameras section of the Zebra GigE Vision driver chapter in the Hardware-specific Notes.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a Trigger-over-Ethernet packet (action command or GigE Vision software trigger) for transmission using a ToE module																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_GC_TRIGGER_SELECTOR +		Inquires the type of GigE Vision trigger that should take place on the camera upon receiving the ToE packet (for example, FrameStart). For the camera to know what should be triggered upon receiving the packet, this control type must match the trigger selector on your camera. To configure the trigger selector on your camera, use MdigControlFeature() with the appropriate GenICam SFNC feature (for example, TriggerSelector). You must specify a combination value from the following table: • To specify which GigE Vision software trigger to inquire															
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*		d													
	AIL_TEXT("*FeatureName*") 1	Specifies the type of trigger; see your GigE Vision camera's documentation for a list of available types.		d													
M_TRIGGER_SOURCE +		Inquires the event that will cause the ToE module to send the specified action command or GigE Vision software trigger as a ToE packet. SET You must specify a combination value from one of the following tables: • To specify which GigE Vision software trigger to inquire • To set which action command to inquire		d													
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d													

	M_AUX_IO*n*	Specifies to use auxiliary input signal *n* as the trigger source, where *n* is the number of the auxiliary signal. INFO		d													
	M_IO_COMMAND_LIST*n* +	Specifies to use the I/O command list *n*, where *n* is the number of the I/O command list. INFO This value is combined with a value from the following table: • To specify which I/O command register bit was used		d													
	M_ROTARY_ENCODER*n*	Specifies to use rotary decoder n as the trigger source, where *n* is the number of the rotary decoder. INFO		d													
	M_SOFTWARE*n*	Specifies to use software as a trigger source to trigger the ToE module, where *n* is the number of the software trigger; *n* can be a value between 1 and 4. INFO		d													
	M_TIMER*n*	Specifies to use the output signal of the specified timer as the trigger source, where *n* is the number of the timer. INFO		d													
M_TRIGGER_STATE +		Inquires the state of the specified action command or GigE Vision software trigger in the ToE module. SET You must specify a combination value from one of the following tables: • To specify which GigE Vision software trigger to inquire • To set which action command to inquire		d													
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*		d													
	M_DISABLE	Specifies the ToE packet is disabled.		d													
	M_ENABLE	Specifies the ToE packet is enabled, and will be transmitted when its associated event (specified using M_TRIGGER_SOURCE) occurs.		d													

[1] If the value was passed in a variable, it won't be returned enclosed in AIL_TEXT().

Combination value for M_DEVICE_NAME; M_DISTRIBUTED_AIL_REMOTE_COMPUTER_NAME; M_SERIAL_NUMBER; M_SYSTEM_DESCRIPTOR; M_DISCOVER_DEVICE_MANUFACTURER_INFO + n; M_DISCOVER_DEVICE_MANUFACTURER_NAME + n; M_DISCOVER_DEVICE_MODEL_NAME + n; M_DISCOVER_DEVICE_USER_NAME + n; M_DISCOVER_DEVICE_VERSION + n; M_GC_TRIGGER_SELECTOR.

You can add the following value to the above-mentioned values to get the string's length.

For getting the string size	
InquireType	Description
UserVarPtr - Possible values returned	
M_STRING_SIZE	Retrieves the length of the string, including the terminating null character ("\0").
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*

Combination value for the values listed in For inquiring about an action command, For inquiring a Trigger-over-Ethernet packet (action command or GigE Vision software trigger) for transmission using a ToE module. (exceptions might exist)

You must add the following value to the above-mentioned values to set which action command to inquire. Alternatively, you can add a value from For specifying which GigE Vision software trigger to inquire.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which action command to inquire																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GC_ACTION*n*	Specifies to inquire about action command *n*, where *n* is a value from 0 to 31.		d		f	g										
	System specific															
	If using this inquire type with Zebra Concord PoE with ToE, it cannot be used with M_GC_TRIGGER_SELECTOR.		d													

Combination value for the values listed in For inquiring a Trigger-over-Ethernet packet (action command or GigE Vision software trigger) for transmission using a ToE module; and for the following value: M_GC_TRIGGER_SELECTOR.

You must add the following value to the above-mentioned values to specify which GigE Vision software trigger to inquire. Alternatively, you can add a value from For specifying which action command to inquire.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which GigE Vision software trigger to inquire																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GC_TRIGGER_SOFTWARE*n*	Specified to inquire about the GigE Vision software trigger n, where n is a value from 0 to 31. This combination value cannot be used with M_GC_ACTION_DEVICE_KEY, M_GC_ACTION_GROUP_KEY, or M_GC_ACTION_GROUP_MASK. If using an Aurora Imaging Library Concord PoE system, this inquire type is only available for Zebra Concord PoE with ToE.		d													

Combination value for [when InquireType=M_IO_SOURCE: M_IO_COMMAND_LISTn;], [when InquireType=M_TIMER_TRIGGER_SOURCE: M_IO_COMMAND_LISTn;], [when InquireType=M_TRIGGER_SOURCE: M_IO_COMMAND_LISTn].

The following value is combined with the above-mentioned values to specify which I/O command register bit was used.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which I/O command register bit was used																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_IO_COMMAND_BIT*n*	Specifies I/O command register bit *n*, where *n* represents the bit number. INFO		d				a	h	i							

Return value

The returned value is the requested information, cast to an *AIL_INT*. If the requested information does not fit into an *AIL_INT*, this function will return **M_NULL** or truncate the information.

Example

The following example uses M_BOARD_TYPE. The returned value is then masked so that only the board type is returned.

```
/* To return only the main board type, and not the sub-board types (for example,
M_CXP or M_CL), mask the return value with M_BOARD_TYPE_MASK.
*/

/* Call MsysInquire with M_BOARD_TYPE to inquire the full board type. */
MsysInquire(AilSystem, M_BOARD_TYPE, &BoardType);

/* Use M_BOARD_TYPE_MASK to verify the main board type. */
if ((BoardType & M_BOARD_TYPE_MASK) == M_RAPIXO)
{
/* Perform a Zebra Rapixo-specific task. */
}
```

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.

MsysInquireFeature

Synopsis

Inquires a feature of the camera.

Syntax

```
void MsysInquireFeature(
    AIL_ID SysId,                  //in
    AIL_INT64 InquireType,         //in
    AIL_CONST_TEXT_PTR FeatureName, //in
    AIL_INT64 UserVarType,         //in
    void *UserVarPtr               //out
)
```

Description

This function allows you to directly inquire various GenICam standard feature naming convention (SFNC) features and manufacturer-specific features of the camera. When used with a GenICam SFNC-compliant camera, it allows you to directly inquire various manufacture-specific features specified with the camera's device description file (XML). Note that for the purpose of this function, a feature can also be of type category or command. Although category and command features lack a feature value, all other attributes of the feature can be inquired. For a list of standard category feature names, see the Standard Feature Naming Convention or your GenICam SFNC-compliant camera's documentation.

The features described in this function are primarily available to check existing settings using Aurora Imaging Library code or to retrieve information from which you can build your own interface for the camera device's description file (XML). Aurora Imaging Library provides two versions of an interface that you can use to interactively inquire the camera's features. At design-time, you can use Aurora Imaging Intellicam's Feature Browser. At runtime, you can launch Feature Browser, using MsysControl() with M_GC_FEATURE_BROWSER. Both versions of Feature Browser provide a list of available feature names and associated possible values, as well as code snippets with the Aurora Imaging Library functions and Aurora Imaging Library constants associated with the settings selected; you can copy the code snippets to your Aurora Imaging Library application code.

Occasionally, the returned information is a string whose length can change between when you inquires its size and when you inquire its value (for example, the temperature can change from 99 to 100 between calls). In this case, when you inquire the information's length to allocate an array of an appropriate size, you might allocate an array that is too small to retrieve the data and experience an application crash. To avoid an application crash, specify the length of the array passed to the UserVarPtr parameter using the M_FEATURE_USER_ARRAY_SIZE() macro; the data will be truncated to fit in the array and an error will be generated.

To control a camera's manufacturer-specific feature, use MsysControlFeature().

Parameters

SysId

Specifies the identifier of the system accessing the camera to inquire. This parameter must be given a valid system identifier, previously allocated using MsysAlloc().

InquireType

Specifies the type of information to inquire about the feature.

See the Parameter associations section for possible values that can be specified.

FeatureName

Specifies the name of the camera feature to inquire. Note that this parameter expects a case-sensitive string.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the name of the feature

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
AIL_TEXT("*FeatureName*") [1]	Specifies the name of the feature. Note that the feature name is case-sensitive. Refer to your camera's documentation for a list of the features available.			e												b
AIL_TEXT("*Root*") [1]	Specifies to inquire the highest-level feature of the XML structure; this feature is of type category and typically, its subfeatures are also of type category. Note that this value is only available when using M_SUBFEATURE_....			e												b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

UserVarType

Specifies the data type of the address pointed to by the UserVarPtr parameter. If not inquiring the feature's value (using M_FEATURE_VALUE), set this parameter to **M_DEFAULT**.

Note that, in the case of a string (or a multi-byte register), the size of the array passed to the UserVarPtr parameter must be equal to or larger than the returned information; otherwise an application crash will occur.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the UserVarPtr's data type																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_TYPE_BOOLEAN	Specifies that UserVarPtr is passed an address of type *AIL_BOOL*.			e												b
M_TYPE_DOUBLE	Specifies that UserVarPtr is passed an address of type *AIL_DOUBLE*.			e												b
M_TYPE_INT64	Specifies that UserVarPtr is passed an address of type *AIL_INT64*.			e												b
M_TYPE_STRING +	Specifies that UserVarPtr is passed an address of type *AIL_TEXT_CHAR*. You must specify a combination value from the following table: • To specify the length of the array			e												b
M_TYPE_UINT8 +	Specifies that UserVarPtr is passed an address of type *AIL_UINT8*. You must specify a combination value from the following table: • To specify the length of the array			e												b

Combination value for M_TYPE_STRING; M_TYPE_UINT8.

You must add the following value to the above-mentioned values to specify the length of the array.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the length of the array																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FEATURE_USER_ARRAY_SIZE(AIL_INT Length)	Specifies the length of the array.			e												b

		Parameters
		Length Sets the length, in bytes.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_BOOL [optionally, in C++: a reference to a std::vector<AIL_BOOL>]
- array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]
- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]
- array of type AIL_TEXT_CHAR [optionally, in C++: a reference to an AIL_STRING]
- array of type AIL_UINT8 [optionally, in C++: a reference to a std::vector<AIL_UINT8>]
- AIL_BOOL
- AIL_DOUBLE
- AIL_INT
- AIL_INT64

Specifies the address of the variable in which to return the requested information.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr** parameter.

- For specifying the type of information to inquire about the feature and the data type returned
- For inquiring about the enumeration entry of a feature's value
- For inquiring about the subfeatures of a feature

The following inquire types are used to specify the type of information about the feature to inquire and the data type returned.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of information to inquire about the feature and the data type returned																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_FEATURE_ACCESS_MODE +	Inquires whether the specified feature is implemented, available, readable, and/or writable. Note that an error is generated if the feature does not exist in the camera's device description file (XML). To learn if the specified feature exists in the file and is implemented on your camera, use M_FEATURE_PRESENT.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e												b

	Bit-encoded access mode value	Specifies a bit-encoded value that details the access mode. To establish the access mode from the encoded value, use the following macros: • To learn whether the specified feature is available, use the **M_FEATURE_IS_AVAILABLE** macro. Note that if a feature is not available, use **M_FEATURE_IS_IMPLEMENTED** to learn whether the specified feature is only temporarily unavailable or actually not implemented on your camera. A feature that is temporarily unavailable typically relies on the setting of another feature (for example, when a trigger feature is set to off, its associated trigger polarity feature would be unavailable). • To learn whether the specified feature is implemented, use the **M_FEATURE_IS_IMPLEMENTED** macro. For example, a Bayer color feature would not be implemented on a monochrome camera because the hardware cannot support it. • To learn whether the specified feature can be read, use the **M_FEATURE_IS_READABLE** macro. • To learn whether the specified feature can be written to, use the **M_FEATURE_IS_WRITABLE** macro.			e											b
M_FEATURE_CACHING_MODE +		Inquires whether a copy of the specified feature's value is stored in Host memory for faster access and retrieval. If so, the feature is known as a cachable feature. A cachable feature is typically only retrieved from the camera whenever the cache is invalidated (such as, before the first read and before the first read to occur after a write operation). A cachable feature is typically a value that only changes based on user-interaction (setting a control value). For example, a temperature feature is not cachable. To return a true/false value for this inquire, use the **M_FEATURE_IS_CACHABLE** macro on the returned result.			e											b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e											b
	M_FEATURE_CACHING_MODE_NONE	Specifies that the feature is not cachable.			e											b
	M_FEATURE_CACHING_MODE_WRITE_AROUND	Specifies that feature's values are written only to the camera, and the cache is updated the next time the feature's value is inquired.			e											b

M_FEATURE_CACHING_MODE_WRITE_THROUGH	Specifies that when information is written to the camera, the cache is updated simultaneously.			e												b
M_FEATURE_CHANGE_HOOK +	Inquires whether an event occurs when the value of the specified feature changes. If enabled, MsysHookFunction() with M_FEATURE_CHANGE hooks a specified function to the feature change event. SET			e												b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e												b
M_DISABLE	Specifies that no event is generated when the value of the specified feature changes. This is the default value.			e												b
M_ENABLE	Specifies that an event is generated when the value of the specified feature changes.			e												b
M_FEATURE_DEPRECATED +	Inquires whether the specified feature is deprecated and should no longer be used.			e												b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e												b
M_FALSE	Specifies that the feature is not depreciated.			e												b
M_TRUE	Specifies that the feature is depreciated.			e												b
M_FEATURE_DESCRIPTION +	Inquires the description of the specified feature.			e												b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e												b
Value	Specifies the description of the feature.			e												b
M_FEATURE_DISPLAY_NAME +	Inquires the specified feature's display name. Note that this can differ slightly from the internally-used feature name (M_FEATURE_NAME).			e												b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e												b
Value	Specifies the display name of the feature.			e												b
M_FEATURE_EXECUTE_COMPLETED +	Inquires whether the specified executable camera feature (command) has finished executing on your camera. Note that to enable command execution-completion polling, set MsysControl() with M_GC_FEATURE_EXECUTE_POLLING_MODE to M_AUTOMATIC.			e												b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_BOOL*			e											b
	M_FALSE	Specifies that the executable camera feature has not finished executing.			e											b
	M_TRUE	Specifies that the executable camera feature has finished executing.			e											b
M_FEATURE_INCREMENT +		Inquires the value by which the feature's value increments. For example, if a feature's possible values are 2, 4, 6, ..., then the feature's value increments by 2.			e											b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e											b
	Value	Specifies the feature's incremental value.			e											b
M_FEATURE_MAX +		Inquires the maximum possible value for the feature.			e											b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *address of an AIL_INT64*			e											b
	Value	Specifies the maximum possible value for the feature.			e											b
M_FEATURE_MIN +		Inquires the minimum possible value for the feature.			e											b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *address of an AIL_INT64*			e											b
	Value	Specifies the maximum possible value for the feature.			e											b
M_FEATURE_NAME +		Inquires the specified internally-used feature name. Note that this can differ slightly from the displayed name of the feature (M_FEATURE_DISPLAY_NAME).			e											b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e											b
	Value	Specifies the feature's name.			e											b
M_FEATURE_POLLING_INTERVAL +		Inquires the interval between inquiring an executable feature's completion status, when automatically polling (using MsysControl() with M_GC_FEATURE_EXECUTE_POLLING_MODE).			e											b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e											b
	Value >= -1	Specifies the polling interval, in msecs.			e											b

M_FEATURE_PRESENT +	Inquires whether the specified feature is present in the camera's device description file. Note that, if the feature is listed in the camera's device description file (XML), use M_FEATURE_ACCESS_MODE and the **M_FEATURE_IS_IMPLEMENTED** macro to determine if the feature is supported. If the feature is not present in the file, M_FEATURE_PRESENT will return M_NO, whereas M_FEATURE_ACCESS_MODE will generate an error.			e												b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_BOOL*			e												b
M_NO	Specifies that the specified feature is not present.			e												b
M_YES	Specifies that the specified feature is present.			e												b
M_FEATURE_REPRESENTATION +	Inquires the display format for the feature's value. The display format dictates the size of the field and its form (for example, check box, slider, text-box) used in a user interface for the camera's device description file (XML), such as Feature Browser (accessible through Aurora Imaging Intellicam or MsysControl() with M_GC_FEATURE_BROWSER). Note that this inquire type can only be used with a feature that has a data type of *AIL_INT64* or *AIL_DOUBLE*.			e												b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e												b
M_FEATURE_REPRESENTATION_BOOLEAN	Specifies that the feature should be displayed as a checkbox.			e												b
M_FEATURE_REPRESENTATION_HEX_NUMBER	Specifies that the feature should be displayed as a hexadecimal value.			e												b
M_FEATURE_REPRESENTATION_IPV4_ADDRESS	Specifies that the feature should be displayed as an IP address in dotted decimal notation.			e												b
M_FEATURE_REPRESENTATION_LINEAR	Specifies that the feature should be displayed as a slider with the appropriate range shown.			e												b
M_FEATURE_REPRESENTATION_LOGARITHMIC	Specifies that the feature should be displayed as a slider with the appropriate logarithmic range shown.			e												b
M_FEATURE_REPRESENTATION_MAC_ADDRESS	Specifies that the feature should be displayed as a hexadecimal MAC address.			e												b

	M_FEATURE_REPRESENTATION_PURE_NUMBER	Specifies that the feature should be displayed as edit text-box with a decimal display.			e											b
M_FEATURE_SELECTOR_COUNT +		Inquires the number of selectors that are ancestors of the feature. Selectors take an enumerator or integer to index which instance of a subfeature should be accessed, when multiple instances of a subfeature exist. For example, the GenICam standard `TriggerSelector` feature is used to select which type of trigger to configure (for example, `FrameStart` and `AcquisitionStart` are both enumerators that can be passed to `TriggerSelector`). All subfeatures of the selector (such as its source) are referenced from the trigger selector (`TriggerSource[AcquisitionStart]`). This inquire type only returns a value greater than 0 if the feature is a subfeature of a selector. It returns 1 if the feature is a subfeature of a selector. It returns 2 if its selector is also referenced by another selector. The M_FEATURE_SELECTOR_COUNT for `TriggerMode`, in this example, would be 1. To inquires the name of the selector associated with the specified feature, use M_FEATURE_SELECTOR_NAME + n.			e											b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e											b
	Value	Specifies the number of ancestor selectors of the feature.			e											b
M_FEATURE_SELECTOR_NAME + *n* +		Inquires the name of the ancestor selector of the specified subfeature, where *n* is the number of generations. So, if inquiring the feature selector name of the 1st selector of `TriggerSource`, the returned name would be `TriggerSelector`. To inquires the number of selectors that are associated with the specified feature, use M_FEATURE_SELECTOR_COUNT.			e											b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e											b
	Value	Specifies the name associated with the selector index.			e											b
M_FEATURE_SIZE +		Inquires the size of the feature's value.			e											b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e											b
	Value	Specifies the feature's size, in bytes.			e											b

M_FEATURE_STREAMABLE +	Inquires whether the specified feature is streamable. A streamable feature can be stored in the local camera's description file (an XML file stored on your computer). Streamable features are typically not persistent values (such as pixel format and image size). In most cases, a streamable feature is capable of being saved and loaded from file at a later date, possibly on a different camera.			e												b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e												b
M_FALSE	Specifies that the feature is not streamable.			e												b
M_TRUE	Specifies that the feature is streamable.			e												b
M_FEATURE_TOOLTIP +	Inquires the tooltip of the specified feature.			e												b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e												b
Value	Specifies the feature's tooltip.			e												b
M_FEATURE_TYPE +	Inquires the specified feature's type.			e												b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e												b
M_TYPE_BOOLEAN	Specifies that the feature's value is a boolean.			e												b
M_TYPE_CATEGORY	Specifies that the feature is a category feature, and its value cannot be inquired.			e												b
M_TYPE_COMMAND	Specifies that the feature is a command and its value cannot be inquired.			e												b
M_TYPE_DOUBLE	Specifies that the feature's value is a floating-point.			e												b
M_TYPE_ENUMERATION	Specifies that the feature's value is an enumeration.			e												b
M_TYPE_INT64	Specifies that the feature's value is a 64-bit integer.			e												b
M_TYPE_REGISTER	Specifies that the feature's value is mapped to a multi-byte register.			e												b
M_TYPE_STRING	Specifies that the feature's value is a string.			e												b

M_FEATURE_VALUE +	Inquires the current value of the feature.			e											b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_BOOL* **|** *address of an AIL_DOUBLE* **|** *address of an AIL_INT64* **|** *array of type AIL_UINT8 [optionally, in C++: a reference to a std::vector<AIL_UINT8>]* **|** *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e											b
Value	Specifies the feature's value.			e											b
M_FEATURE_VALUE_ARRAY	Inquires the array of values of the specified feature. The array of values will be inquired without needing to be indexed by a selector. This is internally handled by Aurora Imaging Library and simplifies implementation.			e											b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_BOOL [optionally, in C++: a reference to a std::vector<AIL_BOOL>]* **|** *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*			e											b
M_FEATURE_VALUE_ARRAY_SIZE	Inquires the number of elements in the specified feature's array of values. For use with M_FEATURE_VALUE_ARRAY.			e											b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e											b
Value >= 0	Specifies the number of elements in the feature array.			e											b
M_FEATURE_VISIBILITY +	Inquires the visibility level of the specified feature. The visibility level indicates the level of expertise required to view or select the feature. When creating a custom user interface for the camera's device description file (XML), the visibility level of the specified feature can be used to potentially hide the specified feature. The current level of visibility should be set within your custom user interface, and the enumeration entry for the visibility level of the specified feature must be compared against it. If the specified enumeration entry has a level of visibility higher than the current level of visibility, it is hidden.			e											b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e											b
M_FEATURE_VISIBILITY_BEGINNER	Specifies that the feature is suggested for beginners.			e											b

	M_FEATURE_VISIBILITY_EXPERT	Specifies that the feature is suggested for experts.			e												b
	M_FEATURE_VISIBILITY_GURU	Specifies that the feature is suggested for advanced experts.			e												b
	M_FEATURE_VISIBILITY_INVISIBLE	Specifies that the feature should not be shown in the user interface for the camera's device description file (XML). Note that an invisible feature can still be set or inquired using MsysControlFeature() and **MsysInquireFeature**(), respectively.			e												b

When the specified feature is of type enumeration, you can use one of the following to inquire about the specified enumeration entry in the supported enumeration list of the feature. When the information pertains to a specific enumeration entry (for example, M_FEATURE_ENUM_ENTRY_ACCESS_MODE + n), the value is described in relation to its position in the enumerated list, where *n* is the index into the enumerated list.

> Note that these values can only be used when the feature type to inquire is an enumeration.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the enumeration entry of a feature's value																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_FEATURE_ENUM_ENTRY_ACCESS_MODE + *n* +		Inquires whether the specified enumeration entry is implemented, available, readable, and/or writable, where *n* is the index into the enumerated list. Note that an error is generated if the enumeration entry does not exist in the camera's device description file (XML). To learn if the specified enumeration entry exists in the file and is implemented on your camera, use the **M_FEATURE_IS_IMPLEMENTED** macro.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e												b

	Bit-encoded access mode value	Specifies a bit-encoded value that details the access mode. To establish the access mode from the encoded value, use the following macros: • To learn whether the specified enumeration entry is available, use the **M_FEATURE_IS_AVAILABLE** macro. Note that if an enumeration entry is not available, use **M_FEATURE_IS_IMPLEMENTED** to learn whether the specified enumeration entry is only temporarily unavailable or actually not implemented on your camera. An enumeration entry that is temporarily unavailable typically relies on the setting of another feature or enumeration entry. • To learn whether the specified enumeration entry is implemented, use the **M_FEATURE_IS_IMPLEMENTED** macro. • To learn whether the specified enumeration entry can be read, use the **M_FEATURE_IS_READABLE** macro.			e												b
M_FEATURE_ENUM_ENTRY_COUNT +		Inquires the total number of enumeration entries in the supported enumerated list of the feature. Note that this inquire type can only be used when the feature to inquire is of type enumeration.			e												b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e												b
	Value	Specifies the total number of enumeration entries.			e												b
M_FEATURE_ENUM_ENTRY_DESCRIPTION + *n* +		Inquires the description of the specified enumeration entry, where *n* is the index into the enumerated list. Note that this inquire type can only be used when the feature type to inquire is an enumeration.			e												b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e												b
	Value	Specifies the enumeration entry's description.			e												b
M_FEATURE_ENUM_ENTRY_DISPLAY_NAME + *n* +		Inquires the internally used name of the specified enumeration entry, where *n* is the index into the enumerated list.			e												b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e												b

	Value	Specifies the enumeration entry's display name.			e											b
M_FEATURE_ENUM_ENTRY_NAME + *n* +		Inquires the internally-used name of the specified enumeration entry of the feature, where *n* is the index into the enumerated list. Note that this can differ slightly from the display name of the entry (M_FEATURE_ENUM_ENTRY_DISPLAY_NAME + n).			e											b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e											b
	Value	Specifies the enumeration entry's name.			e											b
M_FEATURE_ENUM_ENTRY_TOOLTIP + *n* +		Inquires the tool-tip for the specified enumeration entry, where *n* is the index into the enumerated list.			e											b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e											b
	Value	Specifies the enumeration entry's tooltip.			e											b
M_FEATURE_ENUM_ENTRY_VALUE + *n* +		Inquires the enumeration integer value of the specified enumeration entry, where *n* is the index into the enumerated list.			e											b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e											b
	Value	Specifies the enumeration entry's integer value.			e											b
M_FEATURE_ENUM_ENTRY_VISIBILITY + *n* +		Inquires the visibility level of the specified enumeration entry, where *n* is the index into the enumerated list. The visibility level indicates the level of expertise required to view or select the enumeration entry. When creating a custom user interface for the camera's device description file (XML), the visibility level of the specified enumeration entry can be used to potentially hide the specified enumeration entry. The current level of visibility should be set within your custom user interface, and the visibility level of the specified enumeration entry must be compared against it. If the specified enumeration entry has a level of visibility higher than the current level of visibility, it is hidden.			e											b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e											b
	M_FEATURE_VISIBILITY_BEGINNER	Specifies that the enumeration entry is suggested for beginners.			e											b

	M_FEATURE_VISIBILITY_EXPERT	Specifies that the enumeration entry is suggested for experts.			e													b
	M_FEATURE_VISIBILITY_GURU	Specifies that the enumeration entry is suggested for advanced experts.			e													b
	M_FEATURE_VISIBILITY_INVISIBLE	Specifies that the enumeration entry should not be shown in the user interface for the camera's device description file (XML). Note that an invisible feature can still be set or inquired using MsysControlFeature() and **MsysInquireFeature**(), respectively.			e													b

When the specified feature is of type category or "Root", you can use one of the following to inquire about its subfeatures. This is useful for enumerating the features of your camera.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the subfeatures of a feature																		
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																	
M_SUBFEATURE_COUNT +		Inquires the number of subfeatures or subcategories that the specified category feature (node) has.																
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*			e													b
	Value	Specifies the number of subfeatures.			e													b
M_SUBFEATURE_NAME + *n* +		Inquires the name of the specified subfeature, where *n* is the index of subfeature.			e													b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*			e													b
	Value	Specifies the name of the subfeature.			e													b
M_SUBFEATURE_TYPE + *n* +		Inquires the type of the specified subfeature, where *n* is the index of subfeature.			e													b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*			e													b
	M_TYPE_BOOLEAN	Specifies that the feature's value is a boolean.			e													b
	M_TYPE_CATEGORY	Specifies that the feature is a category.			e													b
	M_TYPE_COMMAND	Specifies that the feature is a command to be executed.			e													b
	M_TYPE_DOUBLE	Specifies that the feature's value is a floating-point.			e													b
	M_TYPE_ENUMERATION	Specifies that the feature's value is an enumeration.			e													b

	M_TYPE_INT64	Specifies that the feature's value is a 64-bit integer.			e													b
	M_TYPE_REGISTER	Specifies that the feature is mapped to a multi-byte register.			e													b
	M_TYPE_STRING	Specifies that the feature's value is a string.			e													b

Combination values for the values listed in For inquiring about the enumeration entry of a feature's value, For inquiring about the subfeatures of a feature; and for the following values: M_FEATURE_ACCESS_MODE; M_FEATURE_CACHING_MODE; M_FEATURE_CHANGE_HOOK; M_FEATURE_DEPRECATED; M_FEATURE_DESCRIPTION; M_FEATURE_DISPLAY_NAME; M_FEATURE_EXECUTE_COMPLETED; M_FEATURE_INCREMENT; M_FEATURE_MAX; M_FEATURE_MIN; M_FEATURE_NAME; M_FEATURE_POLLING_INTERVAL; M_FEATURE_PRESENT; M_FEATURE_REPRESENTATION; M_FEATURE_SELECTOR_COUNT; M_FEATURE_SELECTOR_NAME + n; M_FEATURE_SIZE; M_FEATURE_STREAMABLE; M_FEATURE_TOOLTIP; M_FEATURE_TYPE; M_FEATURE_VALUE; M_FEATURE_VISIBILITY.

You can add one of the following values to the above-mentioned values to specify the GenTL configuration file (XML file) that is associated with the feature.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the configuration file associated with the feature																	
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GENTL_INTERFACE_NUMBER(AIL_INT IndexValue)	Specifies which instance of the GenTL interface configuration file is associated with the feature.			e												b
	Parameters															
	IndexValue Specifies the index of the GenTL interface configuration file.			e												b
	0<= *Value*<=*MsysInquire*(M_GENTL_INTERFACE_COUNT) Specifies the index.			e												b
M_GENTL_SYSTEM	Specifies to display the GenTL system configuration information. This is the default value.			e												b

Combination value for M_FEATURE_DESCRIPTION; M_FEATURE_DISPLAY_NAME; M_FEATURE_NAME; M_FEATURE_SELECTOR_NAME + n; M_FEATURE_TOOLTIP; M_FEATURE_VALUE; M_FEATURE_ENUM_ENTRY_DESCRIPTION + n; M_FEATURE_ENUM_ENTRY_DISPLAY_NAME + n; M_FEATURE_ENUM_ENTRY_NAME + n; M_FEATURE_ENUM_ENTRY_TOOLTIP + n; M_SUBFEATURE_NAME + n.

You can add the following value to the above-mentioned values to get the string's length.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To inquire the size of a string																	
InquireType combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																

M_STRING_SIZE	Retrieves the length of the string, including the terminating null character ("\0"). Note that, when used with M_FEATURE_VALUE, you can only use this combination value if the feature is of type string.																
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*			e												b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib.
DLL	Requires ail11.dll.